Convolutional Neural Networks

https://playground.tensorflow.org

Image to grayscale pixel value

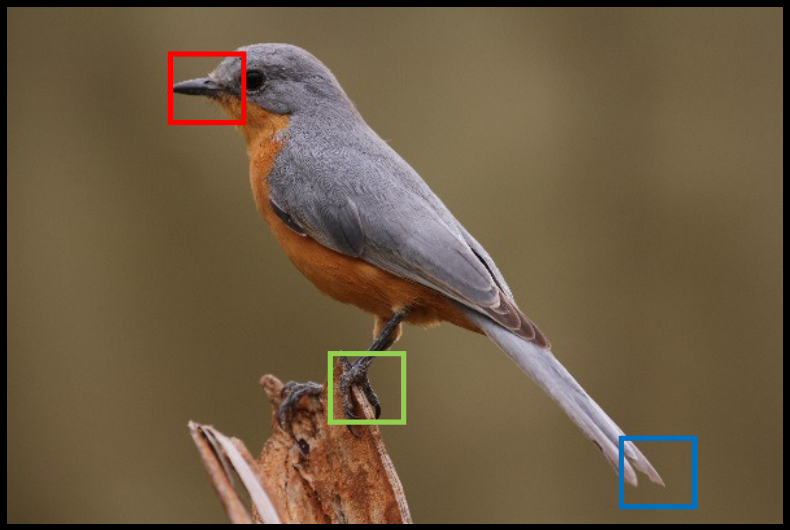


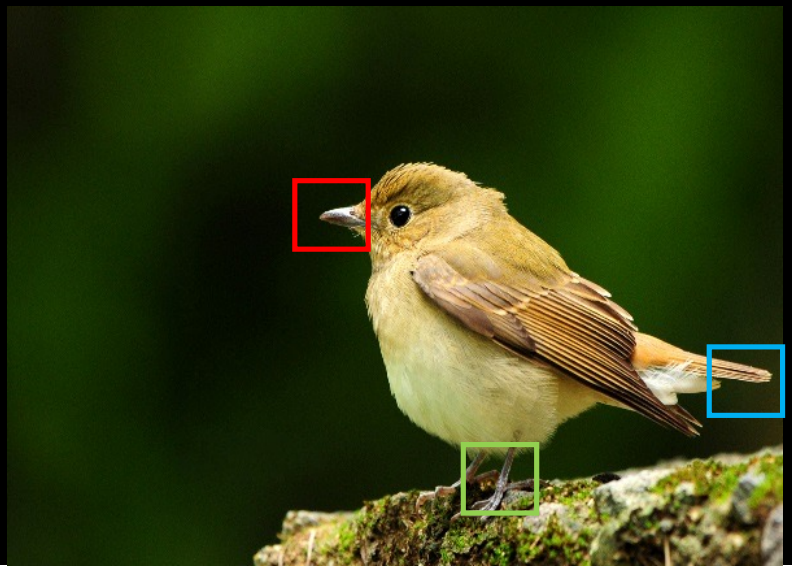
1	0.5	0	0
0.5	0	0	0
0.5	0	0	0
0	0	0.5	1

1
0.5
0
0
0.5
0
0
0
0.5
0
0

numpy.ndarray.flatten

torch.array.view(-1)





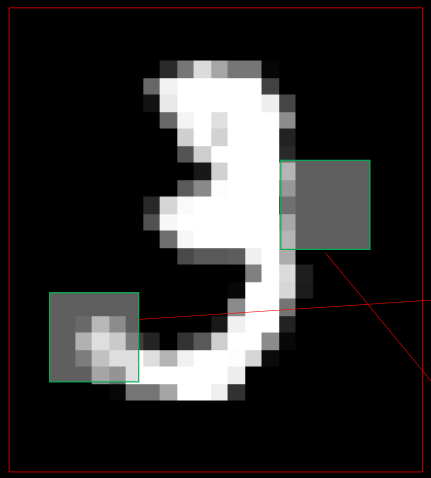
"Convolutional"

Beak (left, right, middle….)

Feet (left, right, middle….)

Tail (left, right, middle….)

Should share the same parameters!



input

1	1	0
1	2	1
0	1	2

1	0	0
1	0	0
1	0	0

*

filter

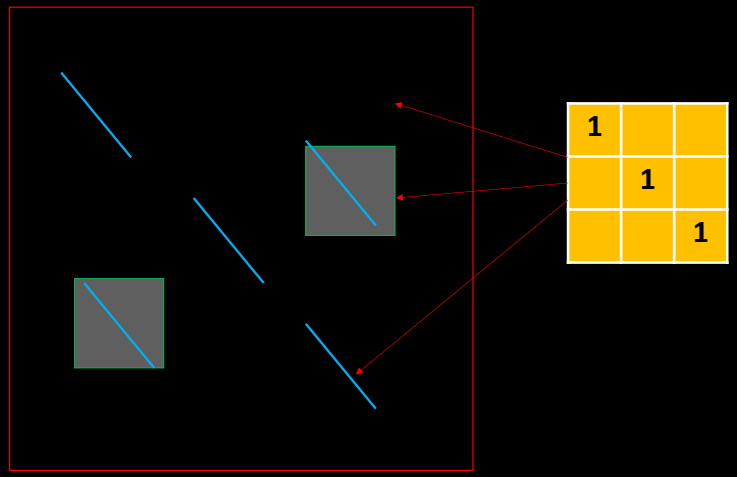
1	0	0
0	1	0
0	0	1

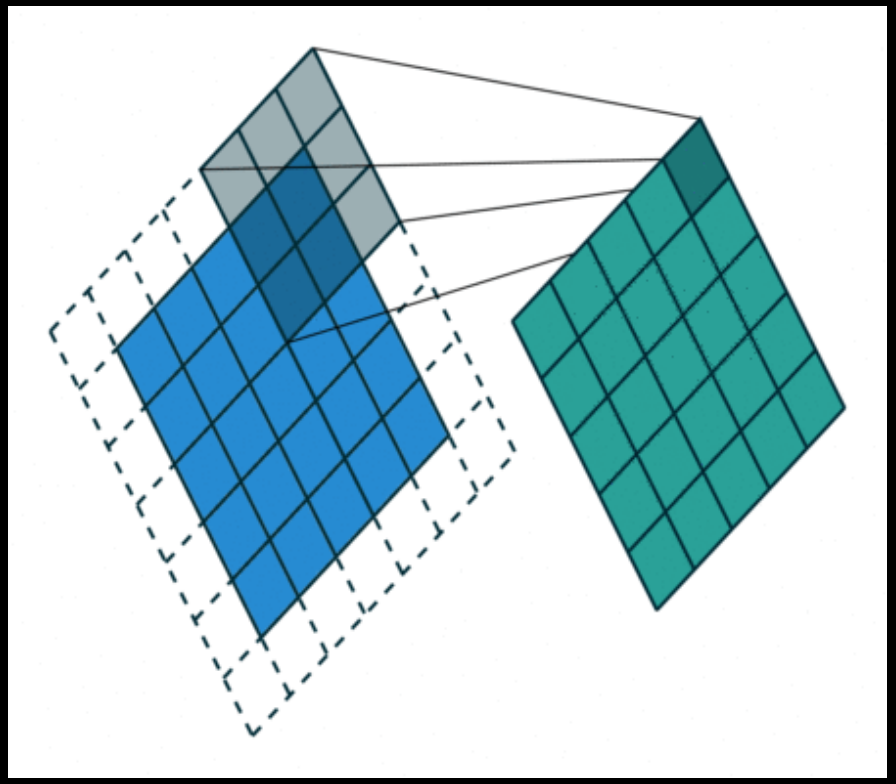
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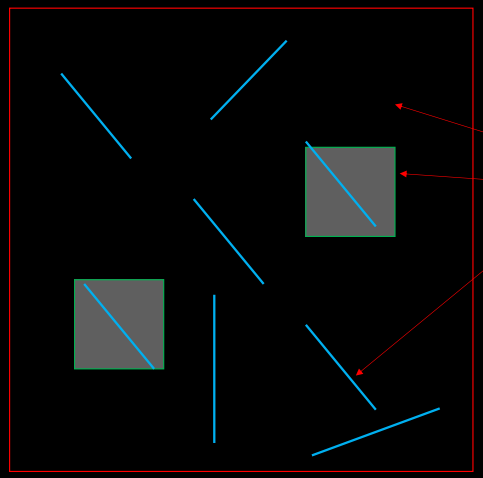
output

1	0	0
0	2	0
0	0	2

1	0	0
0	0	0
0	0	0


1
1
1

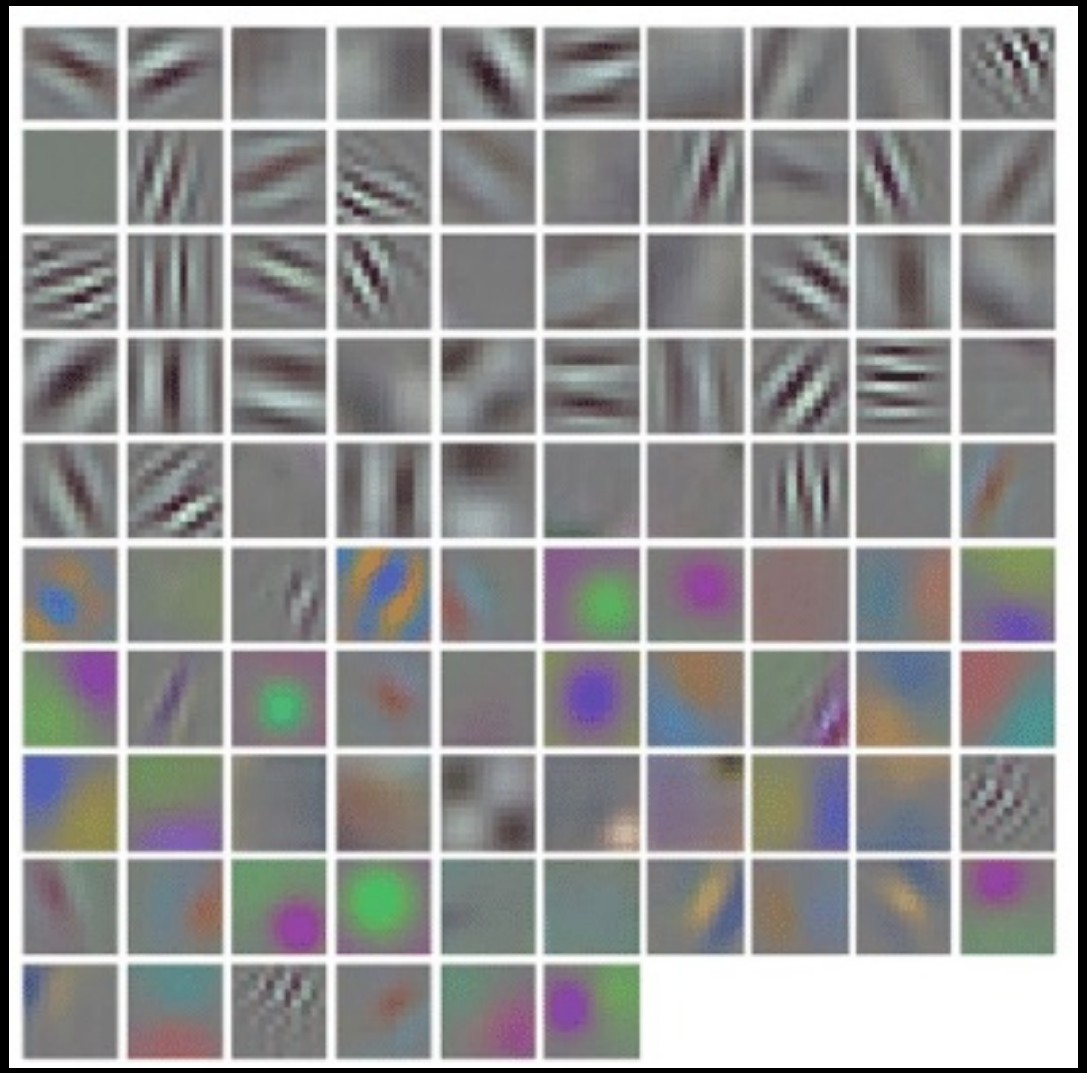


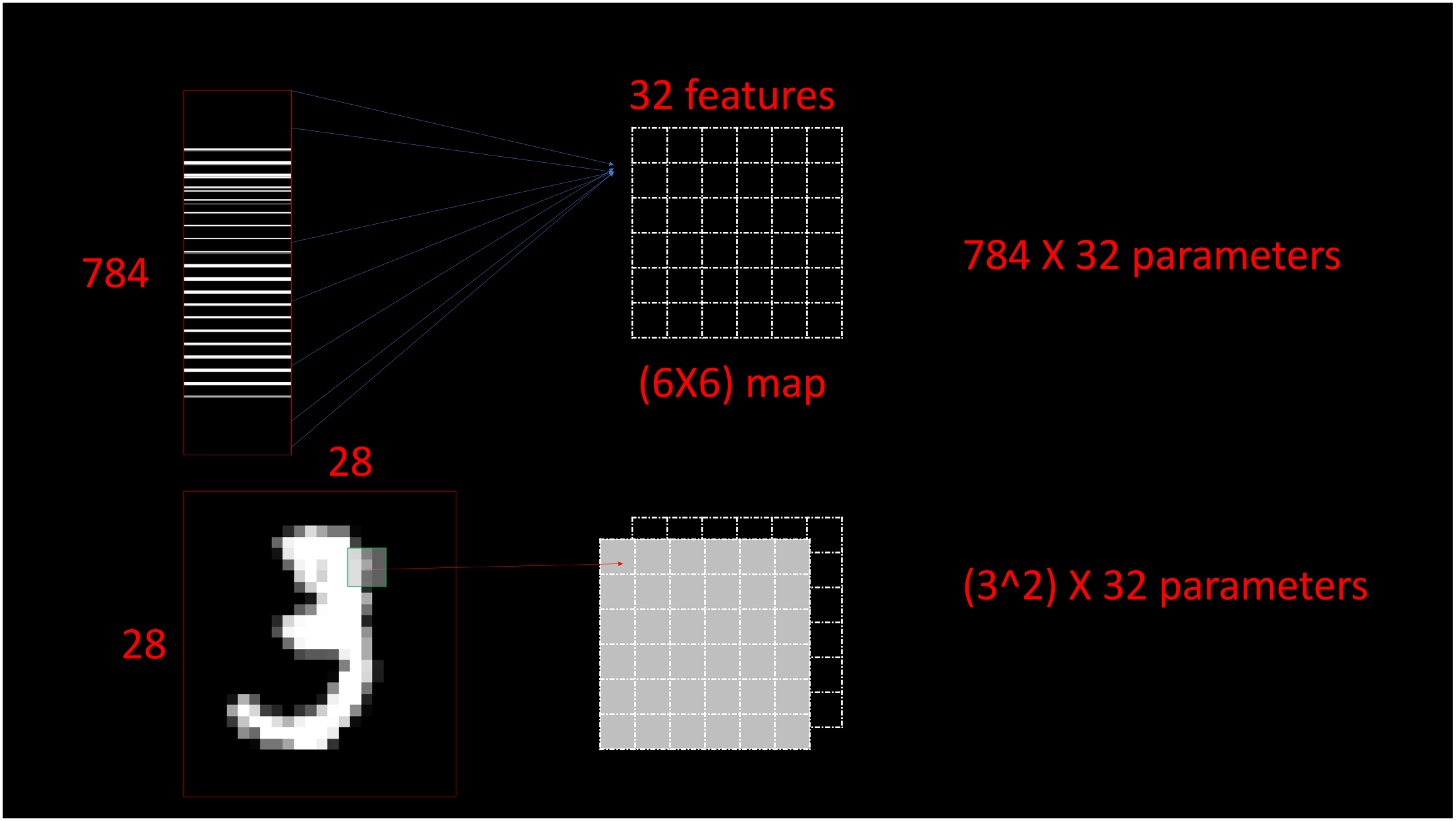


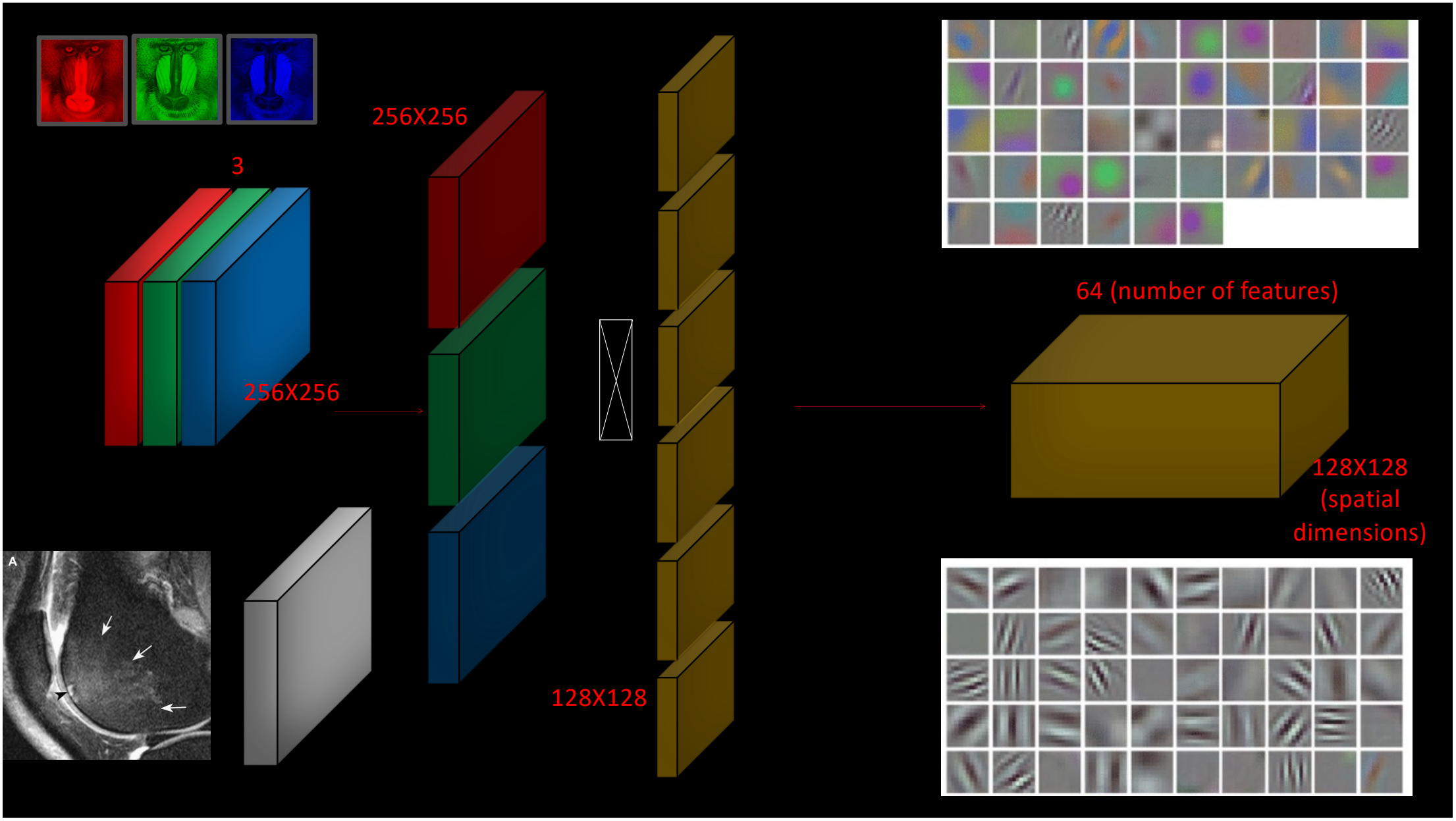
1	0	0
0	1	0
0	0	1

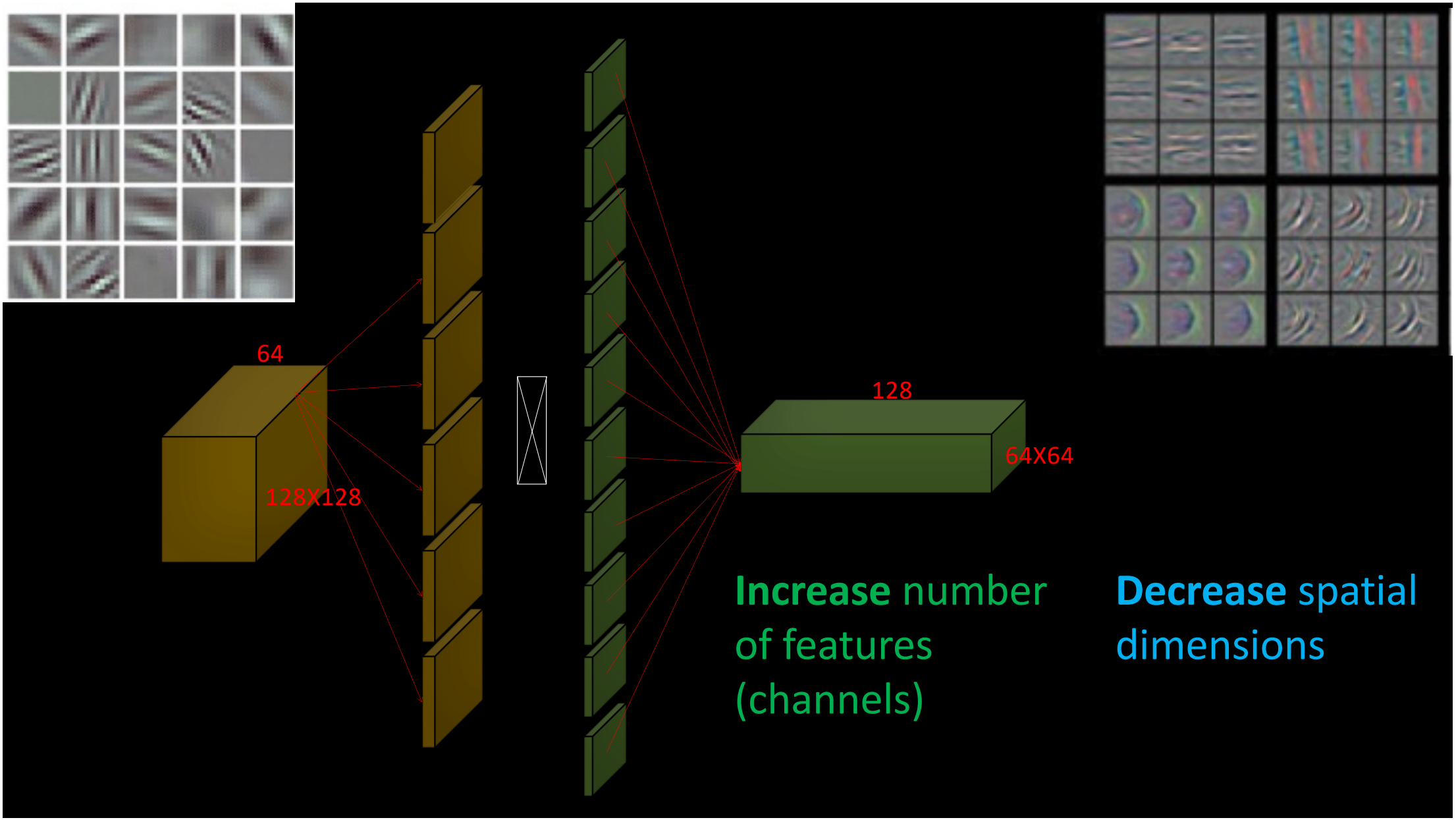
1	0	1
1	0	1
1	0	1

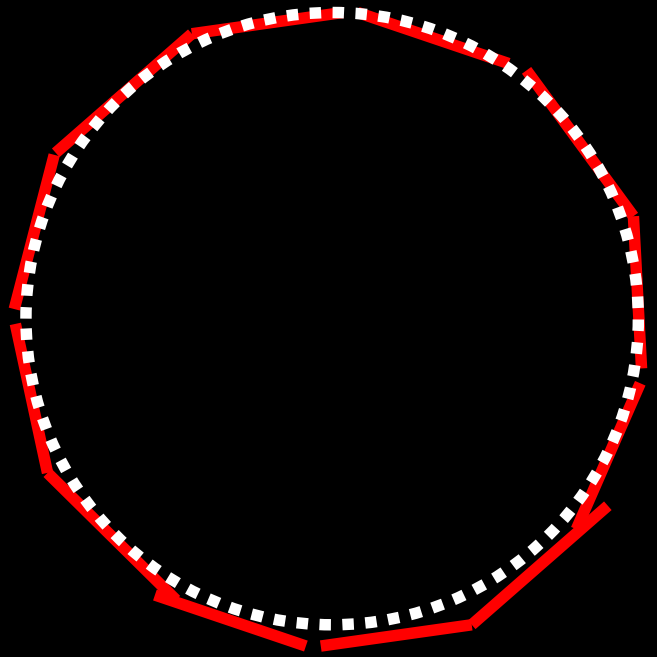
1	1	1
1	1	1
1	1	1

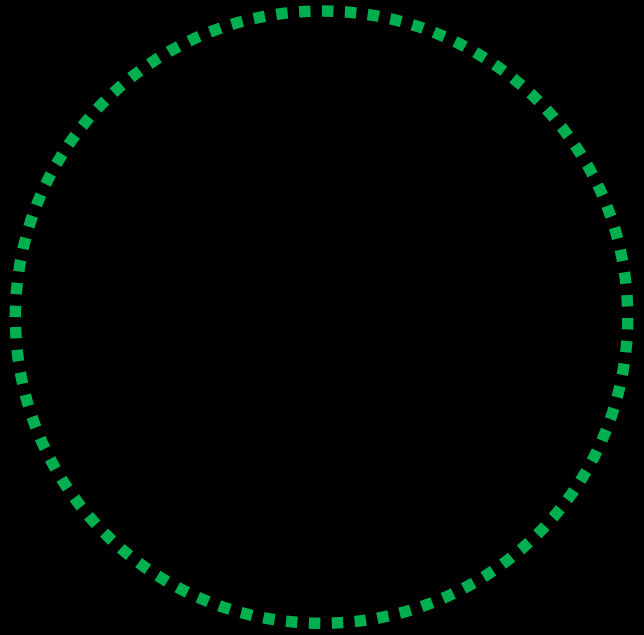



32 features
784
784 X 32 parameters
(6X6) map
28
28
(3^2) X 32 parameters


256X256
3
256X256
128X128
64 (number of features)
128X128
(spatial
dimensions)

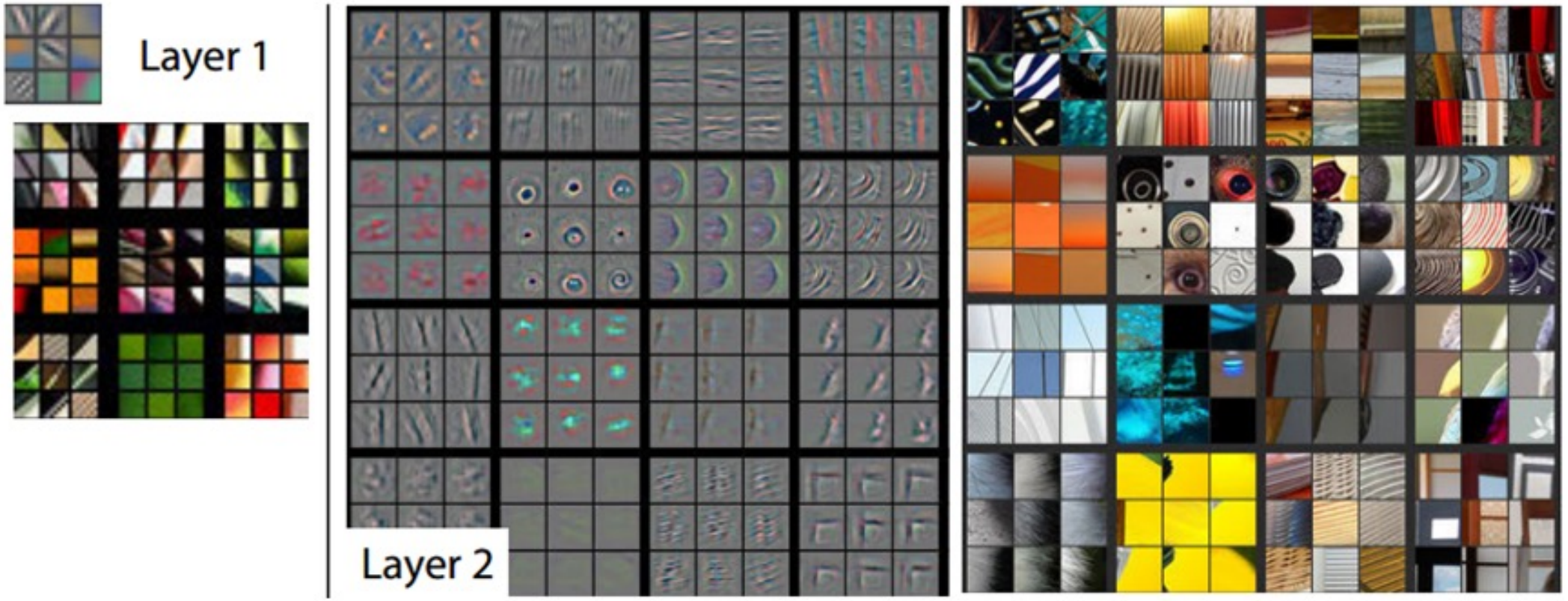

64
128X128
128
64X64
Increase number of features (channels)
Decrease spatial dimensions





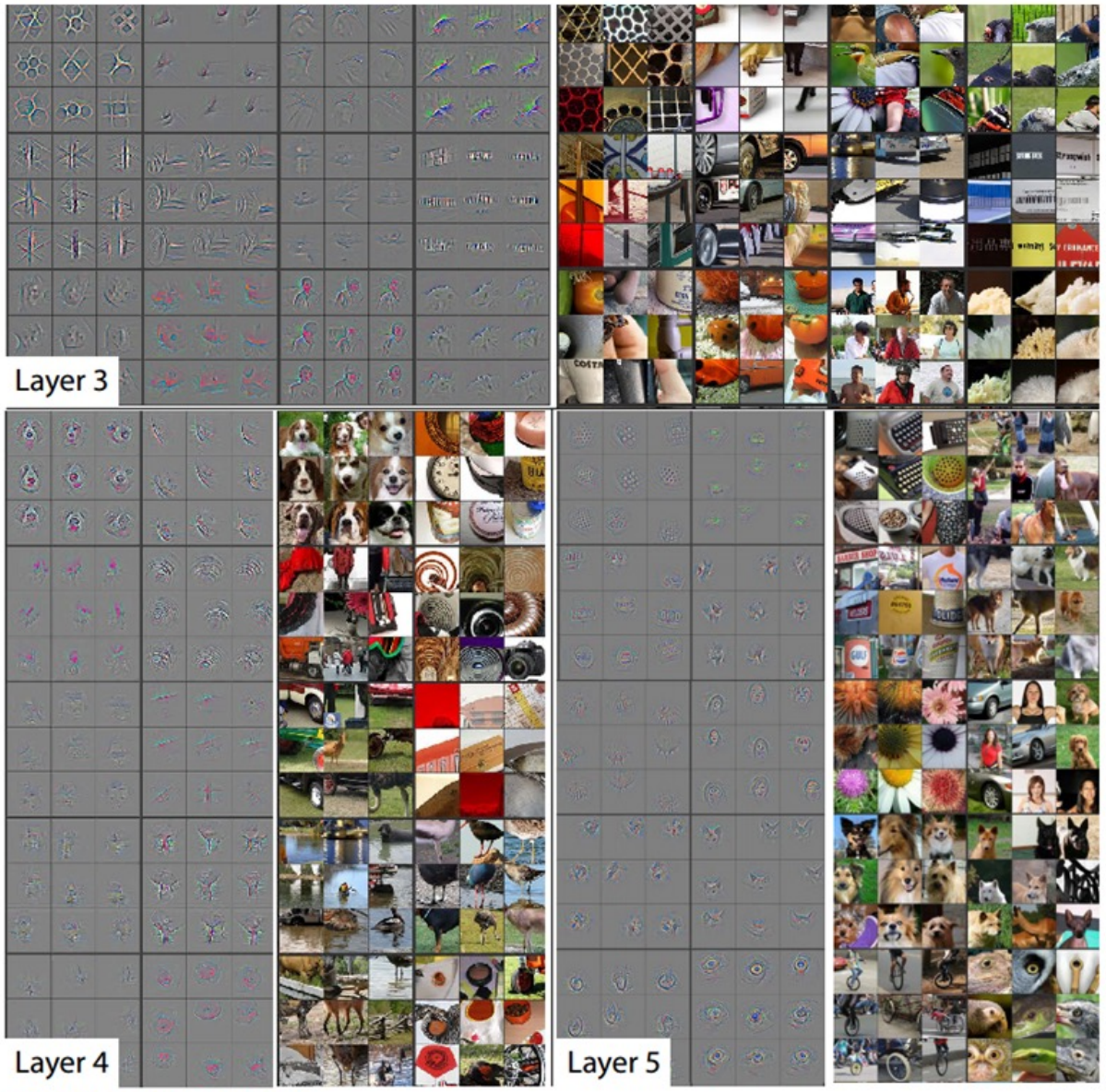
Edges: small spatial dimension, lower level

Edges: larger spatial dimension, higher level

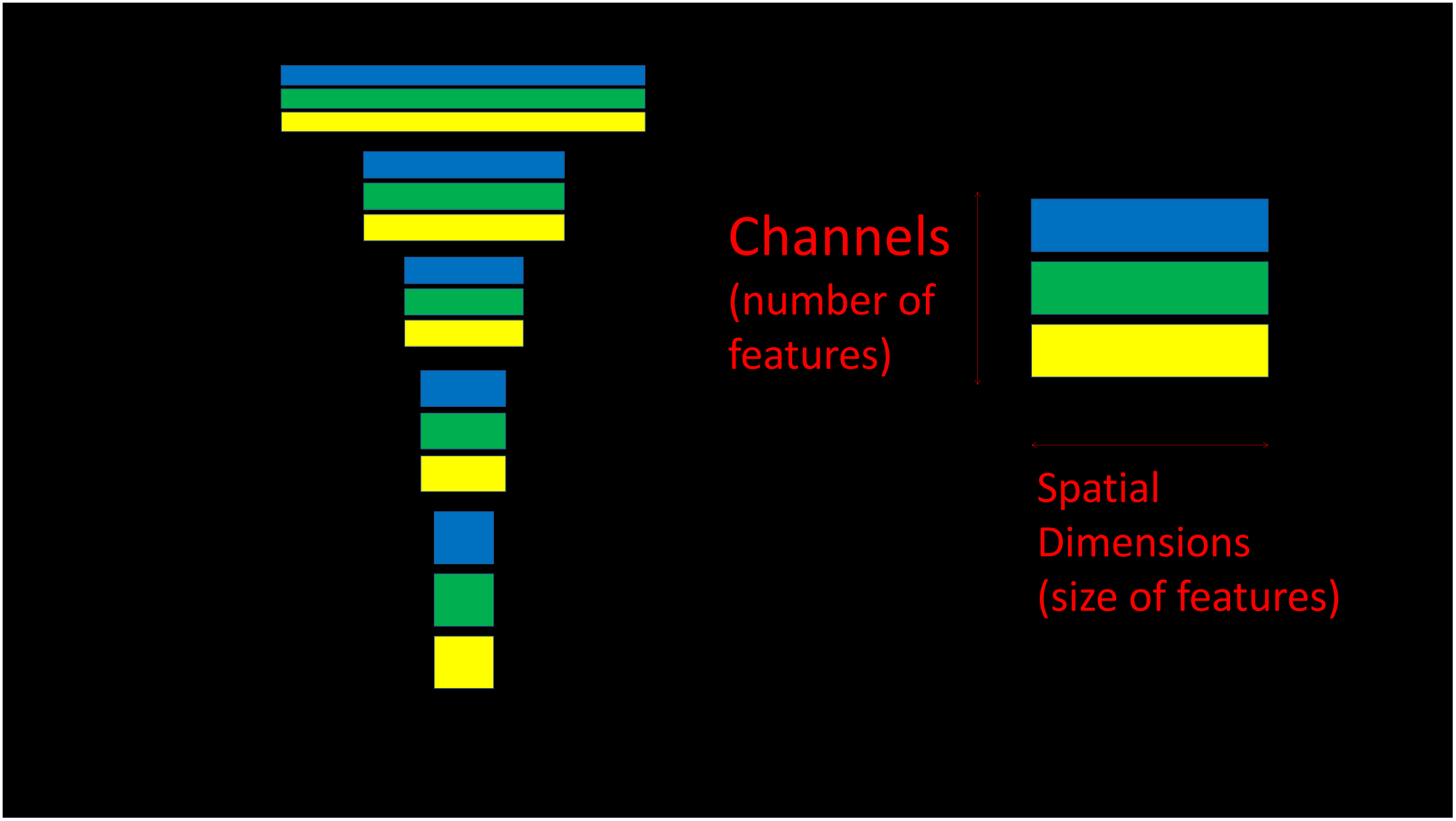


https://towardsdatascience.com/transfer-learning-using-pytorch-part-2-9c5b18e15551

https://towardsdatascience.com/transfer-learning-using-pytorch-part-2-9c5b18e15551

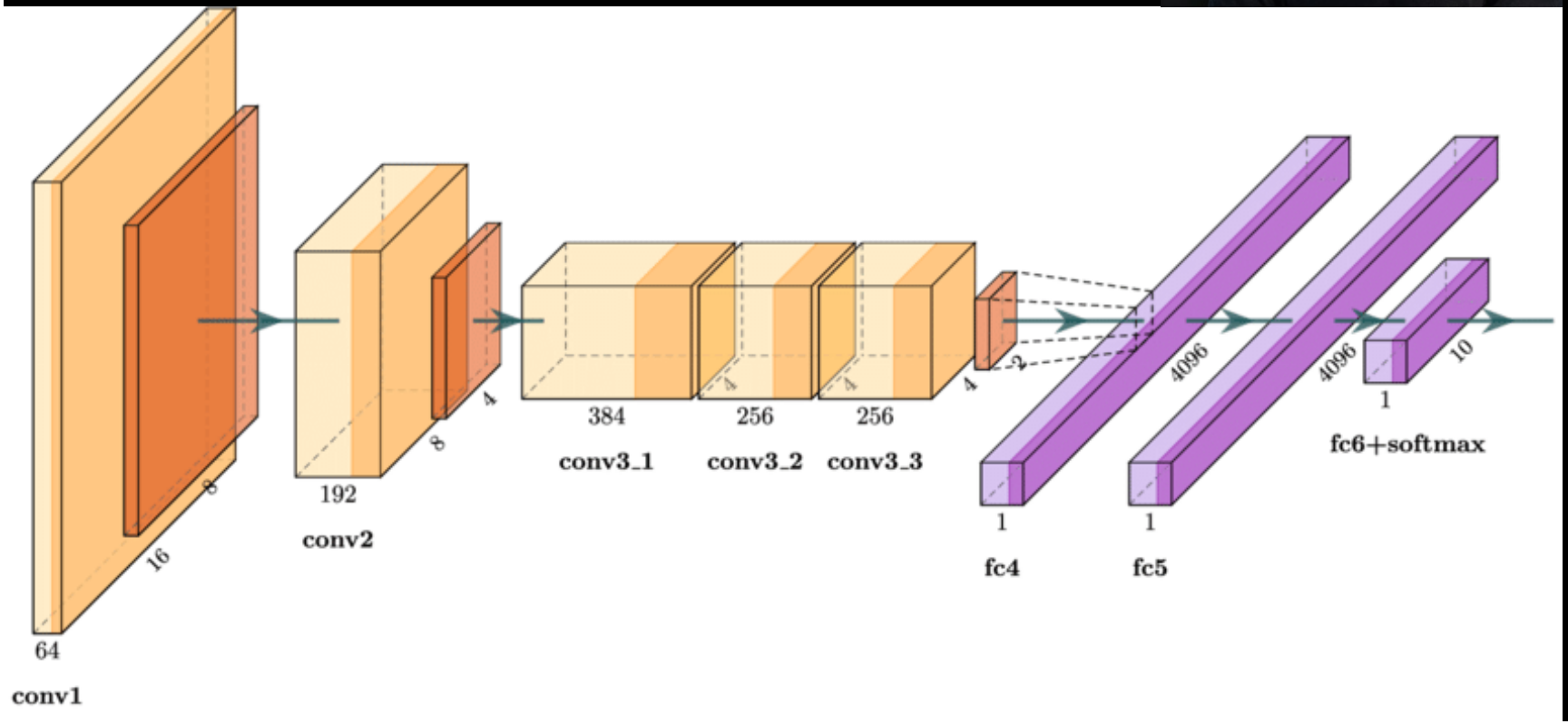


Visualizations of Layers 3, 4, and 5

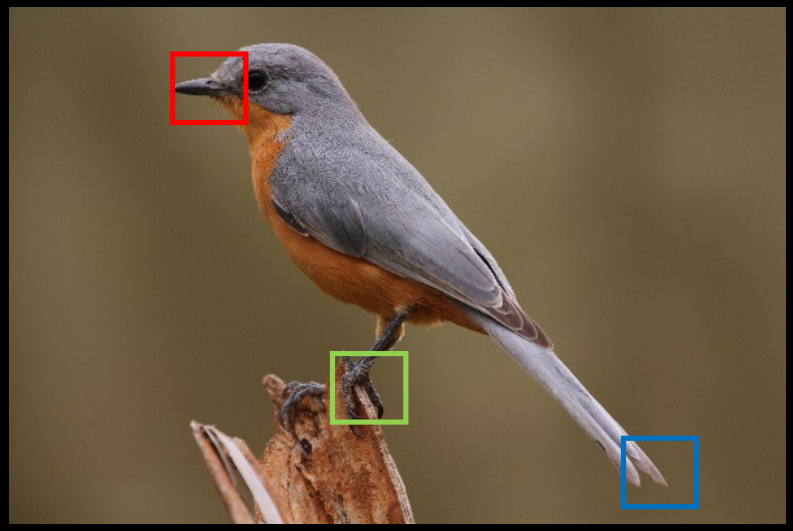

Channels
(number of features)
Spatial Dimensions
(size of features)

Actually an earlier version…
Alexnet





Pooling



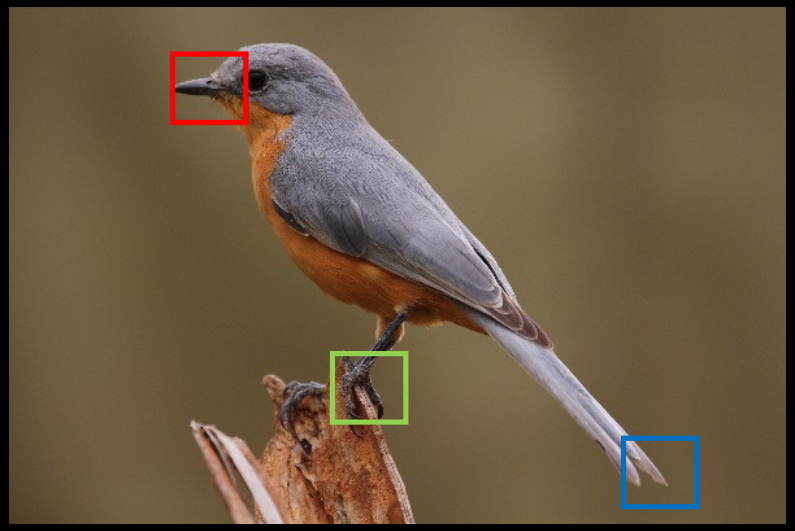


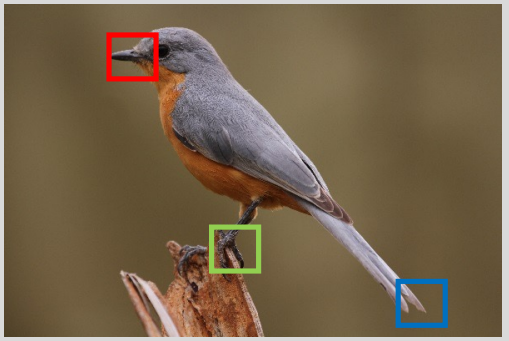
Important features should
be independent of scaling

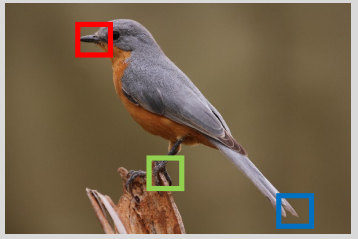
and

Can be in many location

Pooling





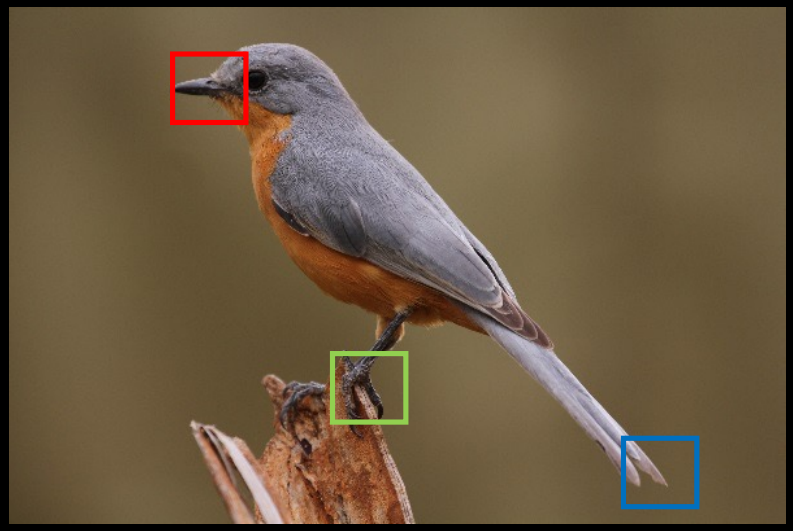


Important features should
be independent of scaling

and

Can be in many location

(Max)Pooling



3	1	0			9
4	5	2			
-5	-2	-1			
					4
	7				

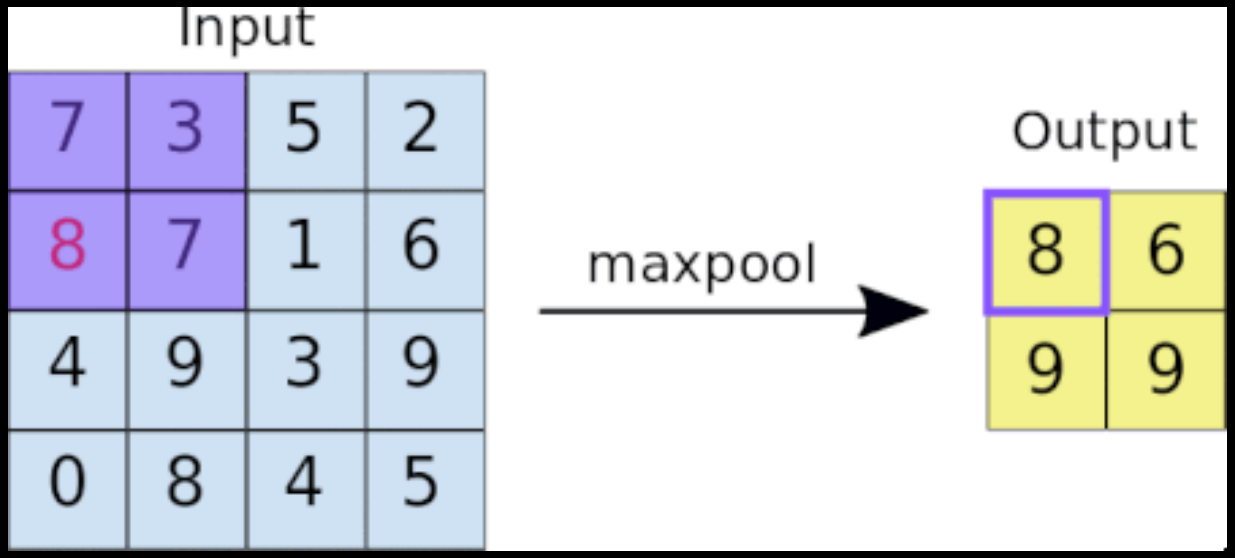
5	9
7	4

Important features should be independent of scaling

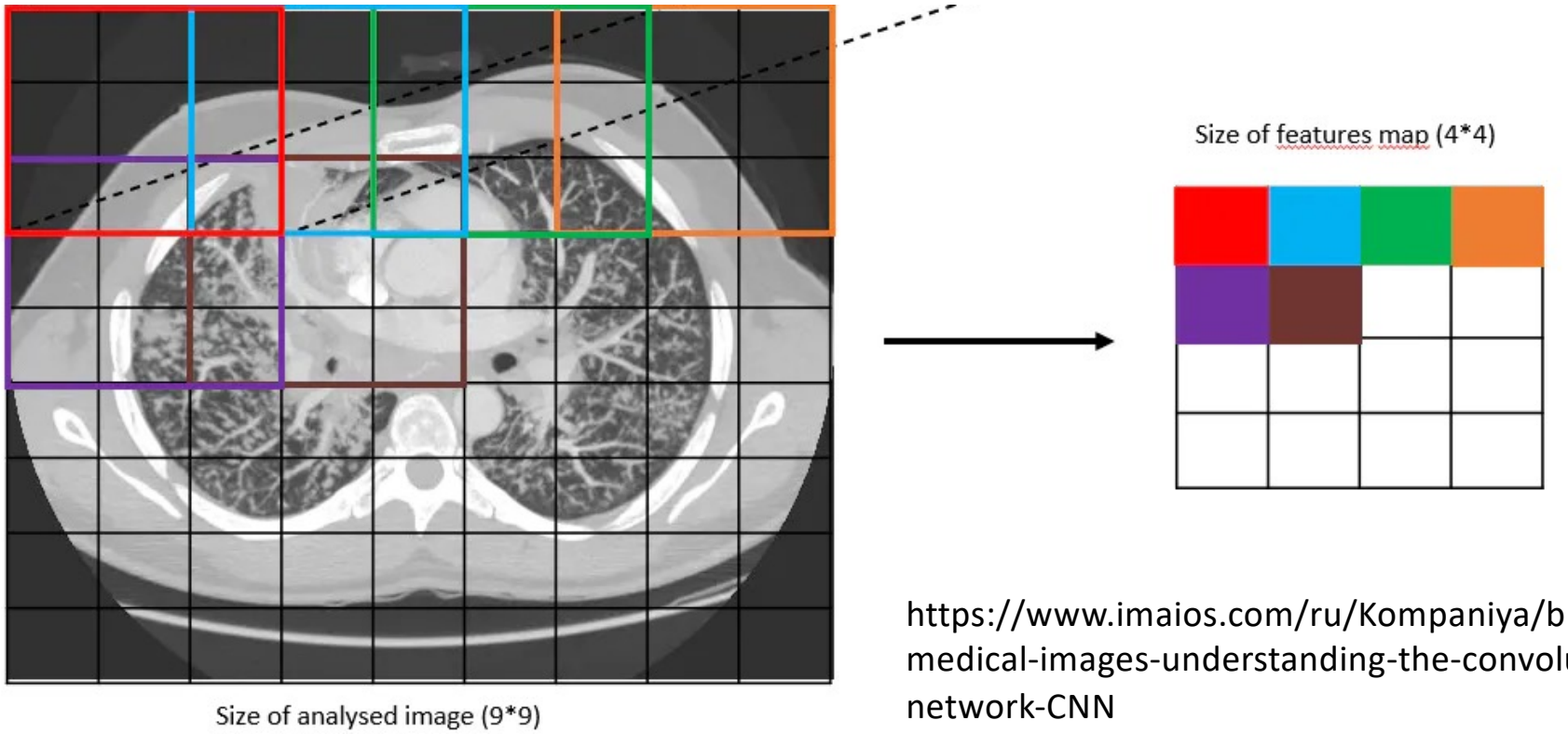
and

Can be in many location

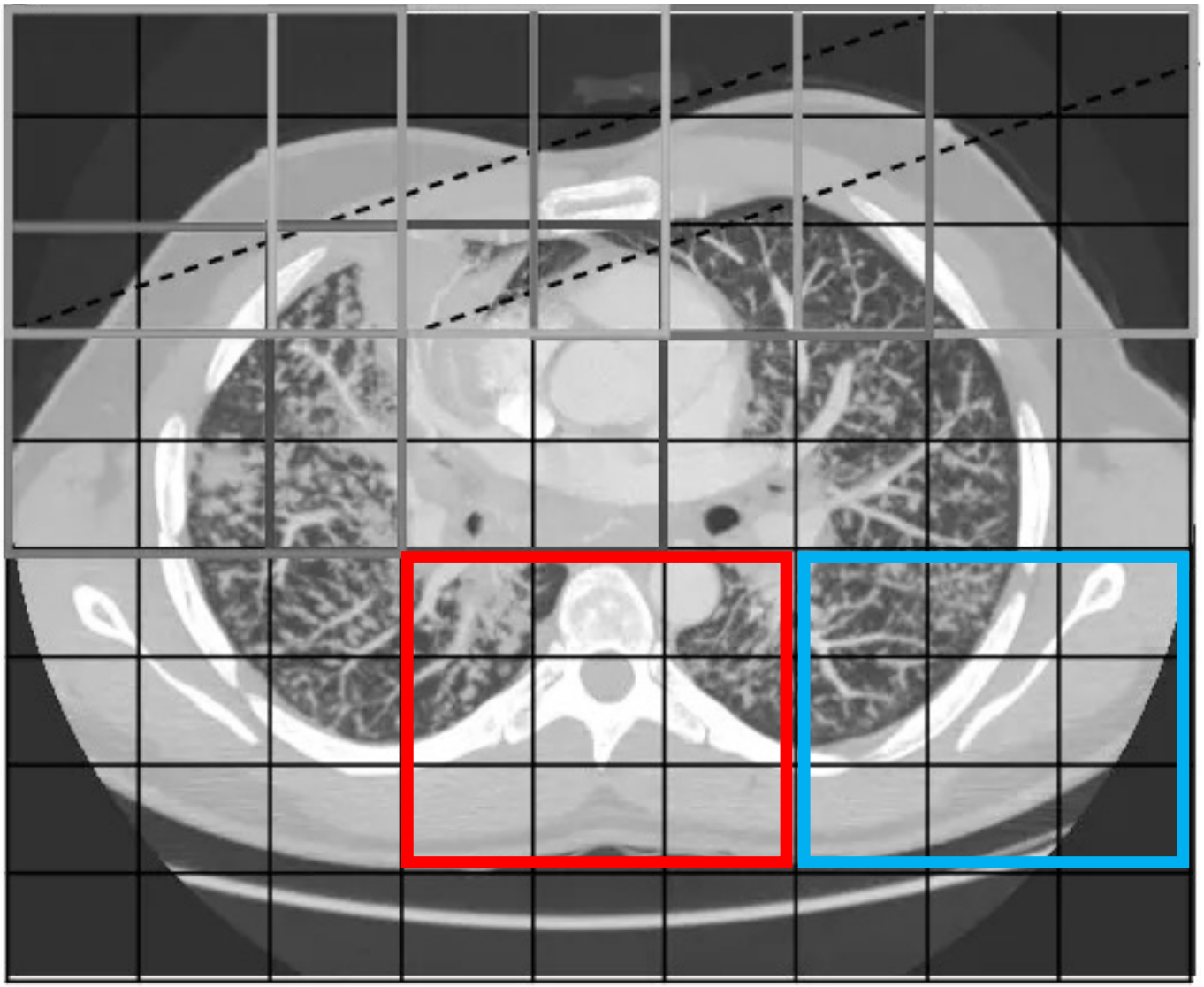
(Max)Pooling



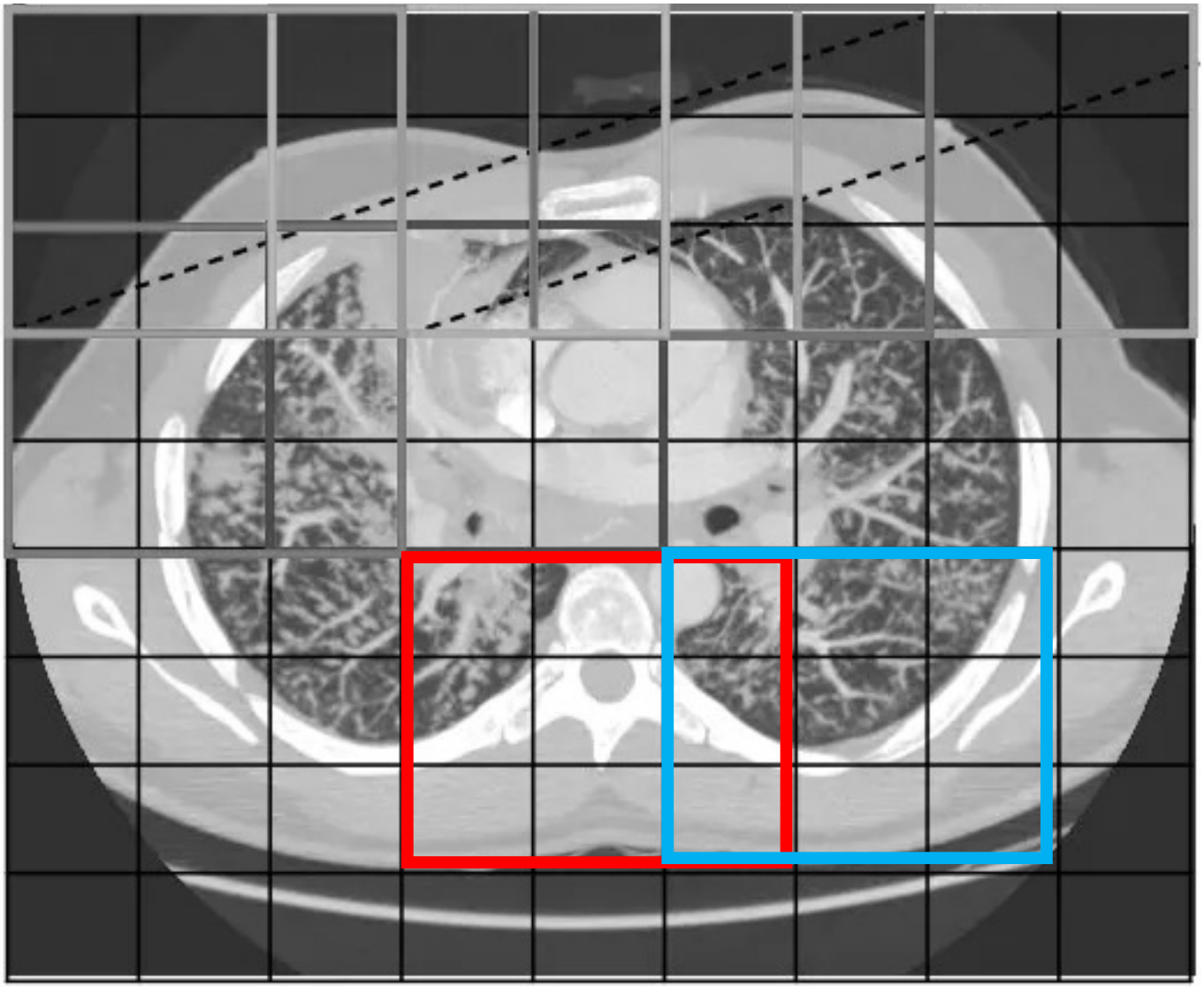
nn.MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)



https://www.imaios.com/ru/Kompaniya/blog/Classification-of-medical-images-understanding-the-convolutional-neural-network-CNN

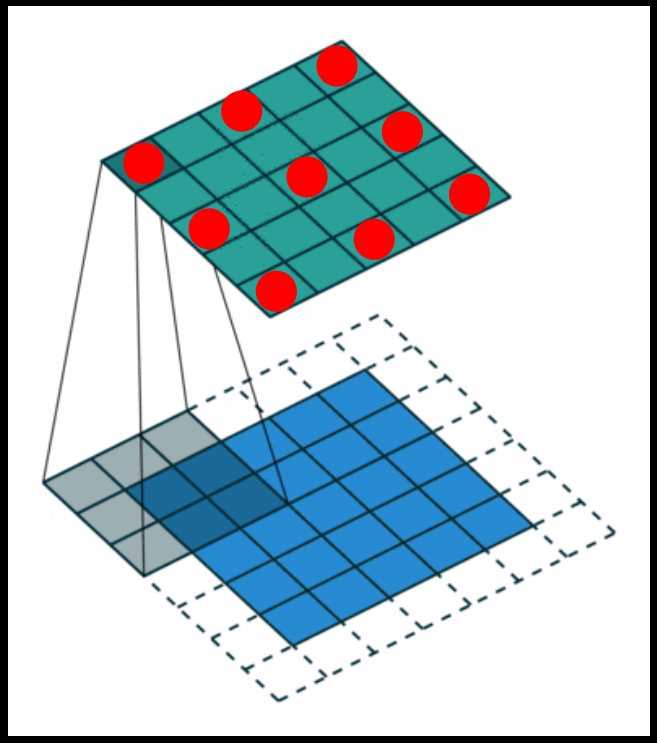


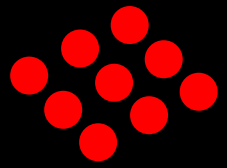
Size of analysed image (9*9)

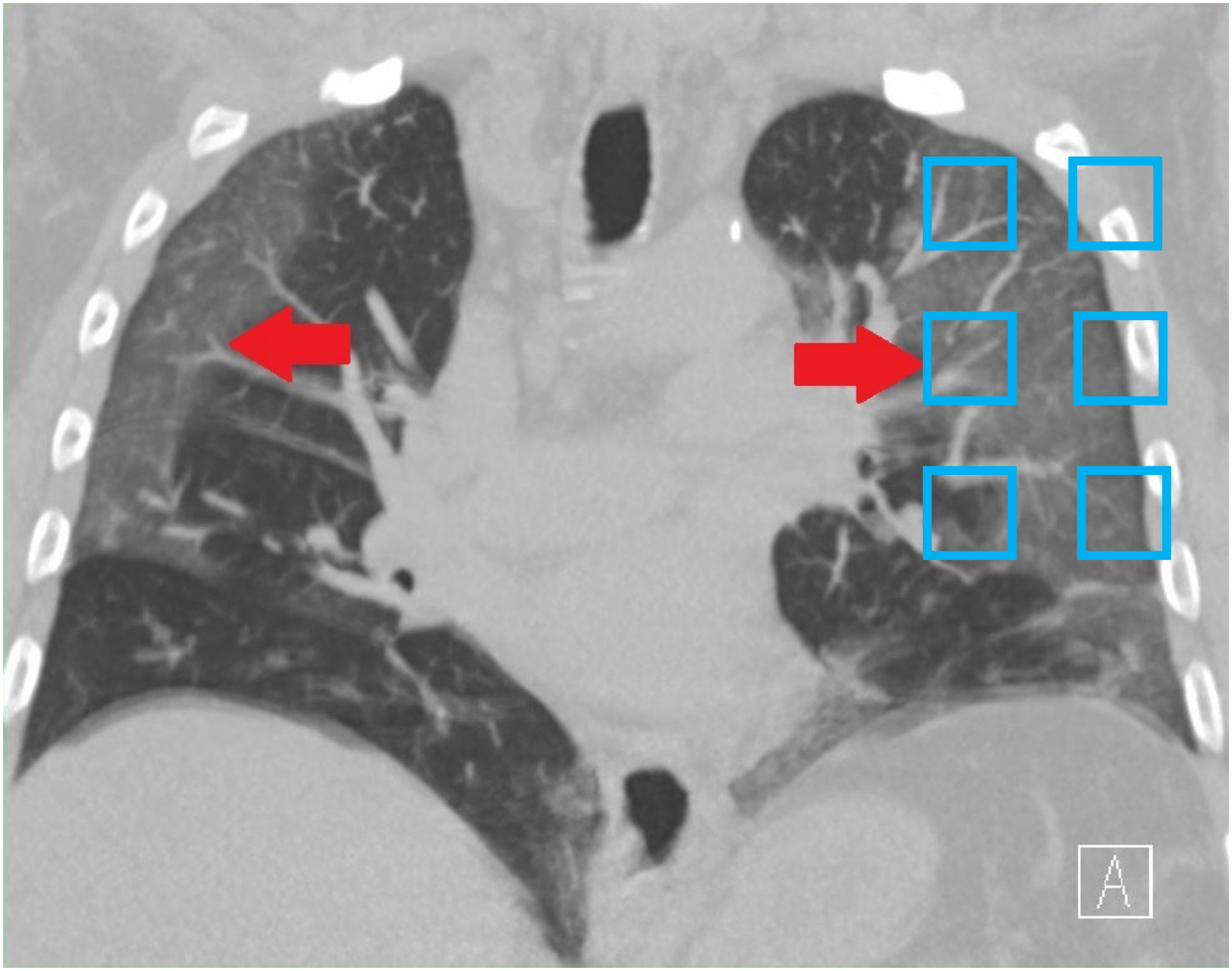


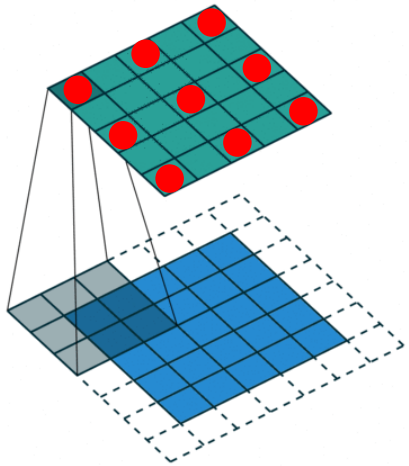
Size of analysed image (9*9)

Stride

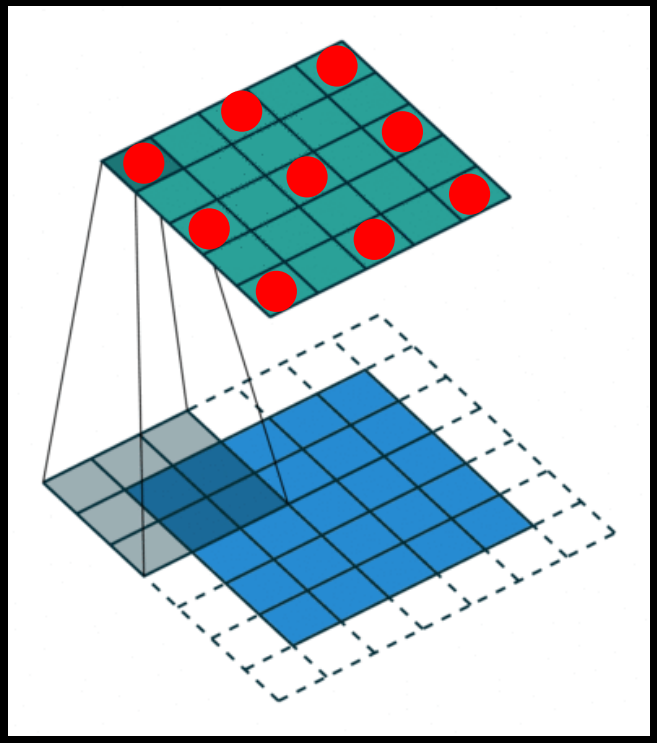


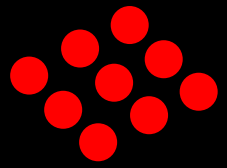



A

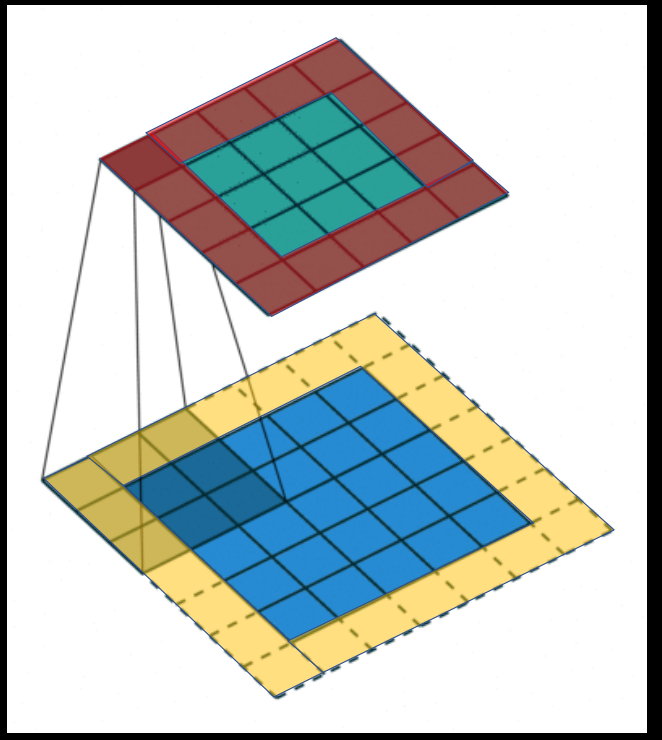


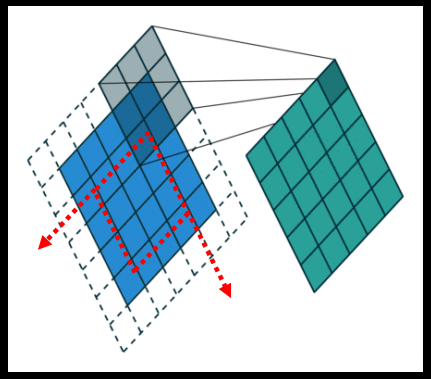
Stride

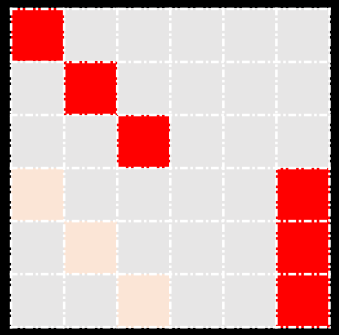


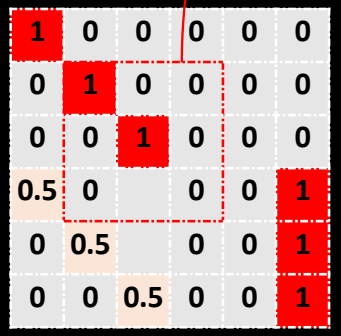


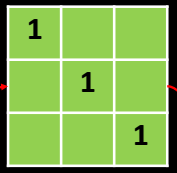
Padding

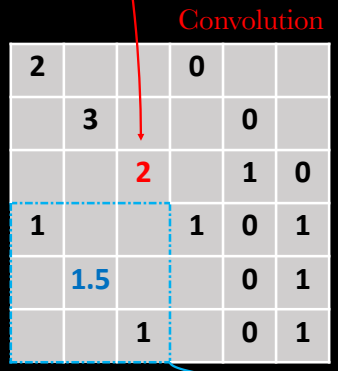




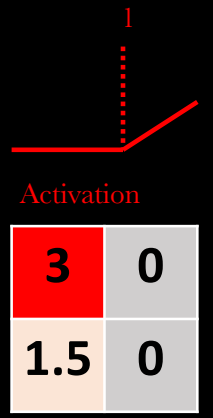












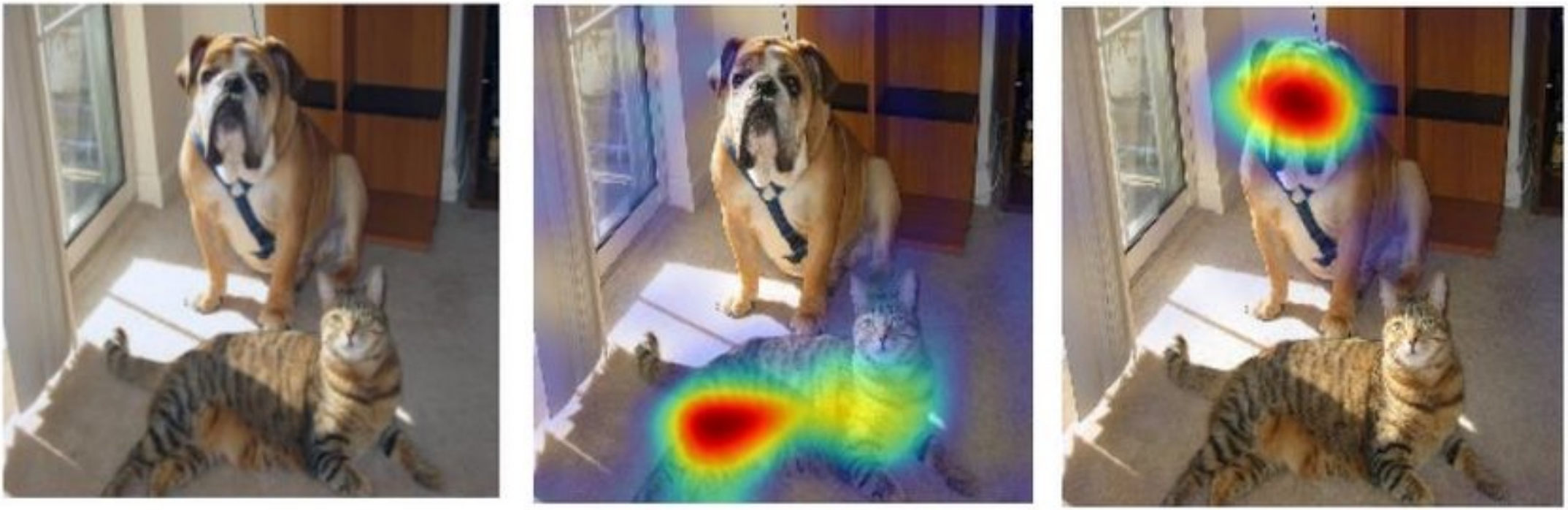
Convolution

Pooling

Nonlinearity

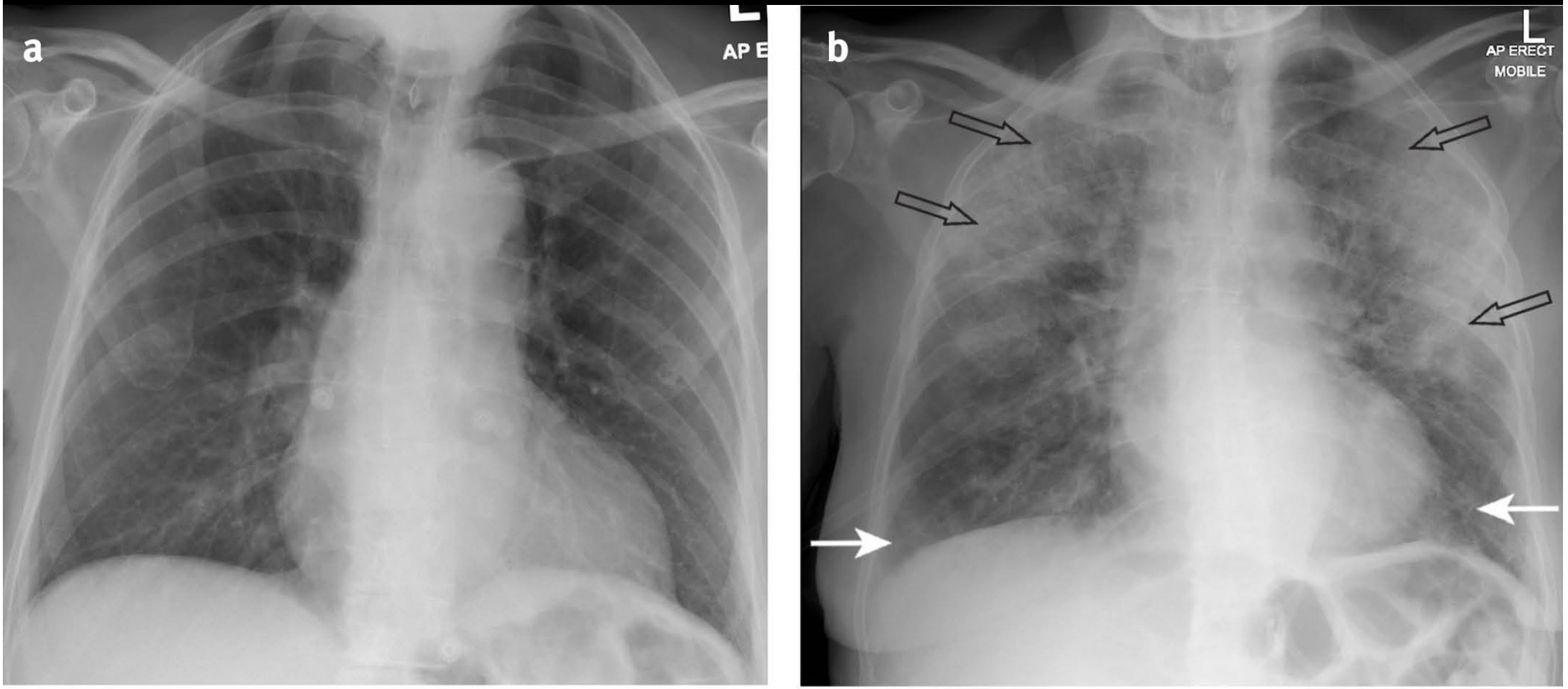
Some considerations for medical imaging vs natural imaging…….

Nature Imaging: Often Large Discriminative Parts



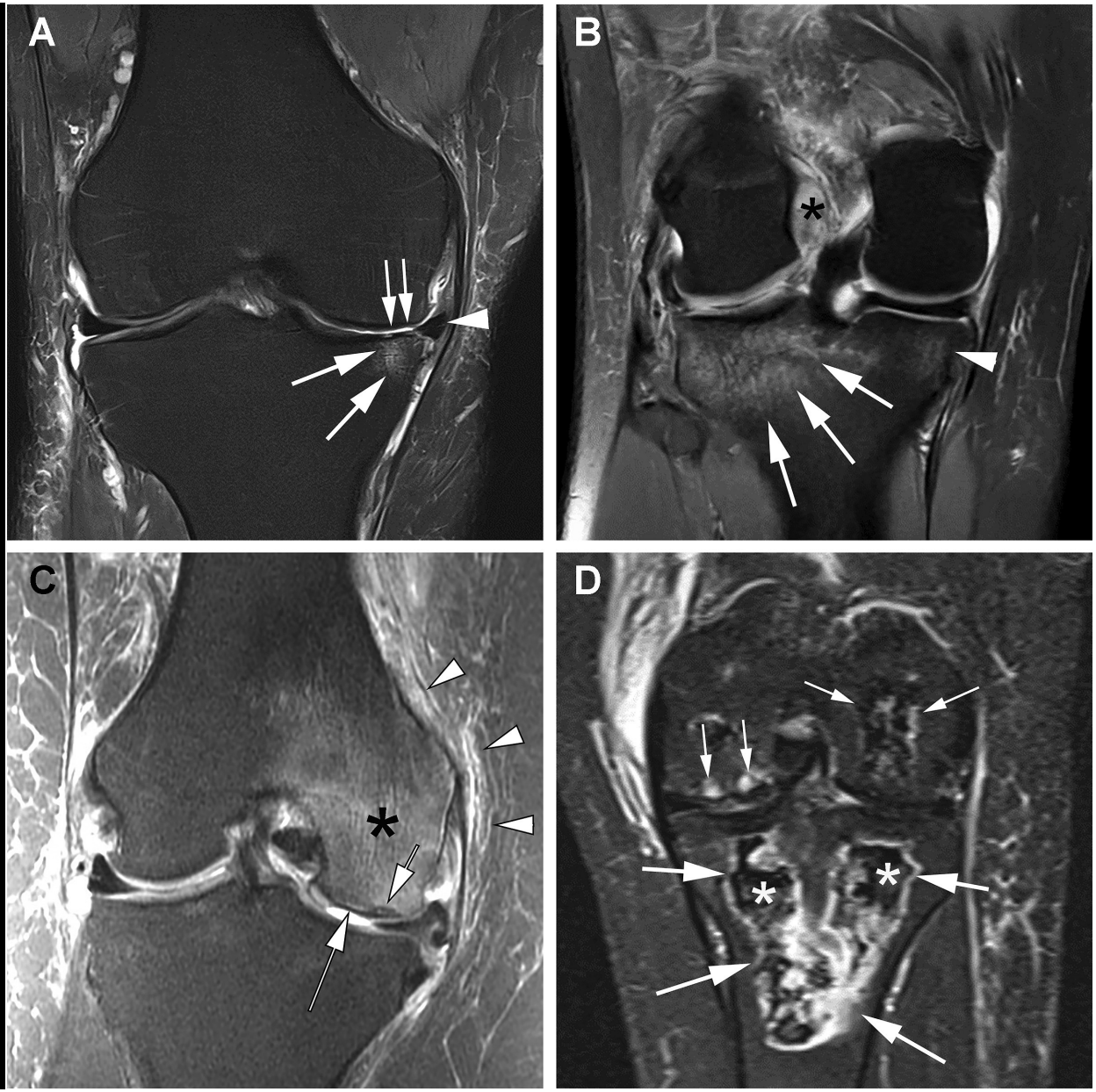
(a) Original Image (b) Grad-CAM ‘Cat’ (c) Grad-CAM ‘Dog’

Medical Imaging: Fine-grained Discriminative Parts

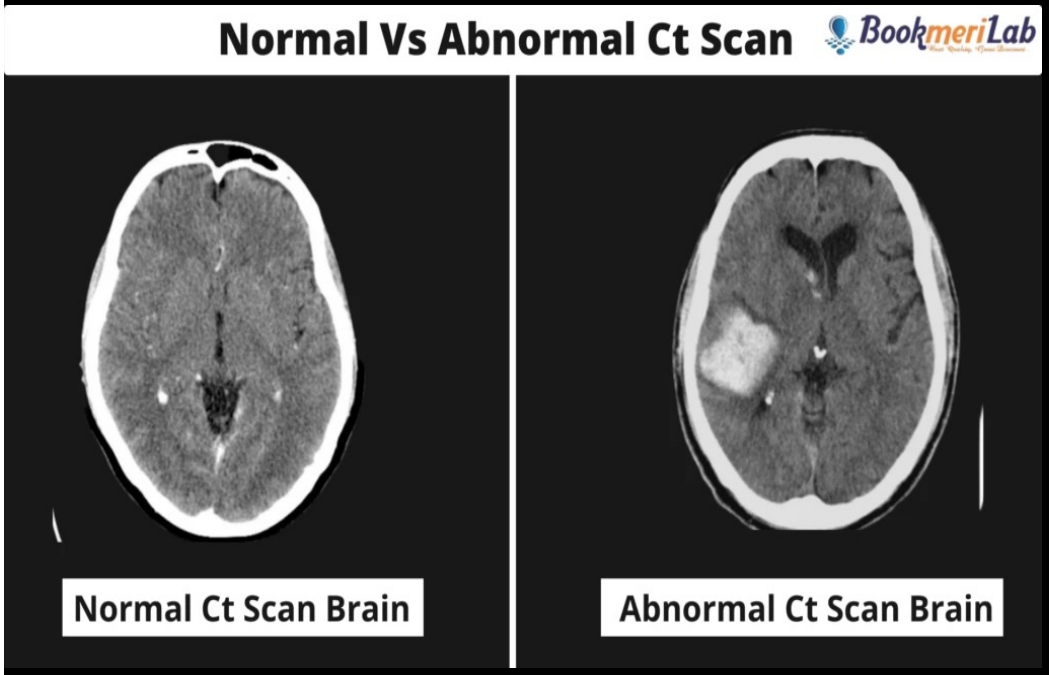


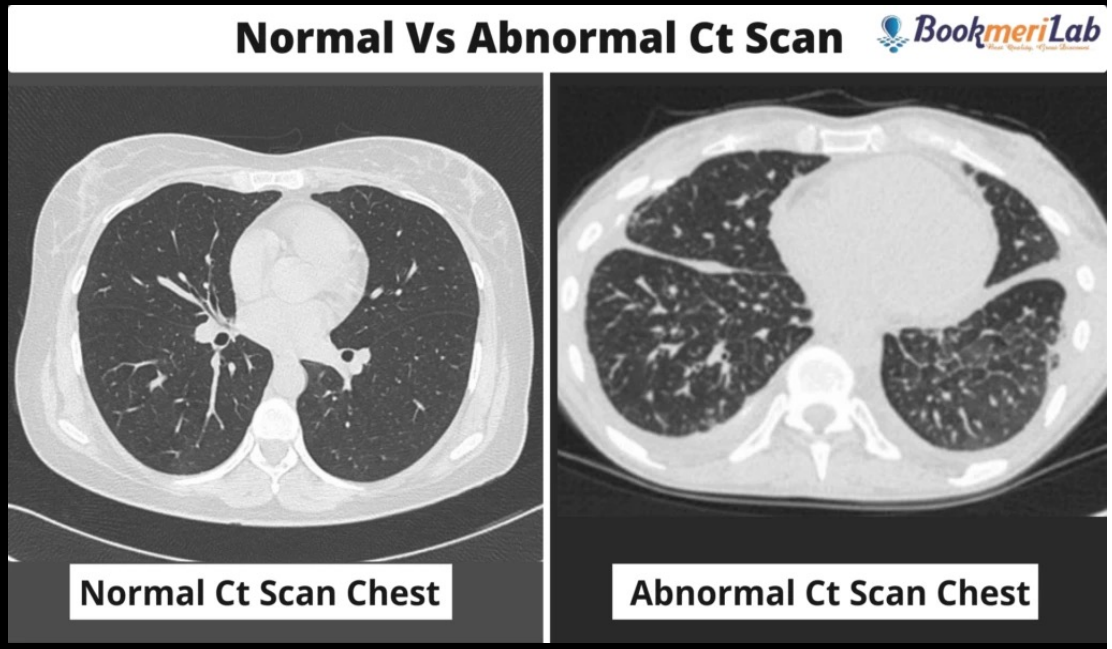
Medical Imaging:
Fine-grained Discriminative Parts

Small in scale != Low level feature

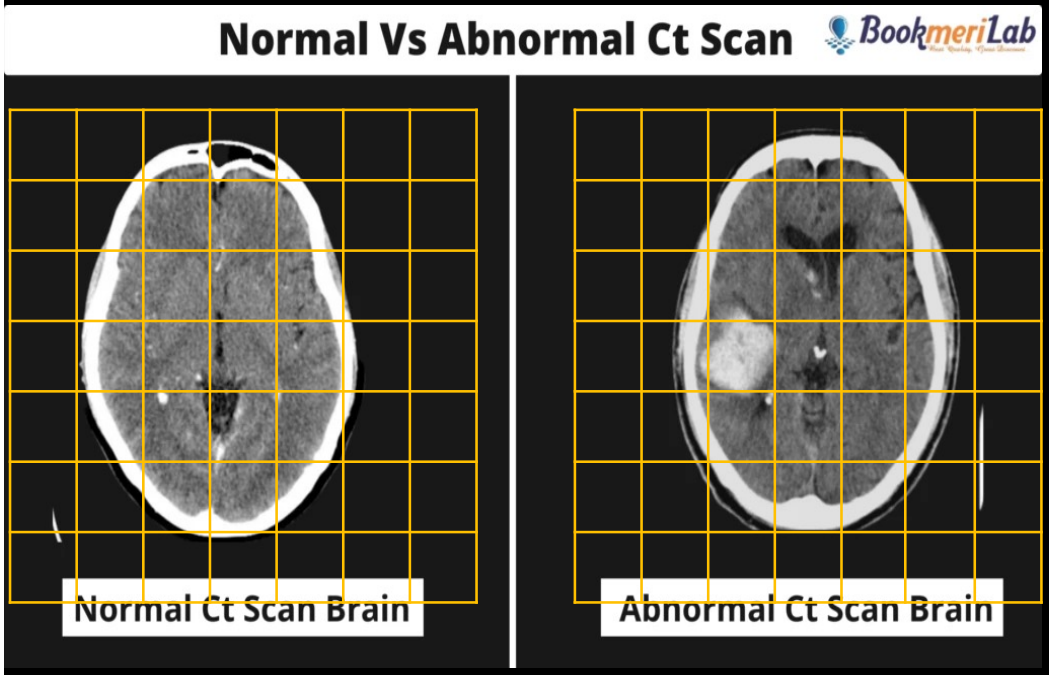


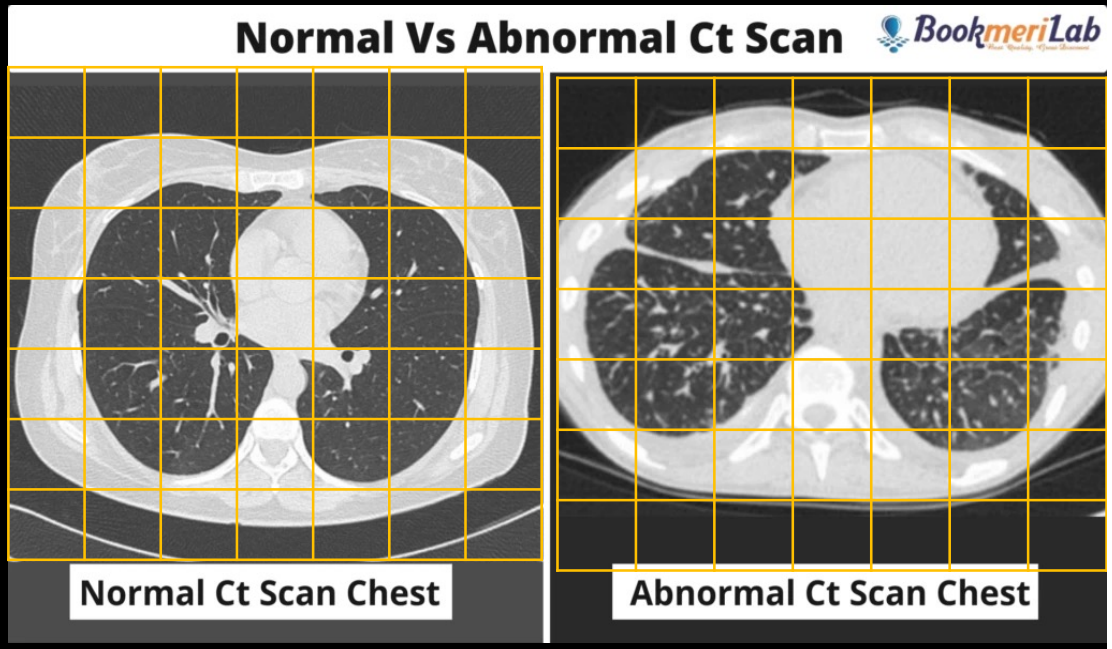
Large strides: risky for which one?



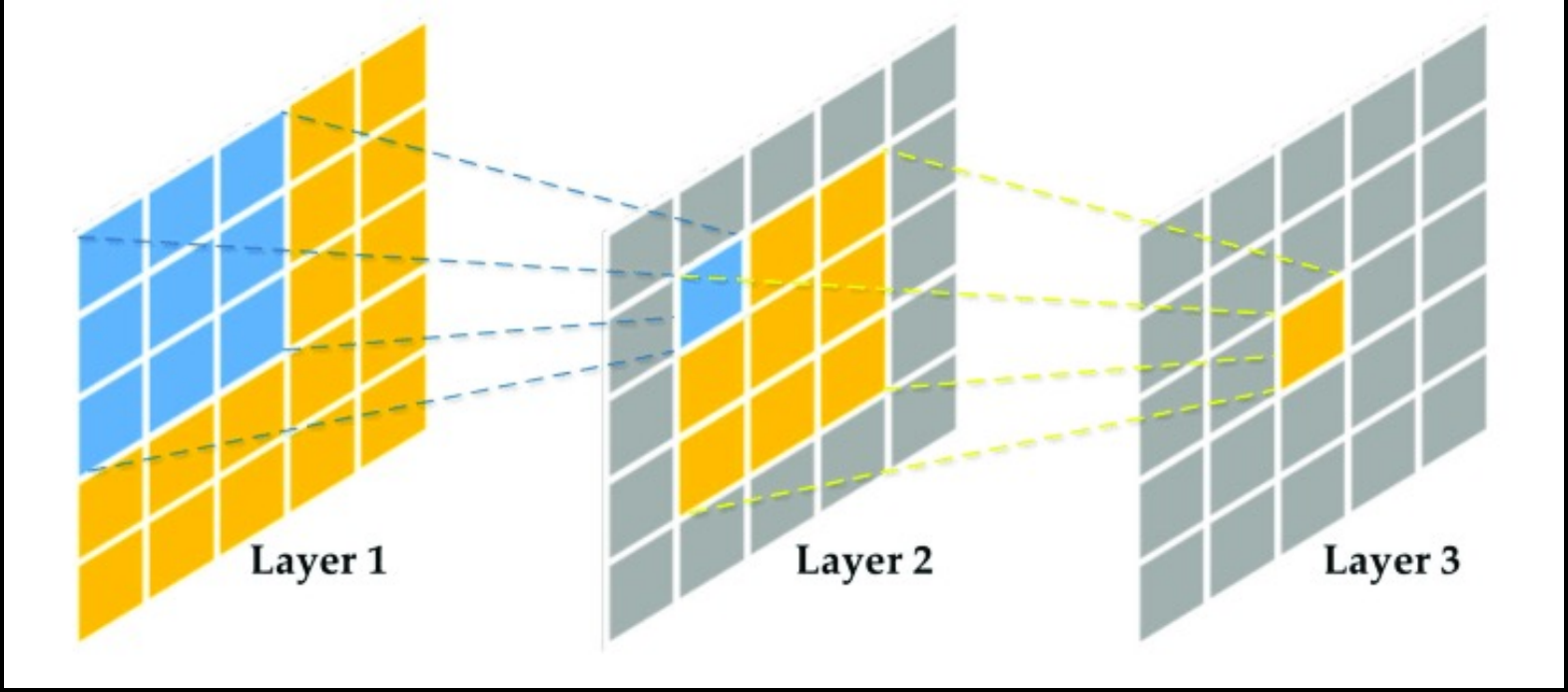


Large strides: risky for which one?

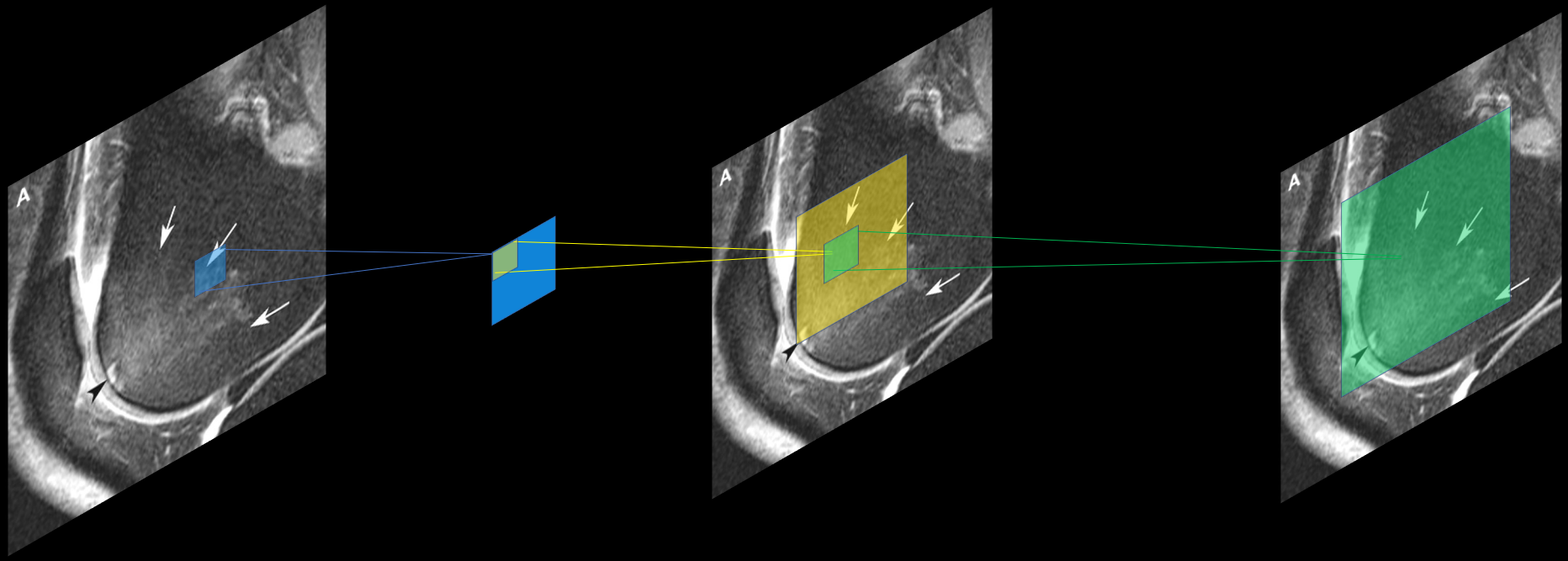


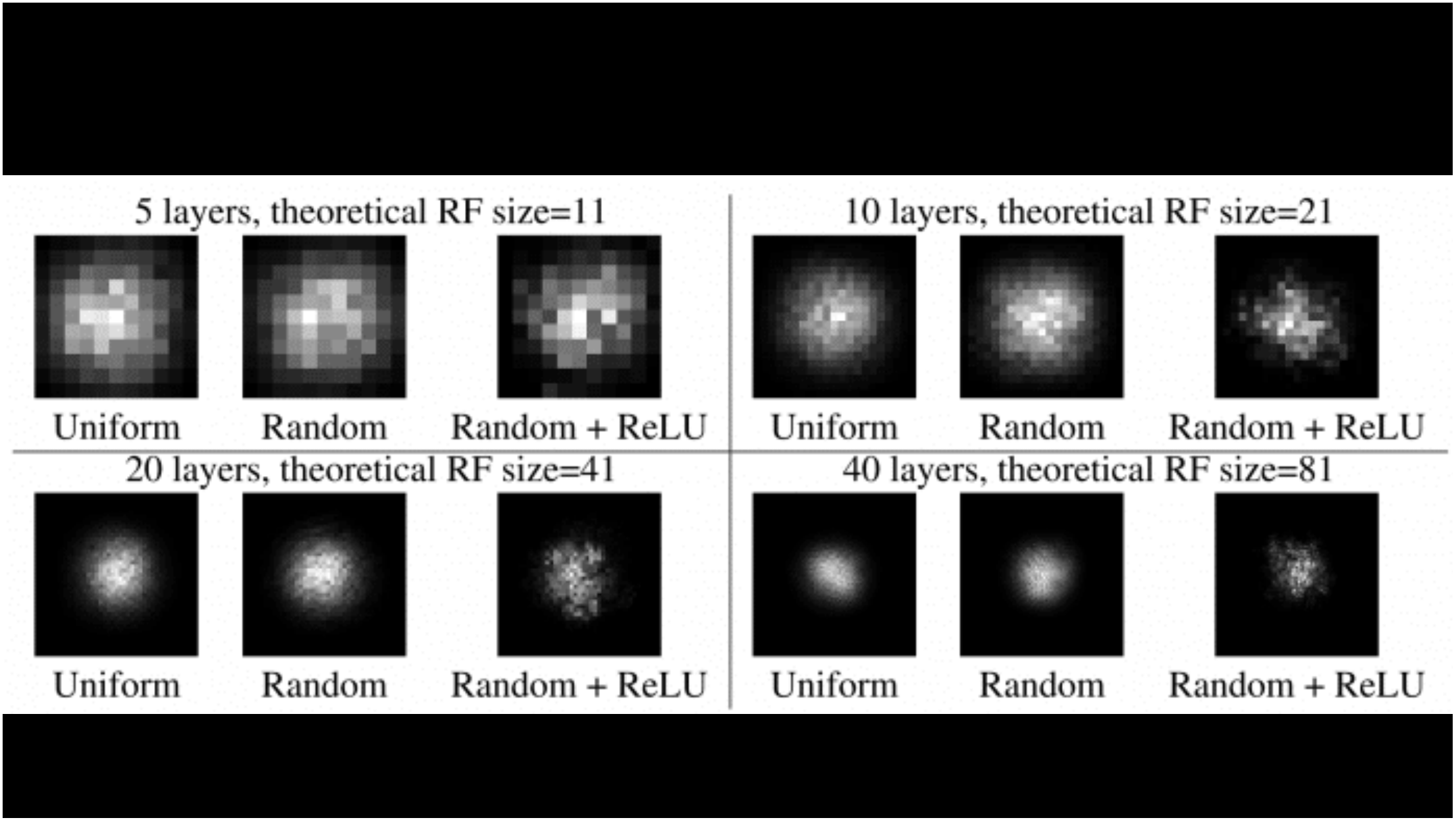


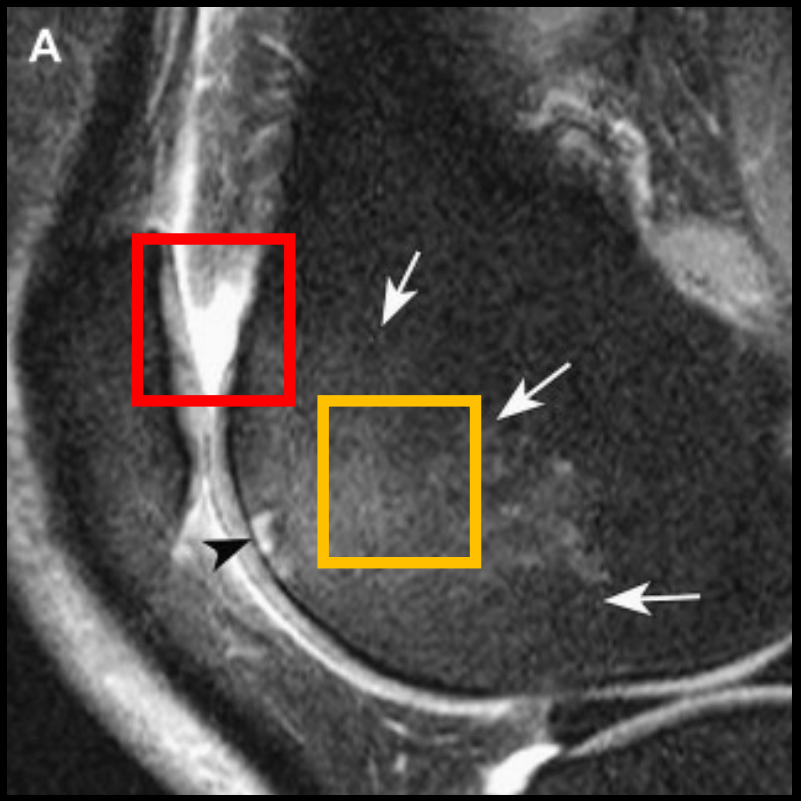
How can we see the neighbors (very far neighbors, dozens of pixels away) without large convolution kernels?)

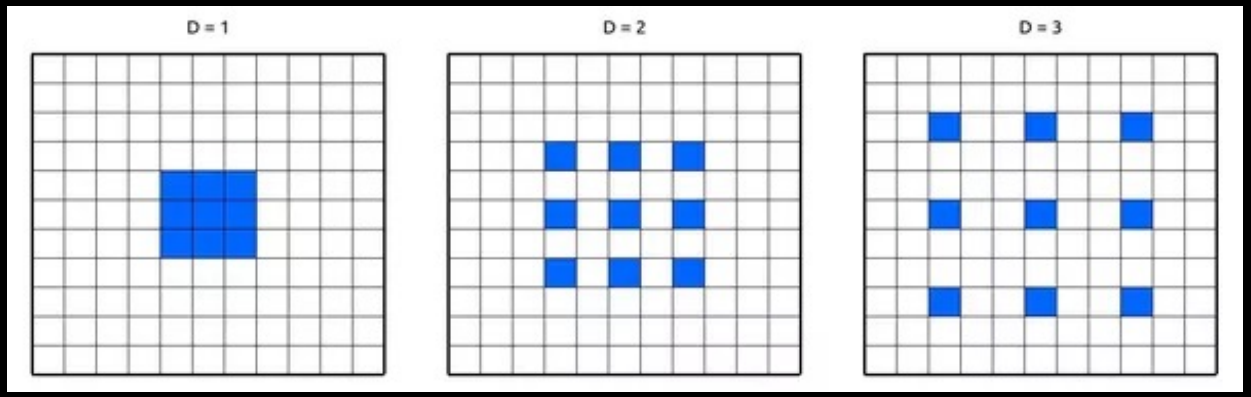


Receptive Field

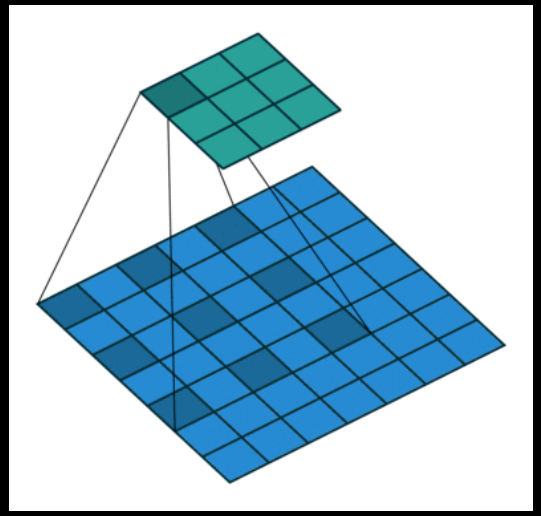


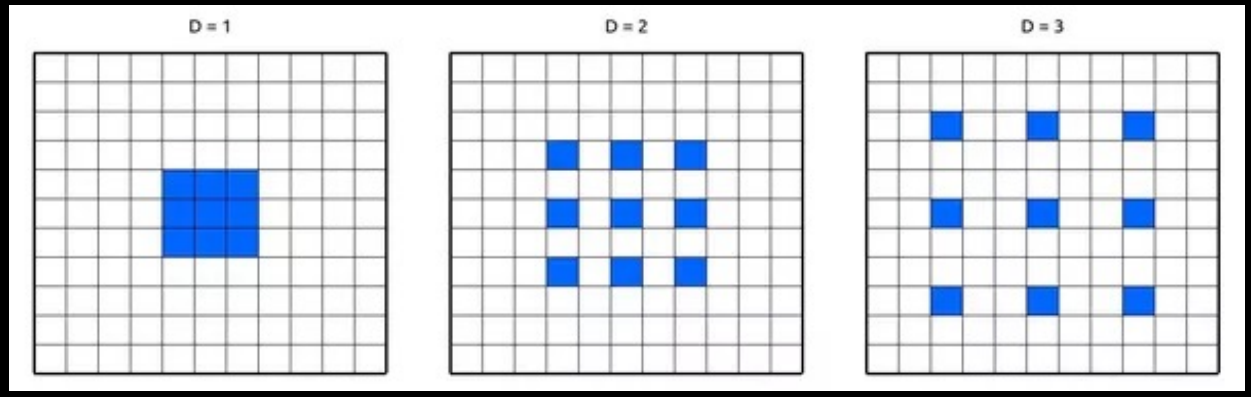

5 layers, theoretical RF size=11
Uniform
Random
Random + ReLU
10 layers, theoretical RF size=21
Uniform
Random
Random + ReLU
20 layers, theoretical RF size=41
Uniform
Random
Random + ReLU
40 layers, theoretical RF size=81
Uniform
Random
Random + ReLU


A


D = 1
D = 2
D = 3

Dilation



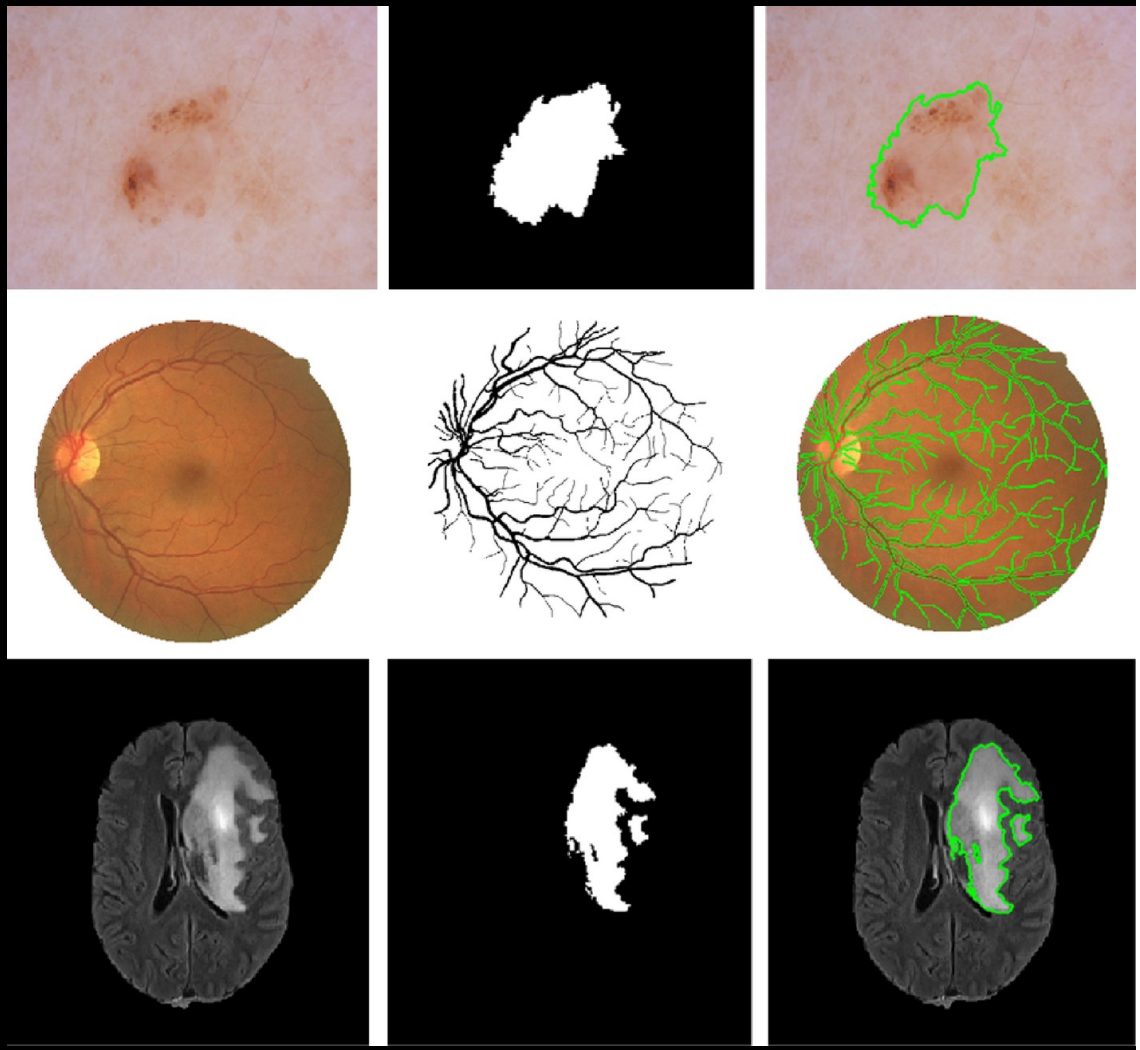


We need features possibly

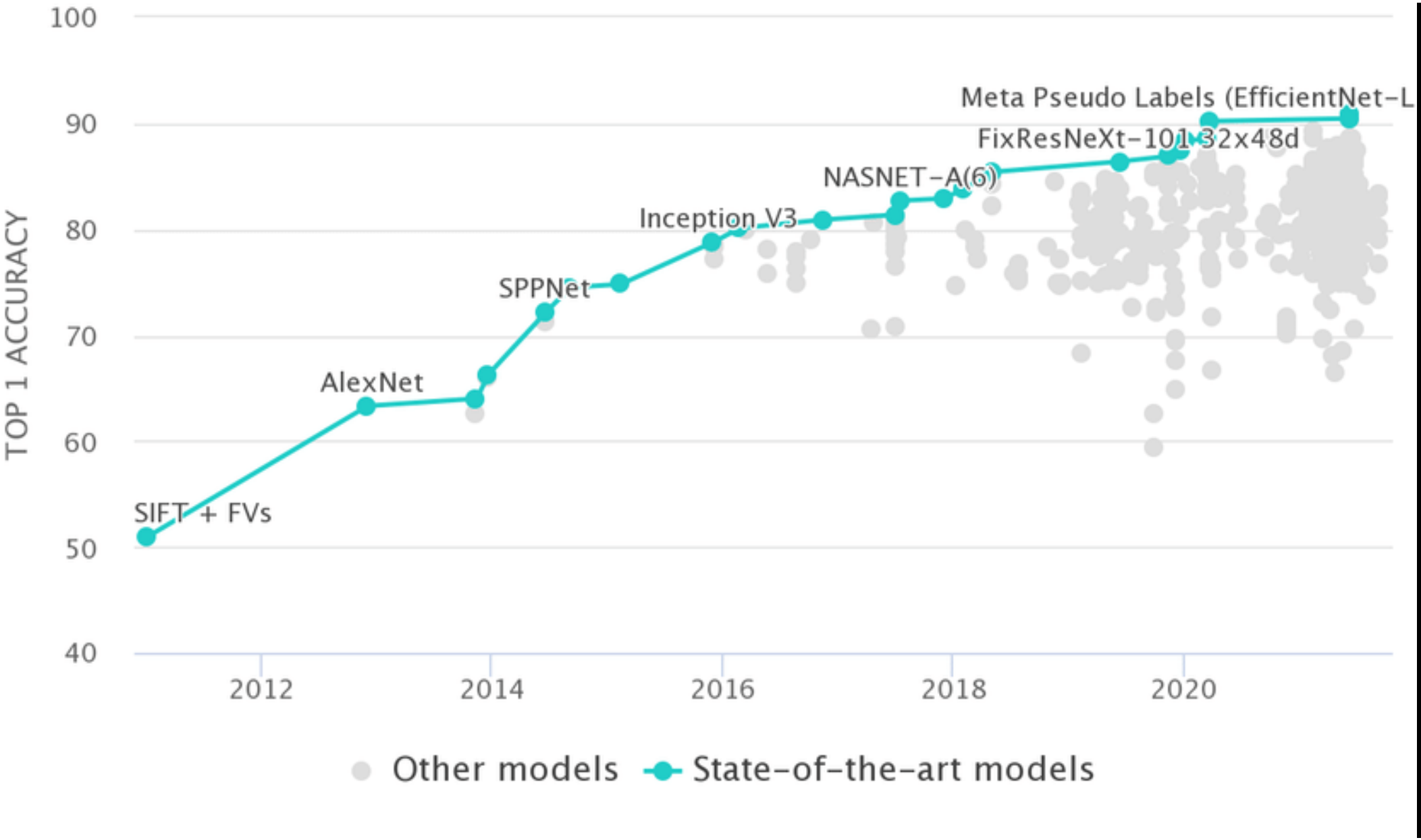
from different scales

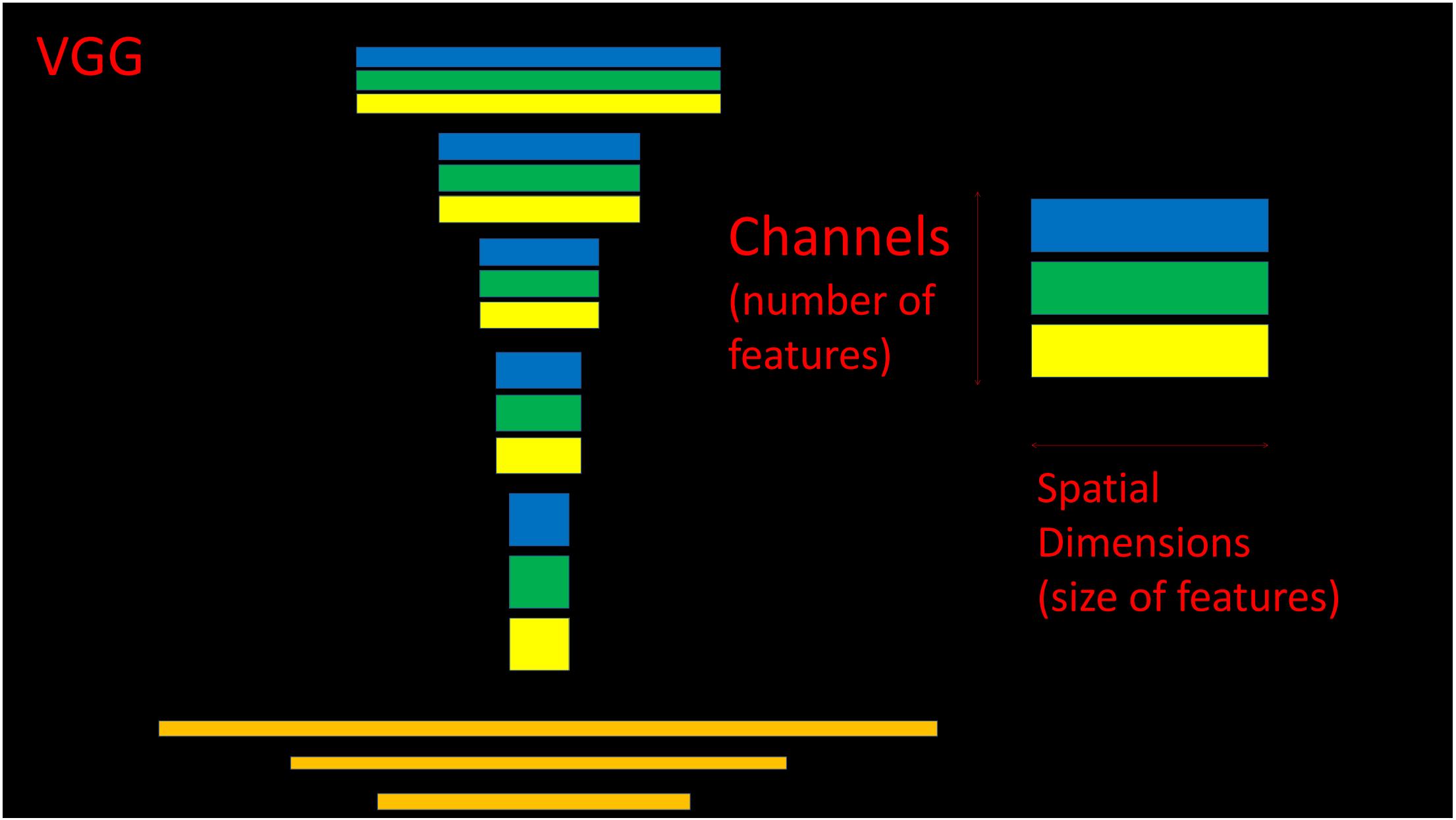
can be high level or low level

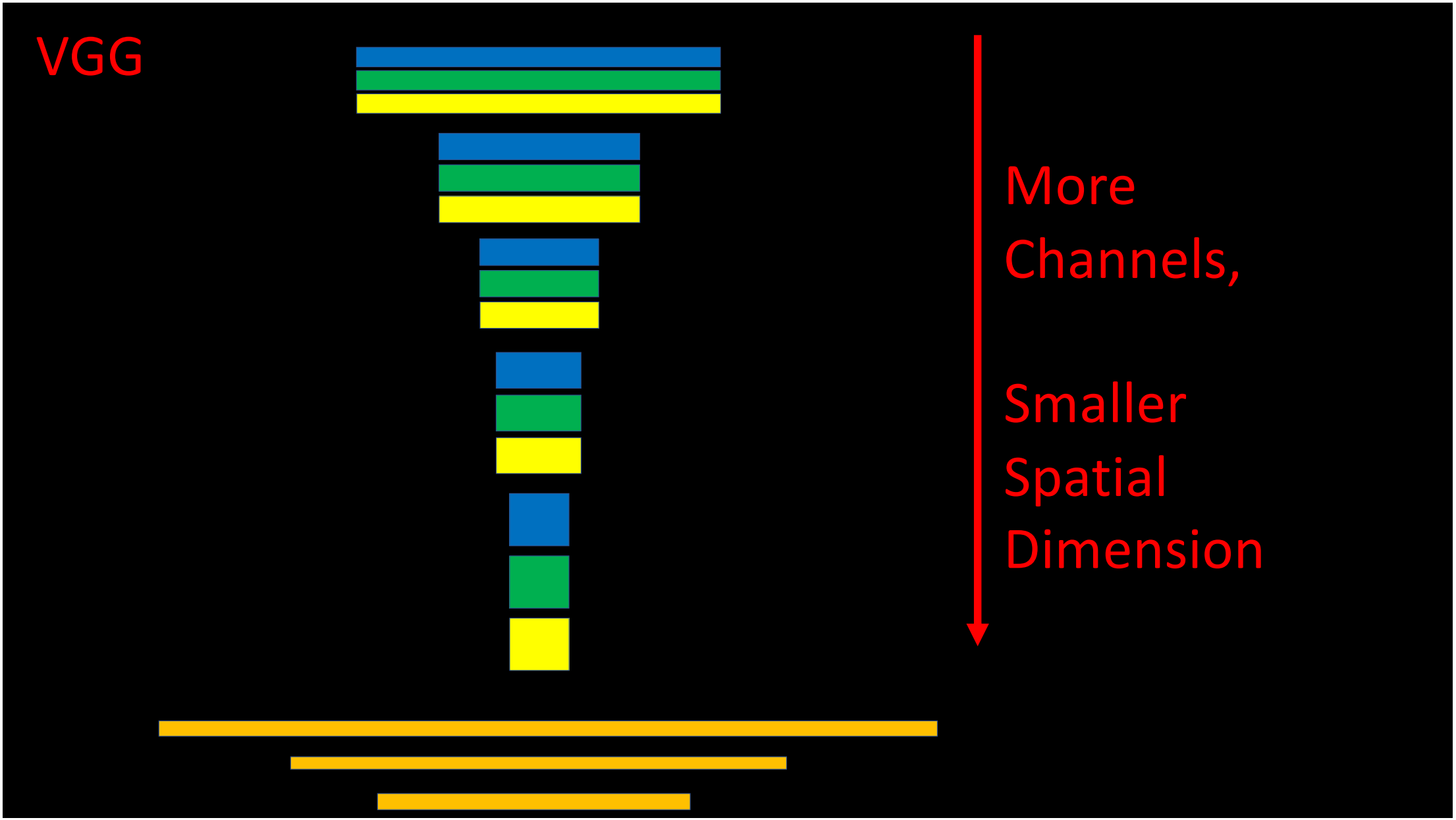
can be vastly different between modalities / applications

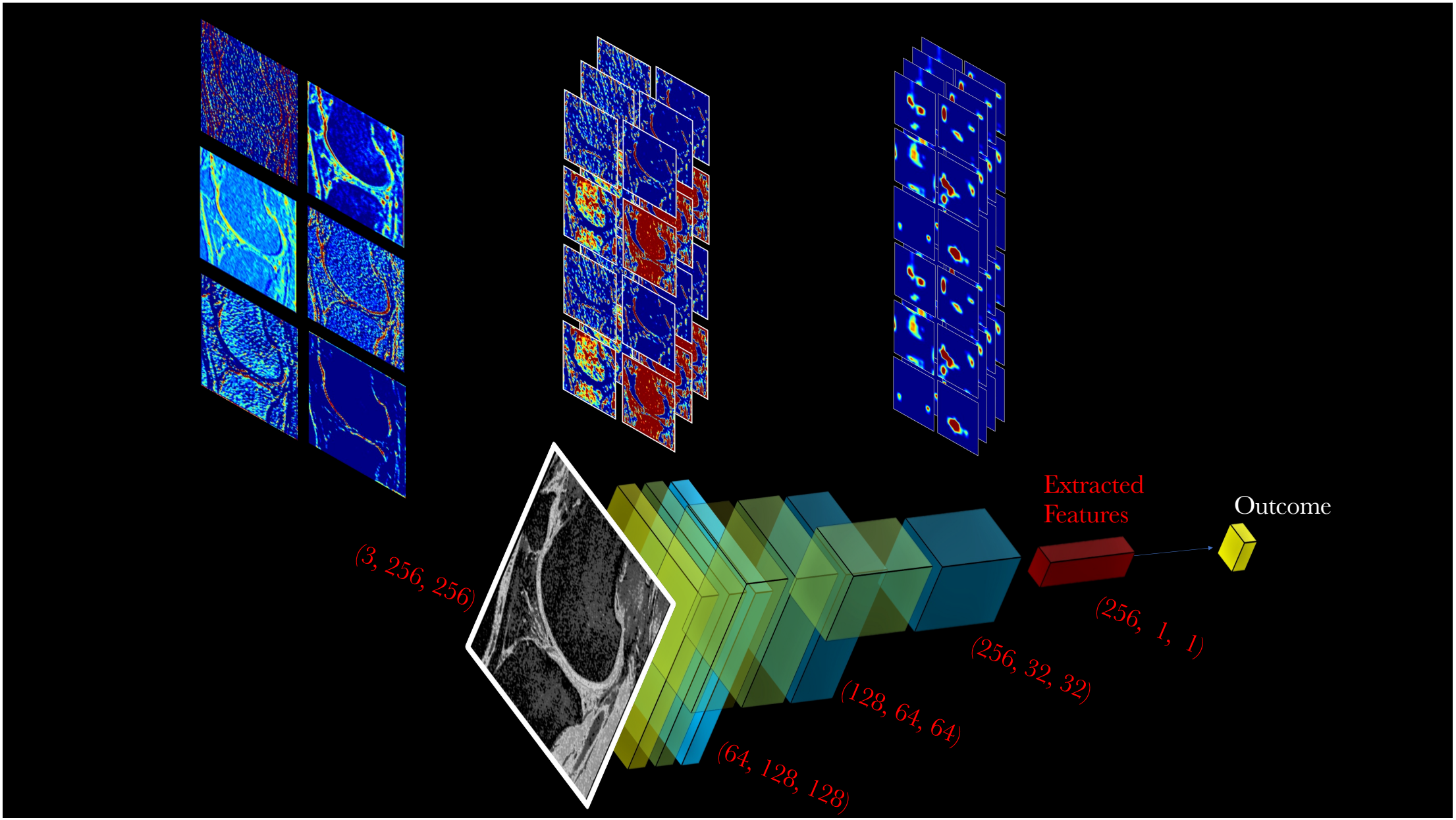


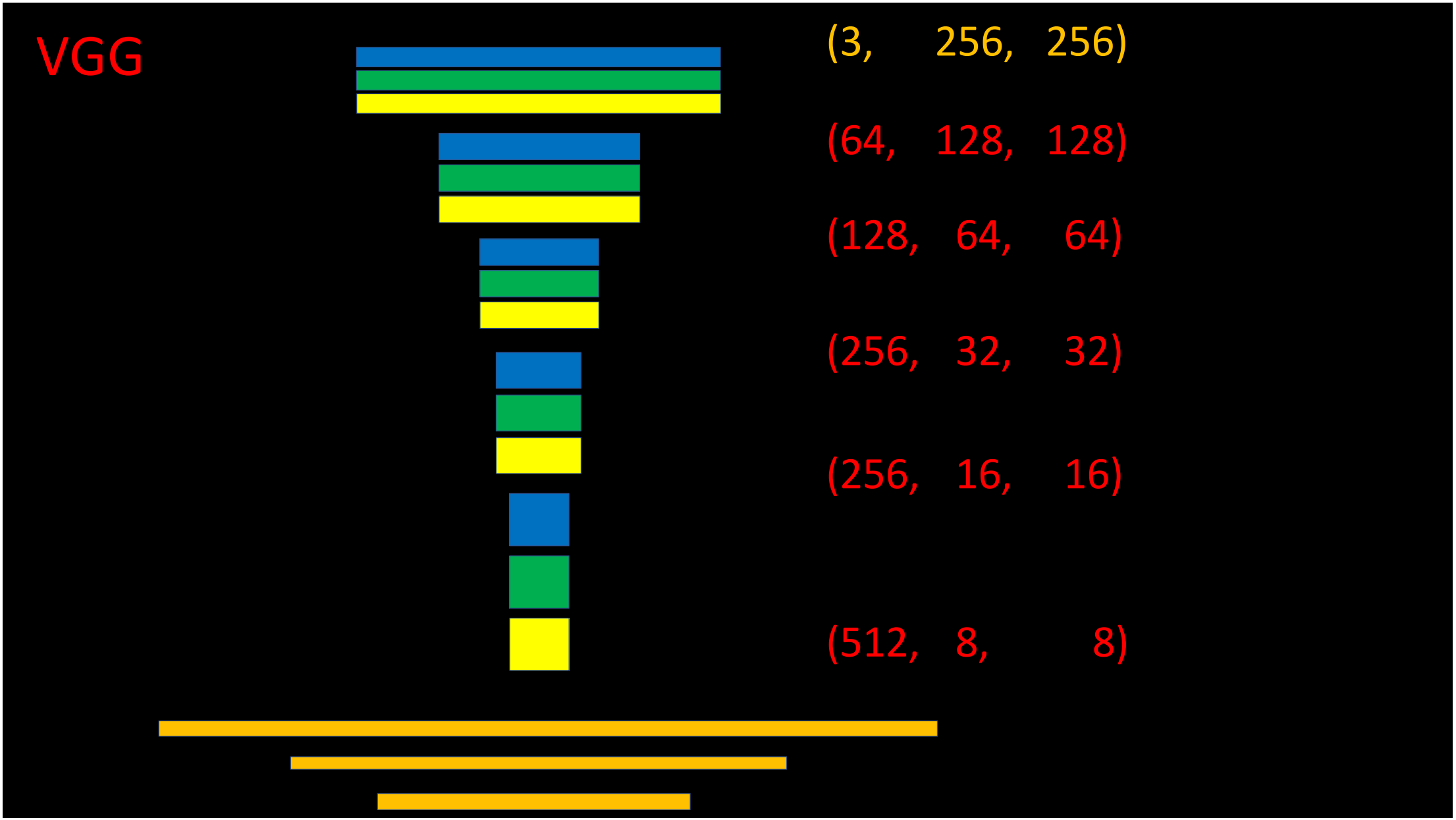
Overview of some popular CNN architectures

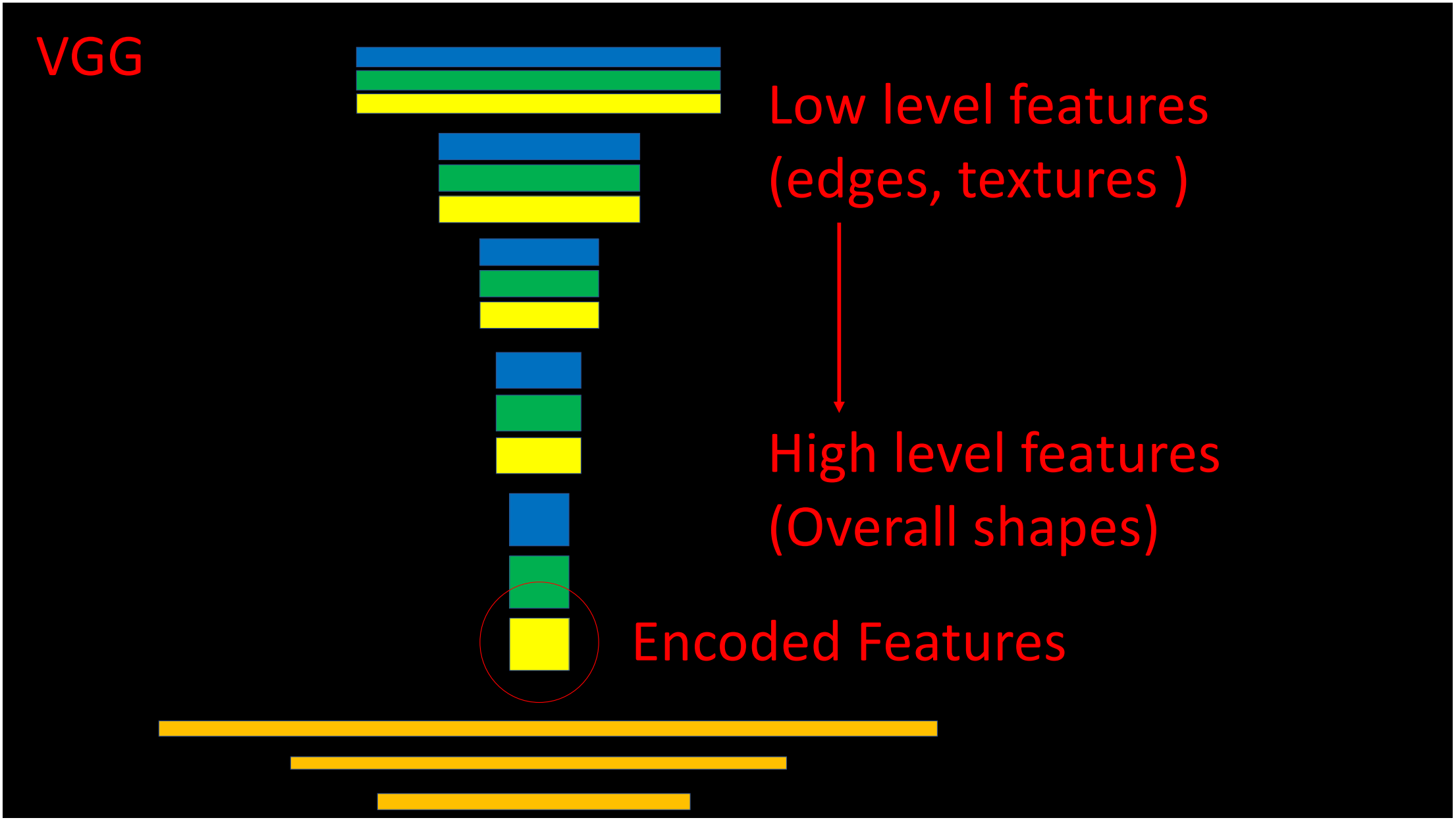

100
90
80
70
60
50
40
TOP 1 ACCURACY
2012
2014
2016
2018
2020
SIFT + FVs
AlexNet
SPPNet
Inception V3
NASNET-A(6)
FixResNeXt-101 32x48d
Meta Pseudo Labels (EfficientNet-L
Other models
State-of-the-art models

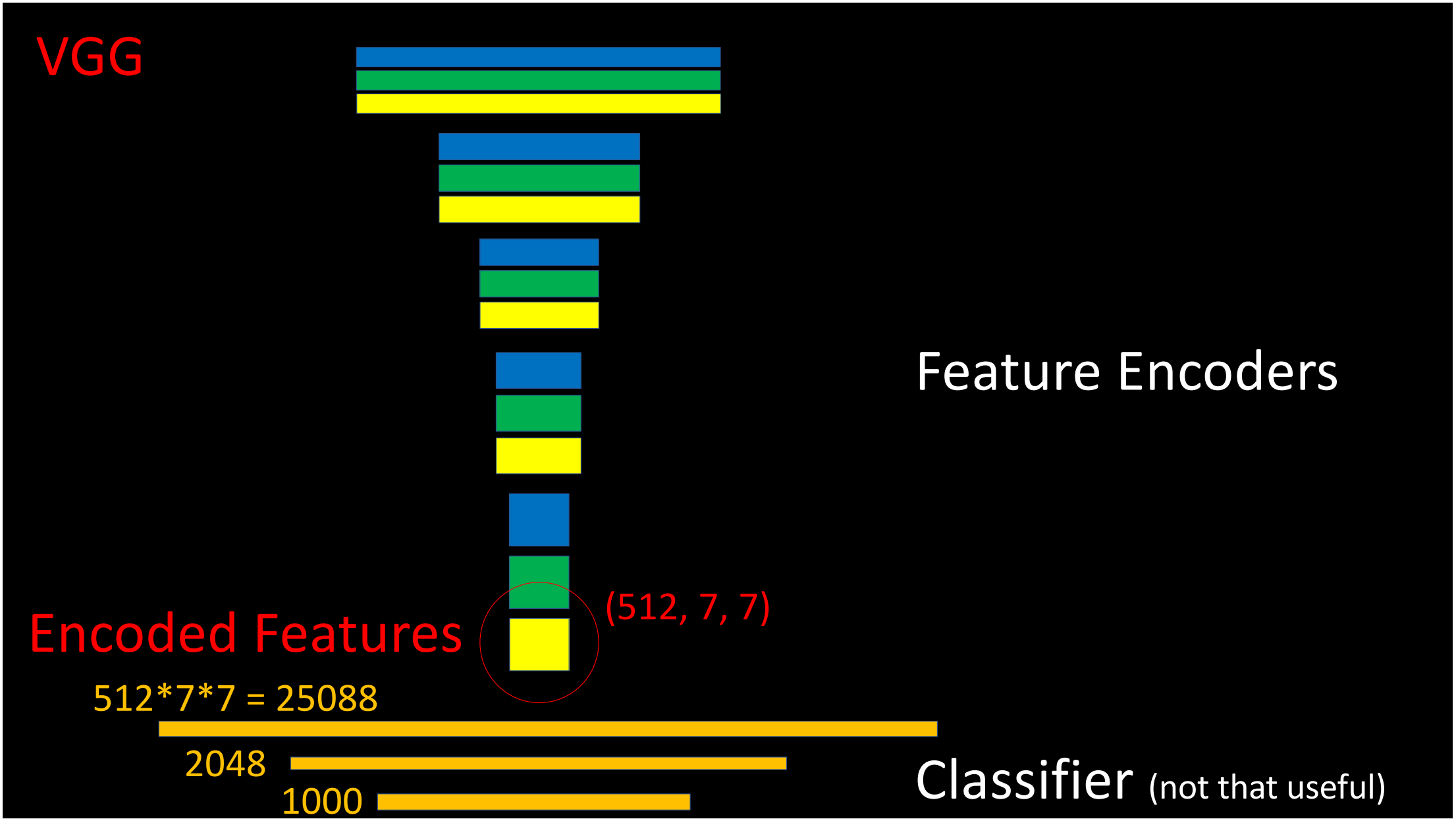

VGG
Channels
(number of
features)
Spatial
Dimensions
(size of features)


VGG
More
Channels,
Smaller
Spatial
Dimension

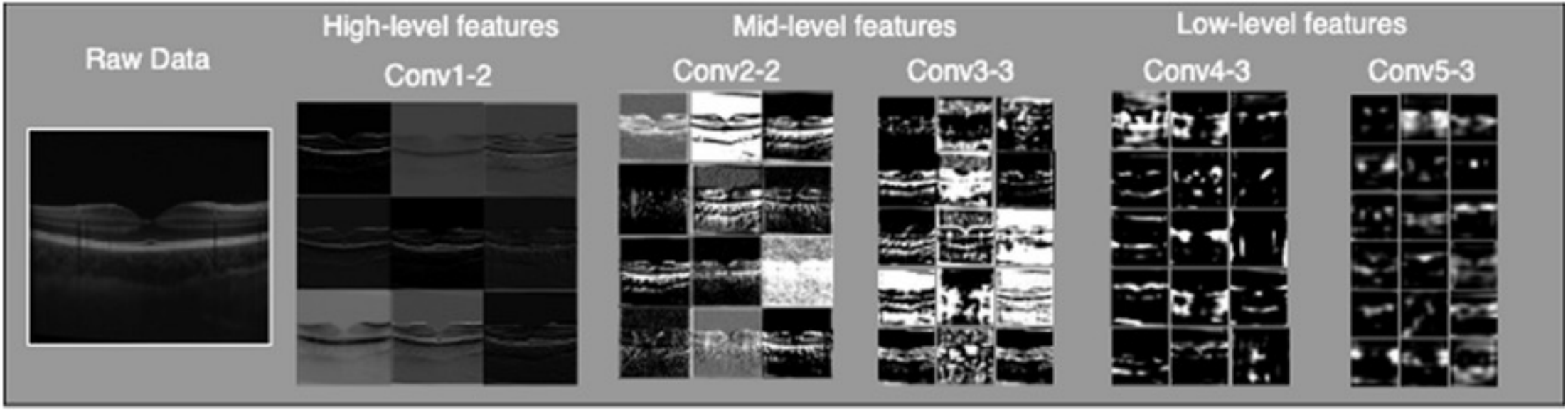

(3, 256, 256)
(64, 128, 128)
(128, 64, 64)
(256, 32, 32)
(256, 1, 1)
Extracted Features
Outcome

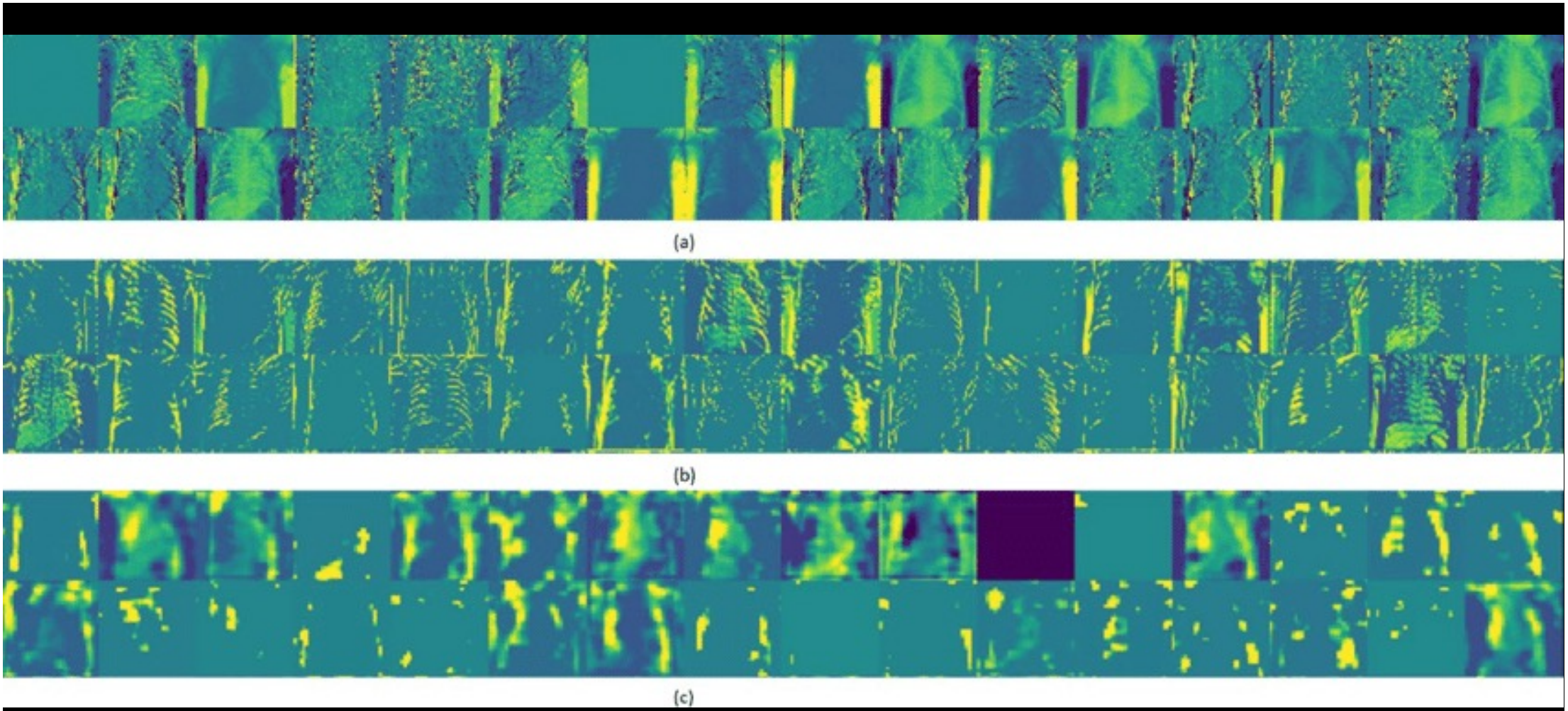

VGG
(3, 256, 256)
(64, 128, 128)
(128, 64, 64)
(256, 32, 32)
(256, 16, 16)
(512, 8, 8)


VGG
Low level features
(edges, textures)
High level features
(Overall shapes)
Encoded Features


VGG
Feature Encoders
(512, 7, 7)
Encoded Features
512*7*7 = 25088
2048
1000
Classifier (not that useful)

Deep feature loss to denoise OCT images using deep neural networks

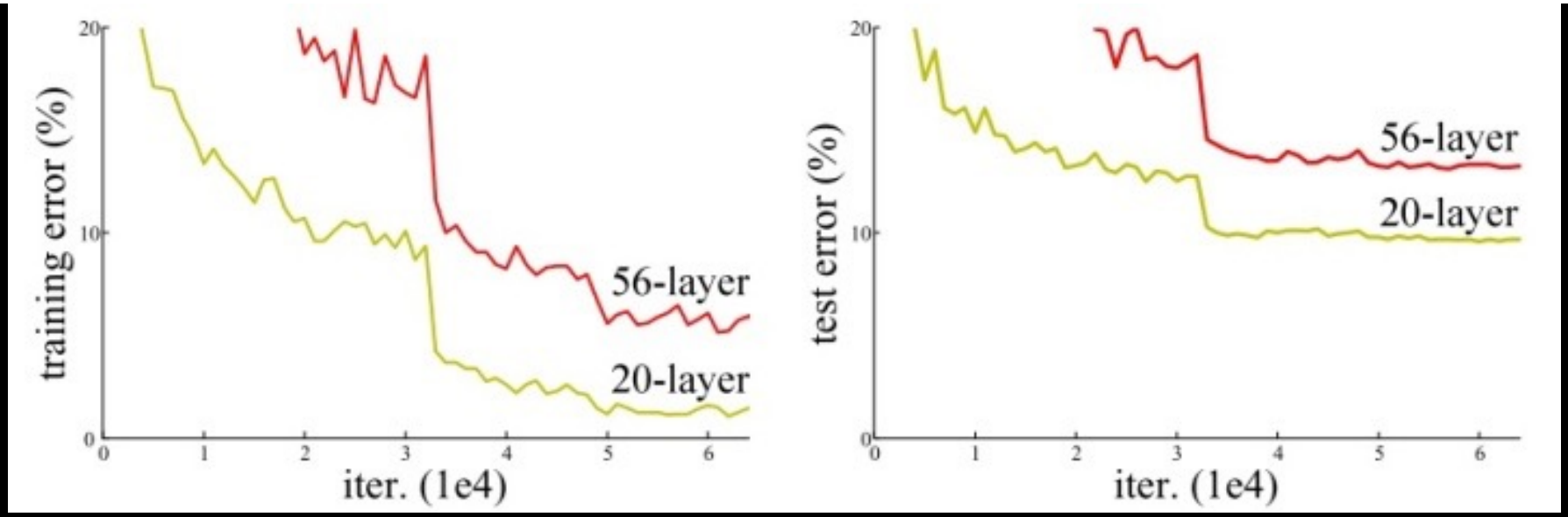




Visualizing and explaining deep learning predictions for pneumonia detection in pediatric chest radiographs

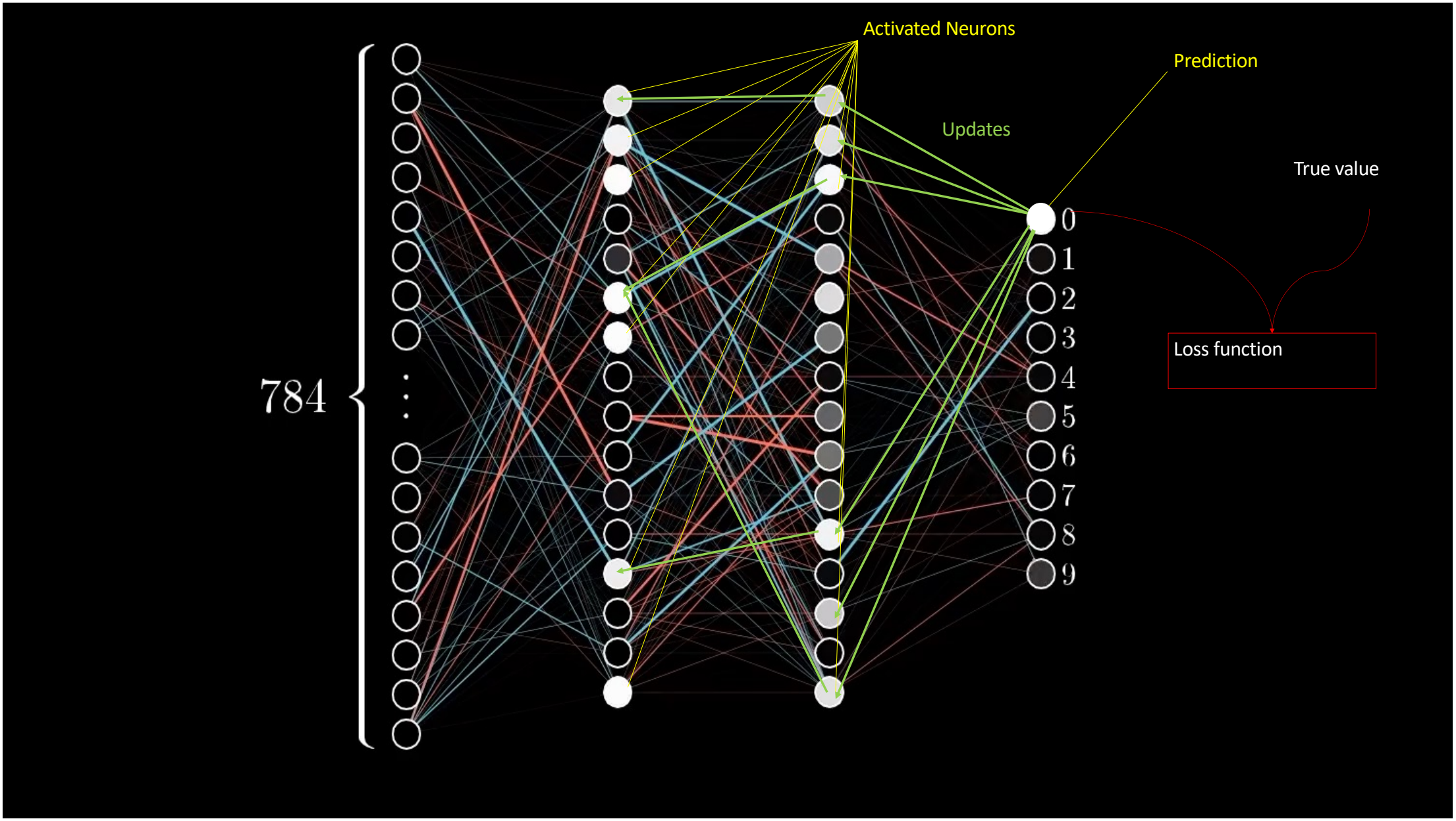
```
torchvision.models.vgg11()
VGG(
  (features): Sequential(
    (0): Conv2d(3, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU(inplace=True)
    (2): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
    (3): Conv2d(64, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (4): ReLU(inplace=True)
    (5): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
    (6): Conv2d(128, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (7): ReLU(inplace=True)
    (8): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (9): ReLU(inplace=True)
    (10): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
    (11): Conv2d(256, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (12): ReLU(inplace=True)
    (13): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (14): ReLU(inplace=True)
    (15): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
    (16): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (17): ReLU(inplace=True)
    (18): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (19): ReLU(inplace=True)
    (20): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
  )
```

Residual Networks (Resnet)

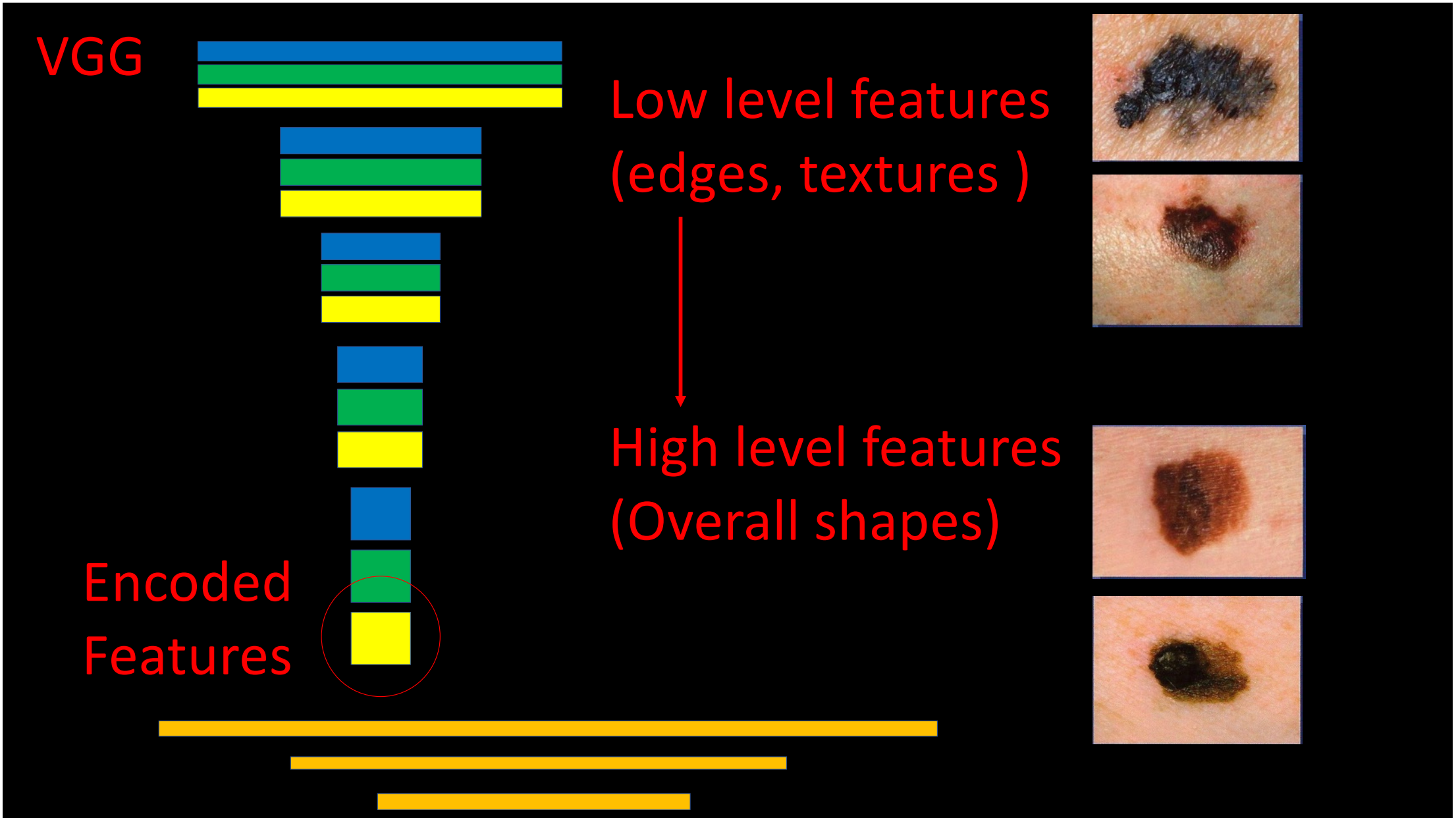


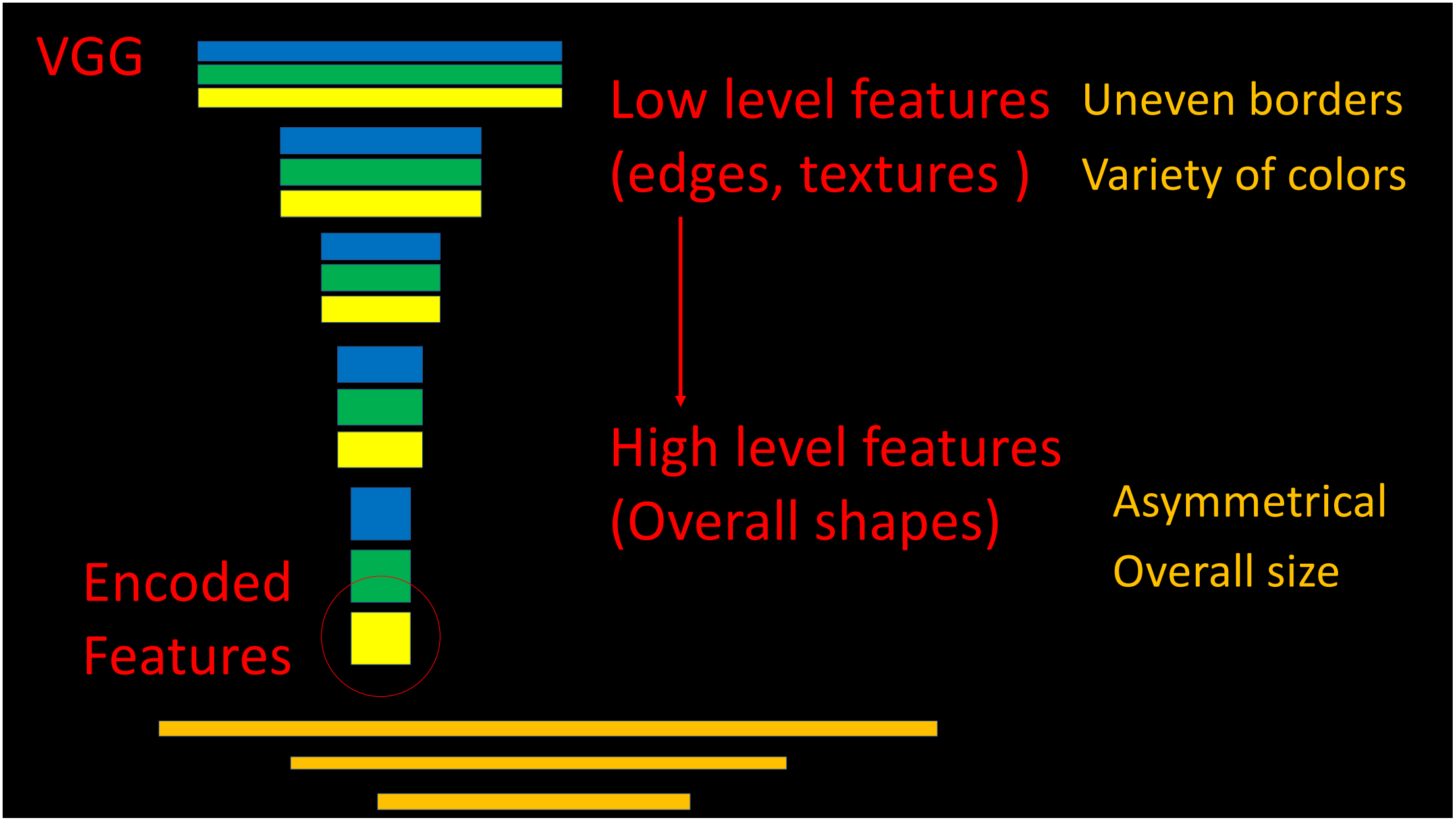


Deeper the better?


Activated Neurons
Prediction
Updates
True value
784
0
1
2
3
4
5
6
7
8
9
Loss function

ResNet 34….up to 101
(most commonly used)


VGG
Low level features
(edges, textures)
High level features
(Overall shapes)
Encoded
Features


VGG
Low level features
(edges, textures)
Uneven borders
Variety of colors
High level features
(Overall shapes)
Asymmetrical
Overall size
Encoded
Features







I want to have the option to keep updating it,

OR keep it the same

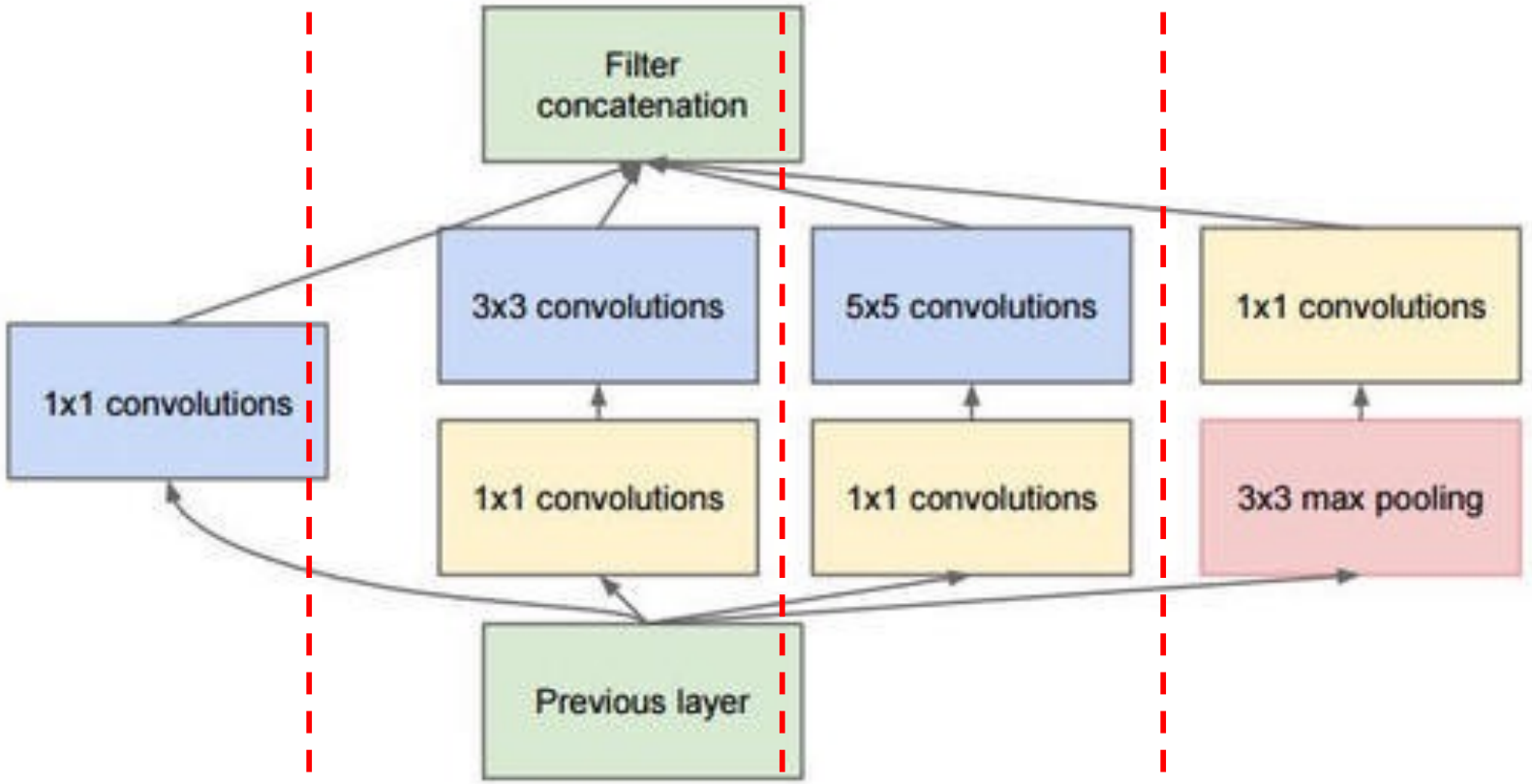
High level may appear later / earlier!

CONVOLUTION WITH BRANCHES

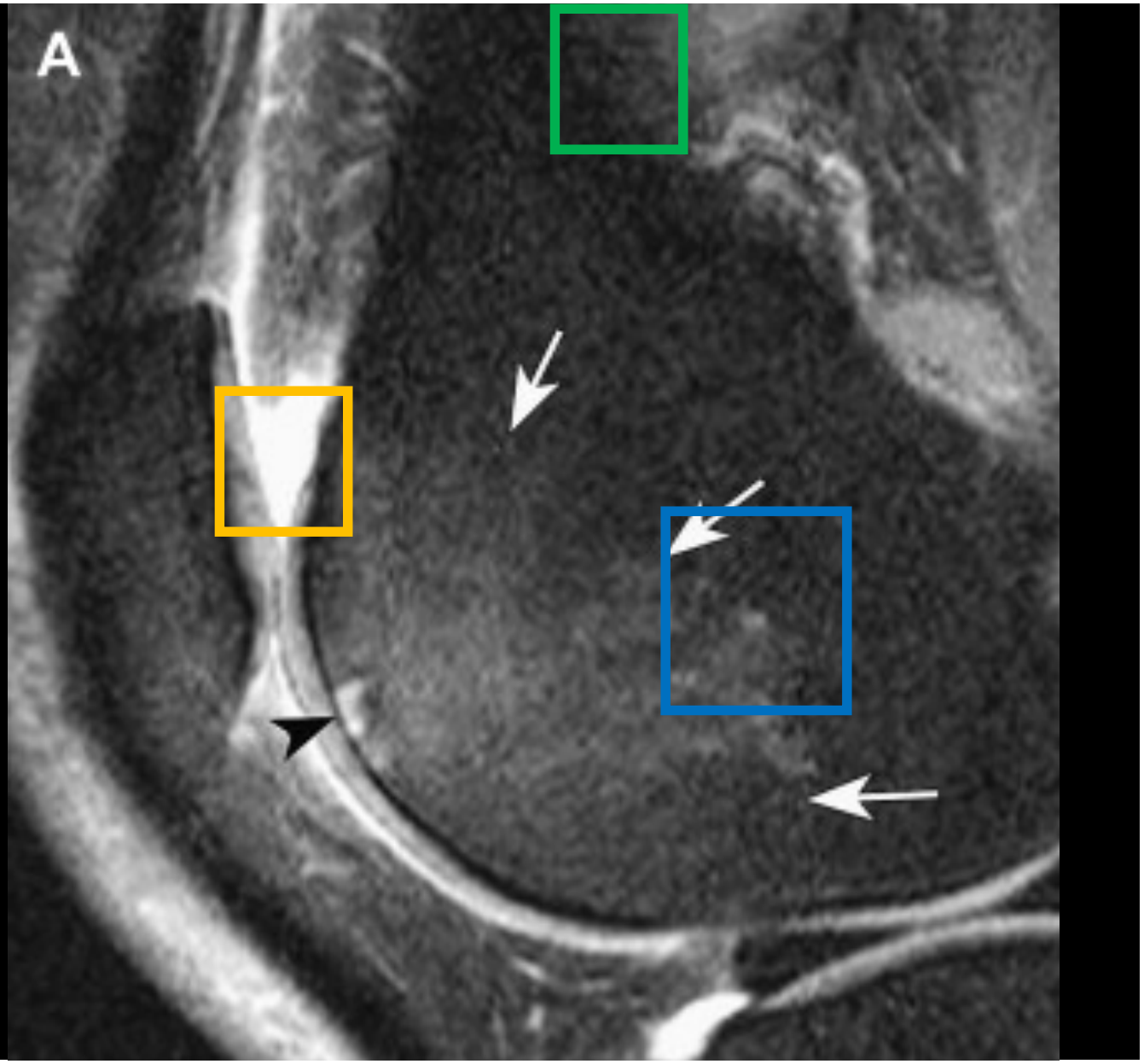
Need to know more From my neighbor (and maybe change my mind)

Need to know from my FURTHER neighbor

Just Pool over my neighbor

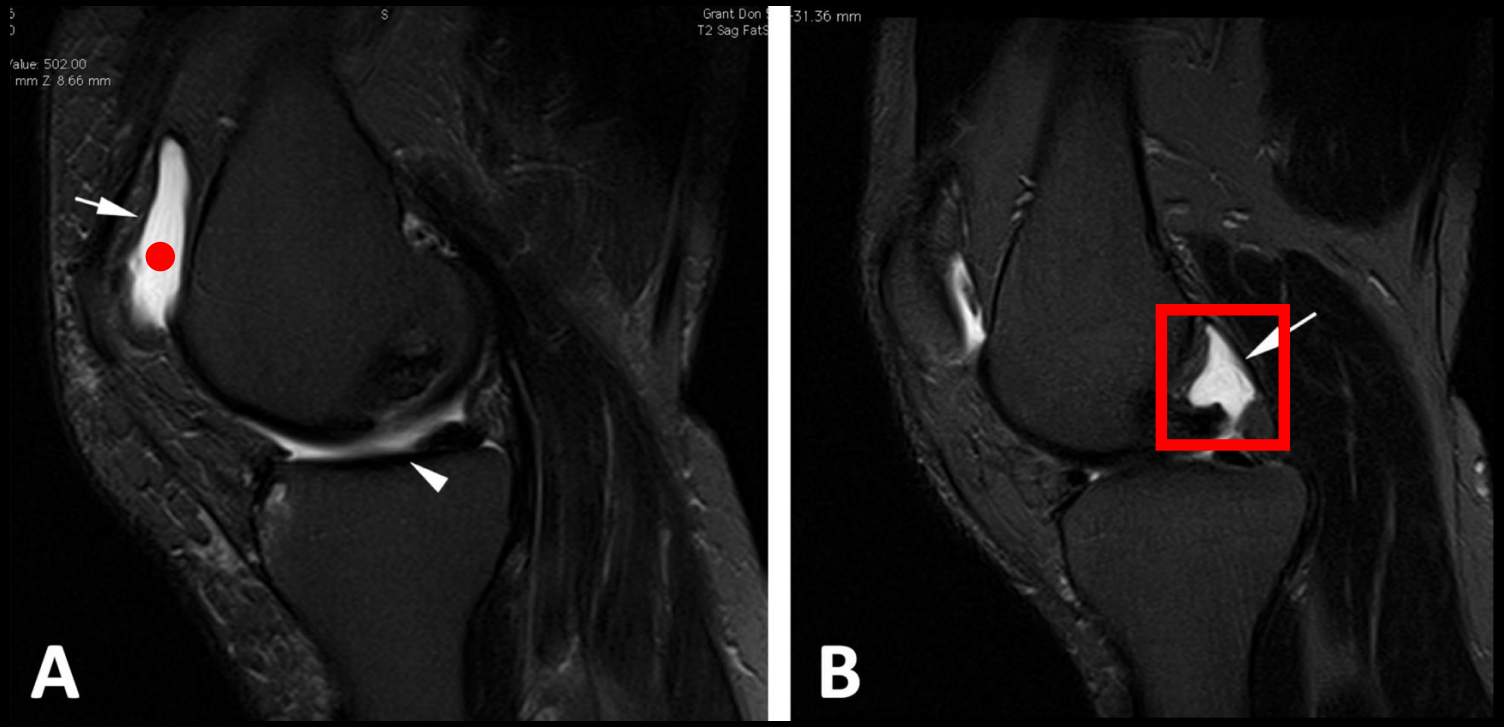


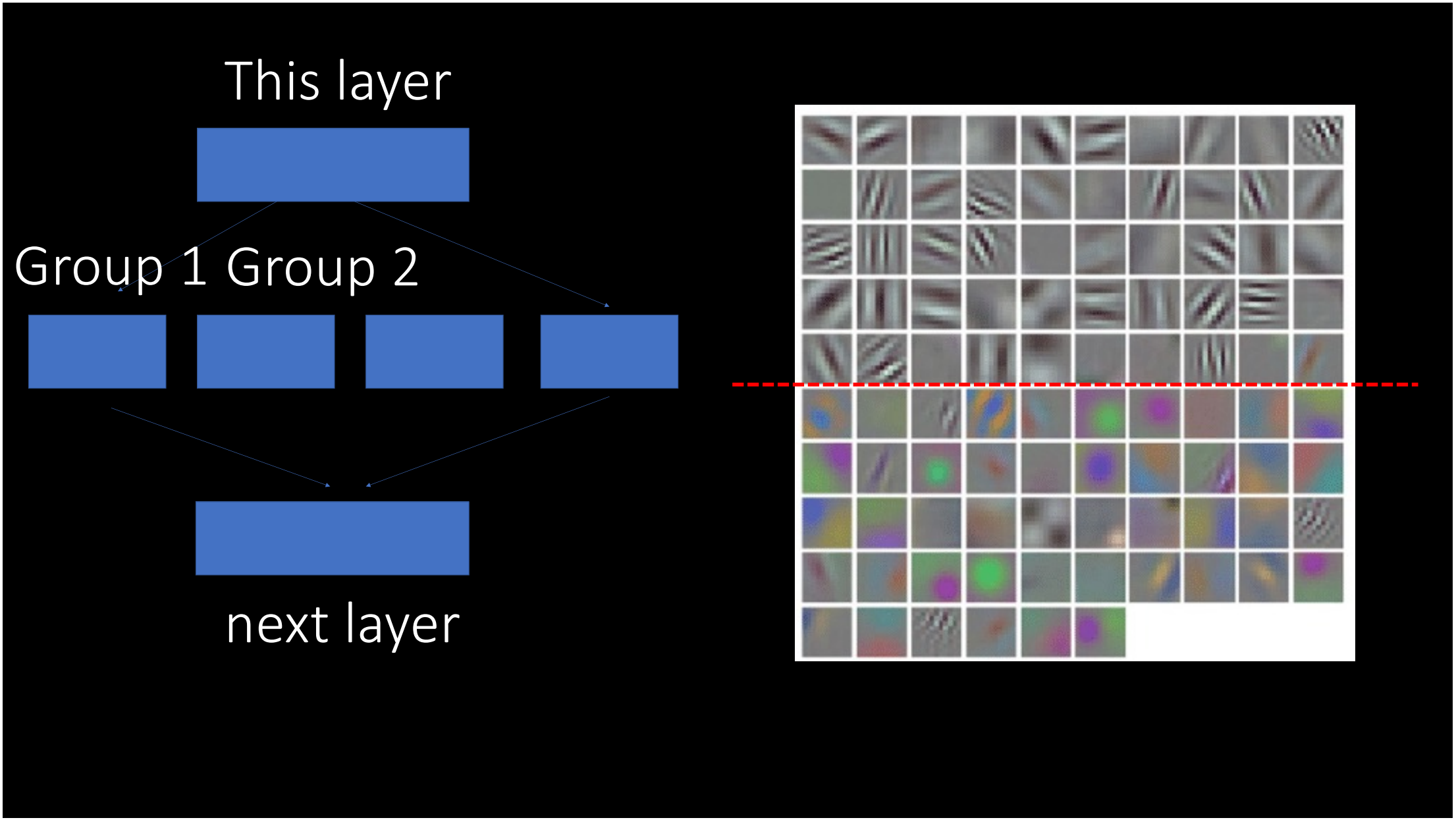
Q: is this abnormal?



Q1: is this effusion?

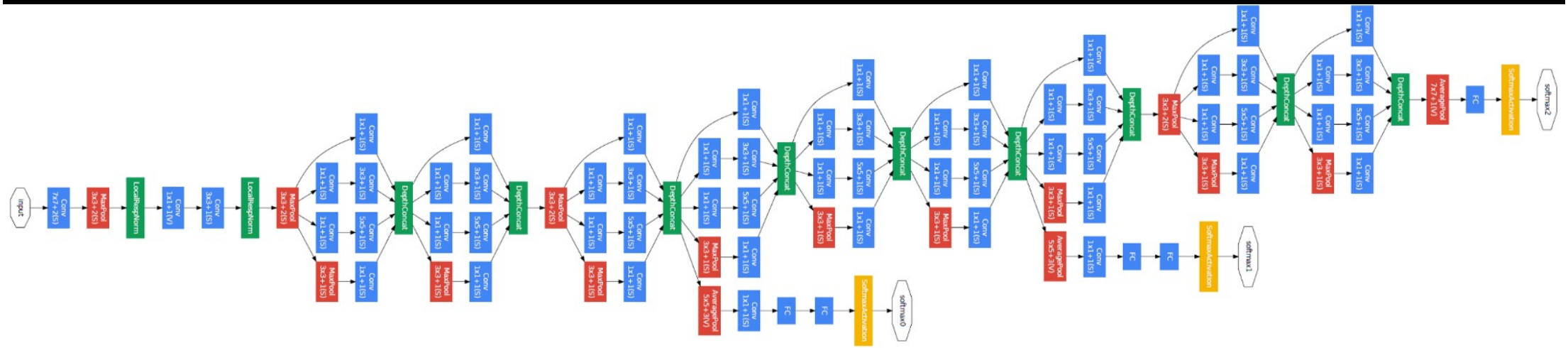
Q2: where is the effusion?

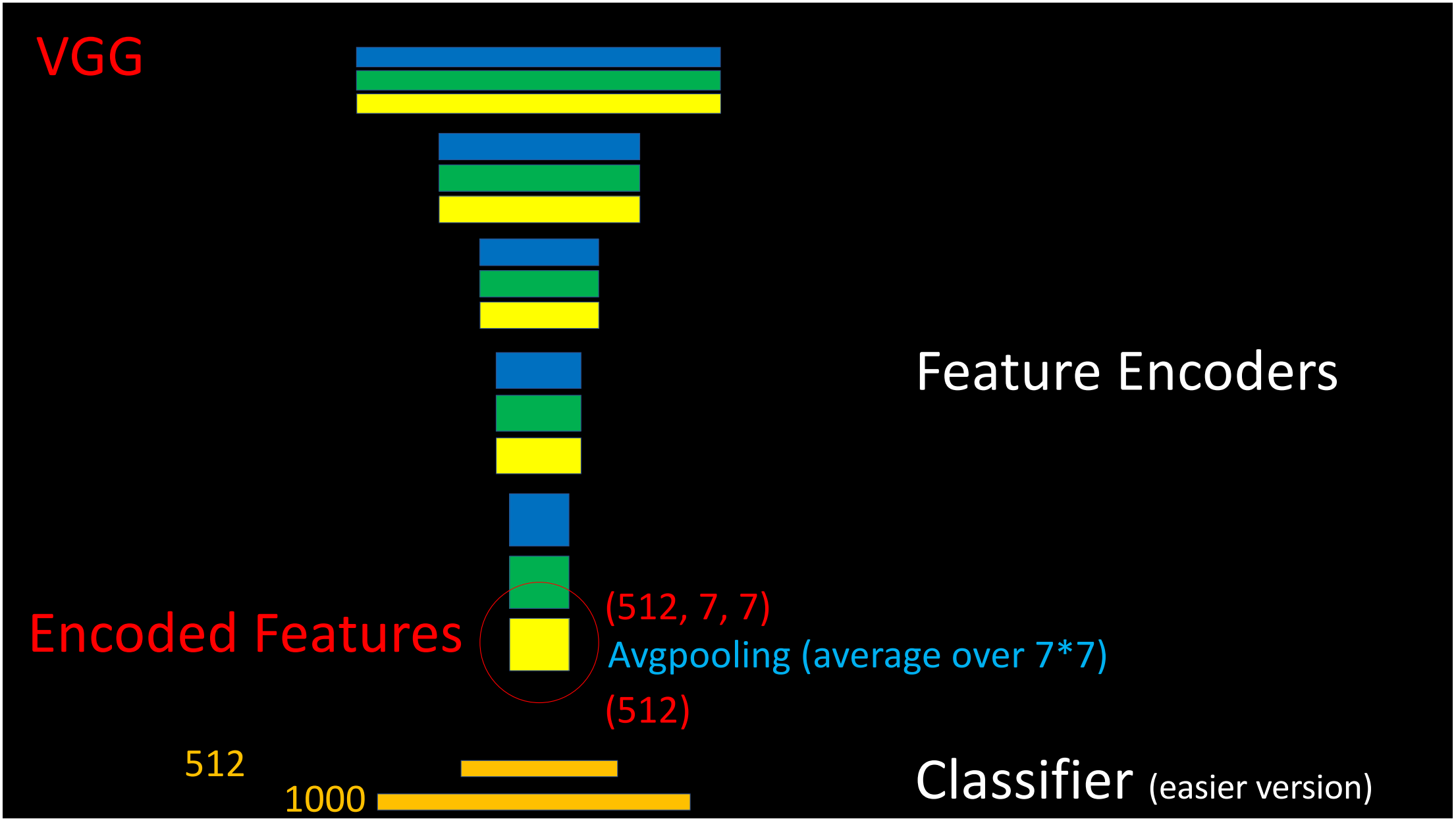



This layer
Group 1 Group 2
next layer



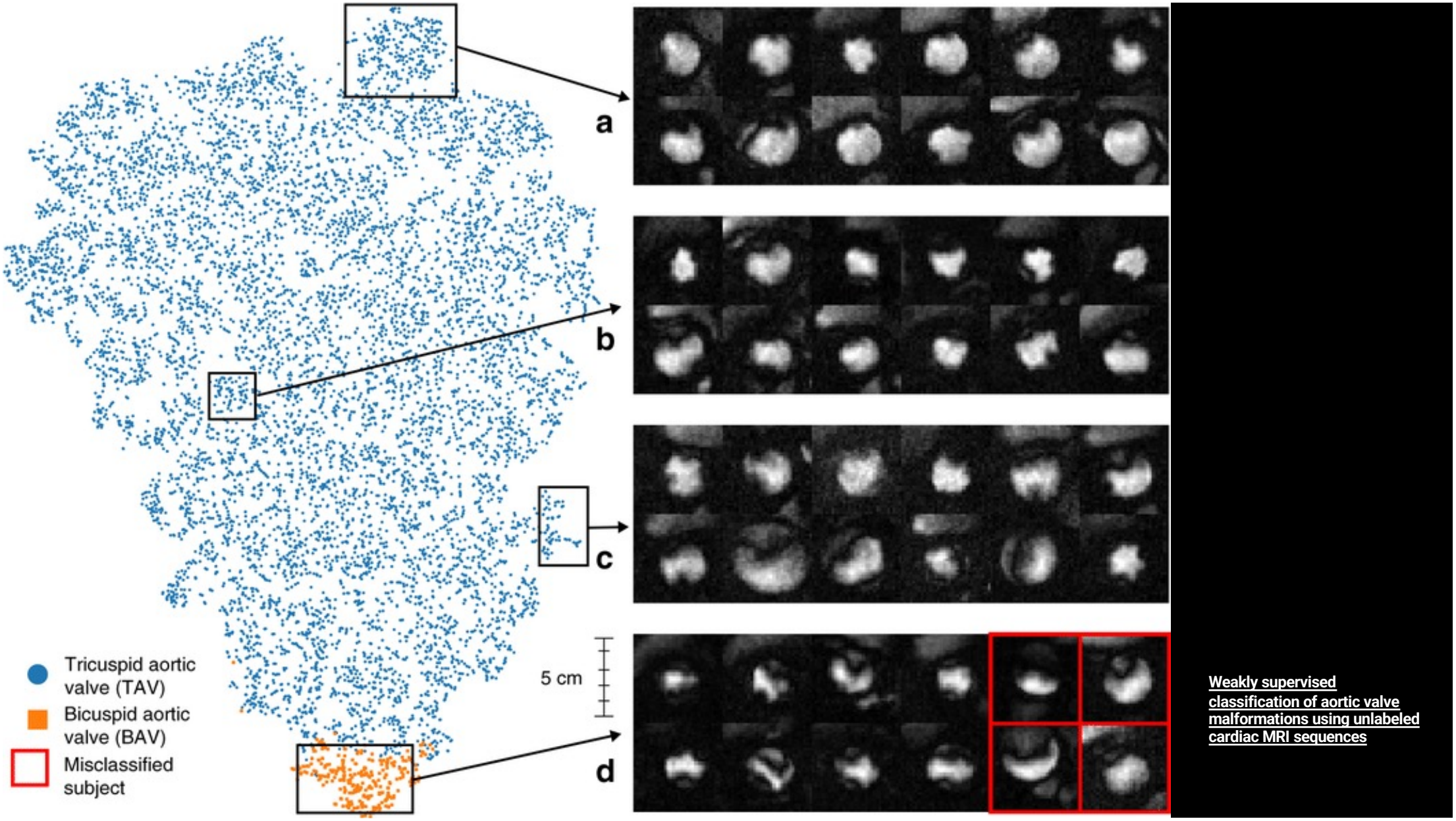
Inception (2014)




VGG
Feature Encoders
Encoded Features
(512, 7, 7)
Avgpooling (average over 7*7)
(512)
512
1000
Classifier (easier version)

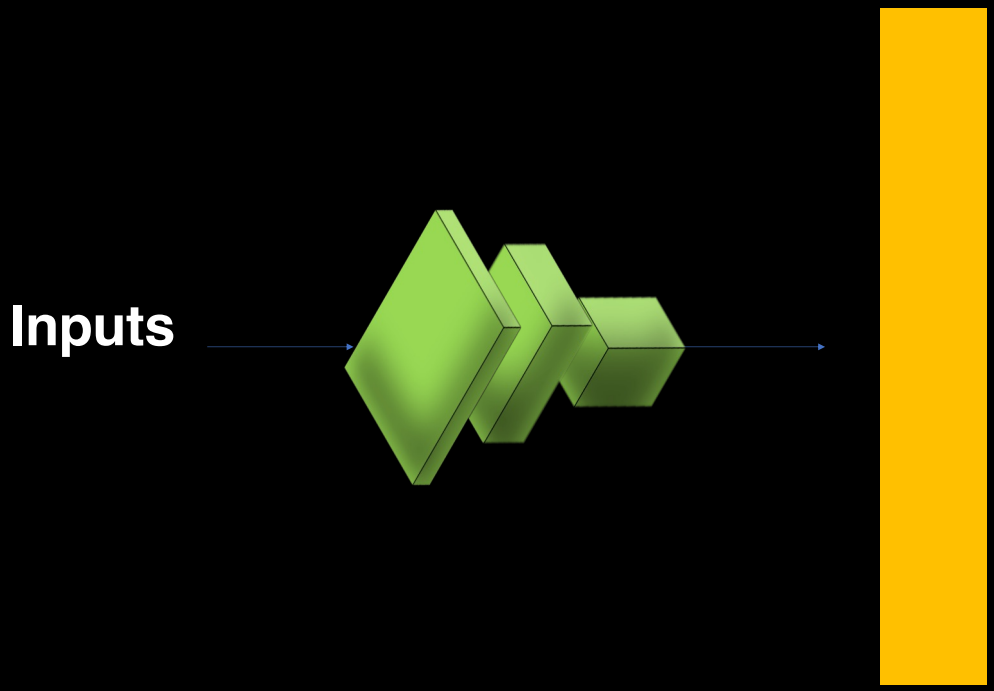
“latent space”

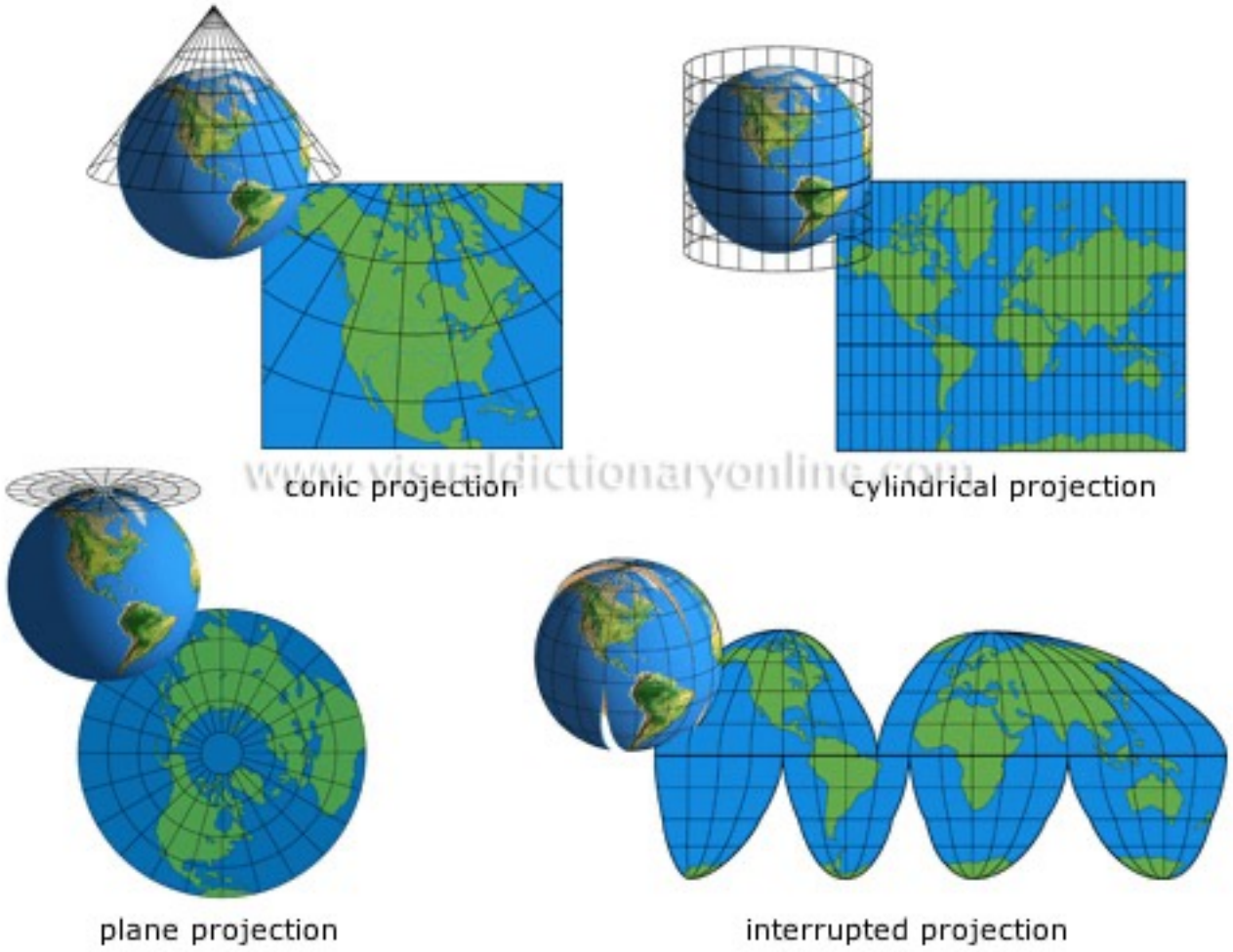




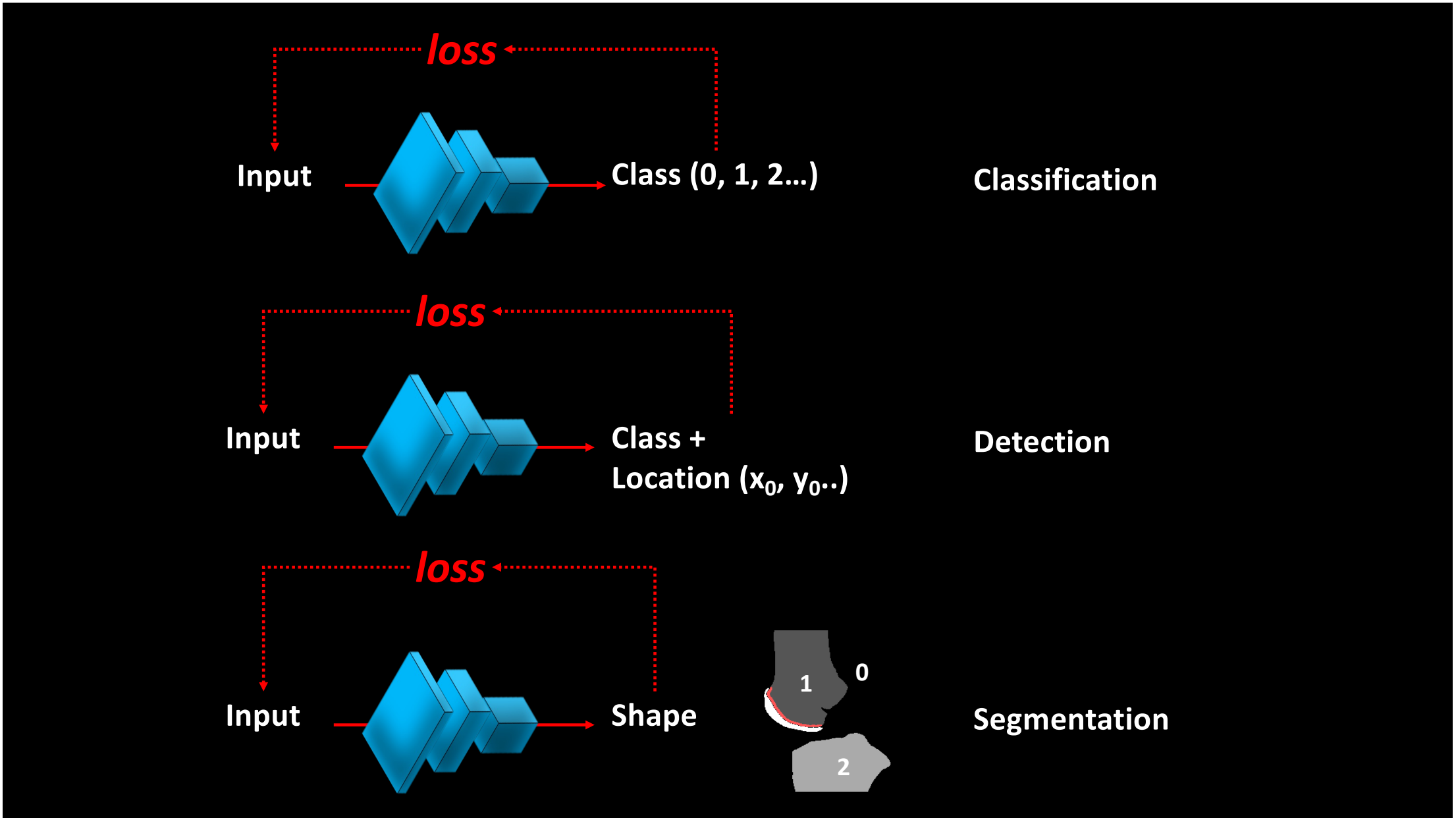
Weakly supervised classification of aortic valve malformations using unlabeled cardiac MRI sequences

A good latent space include the essential information about the data

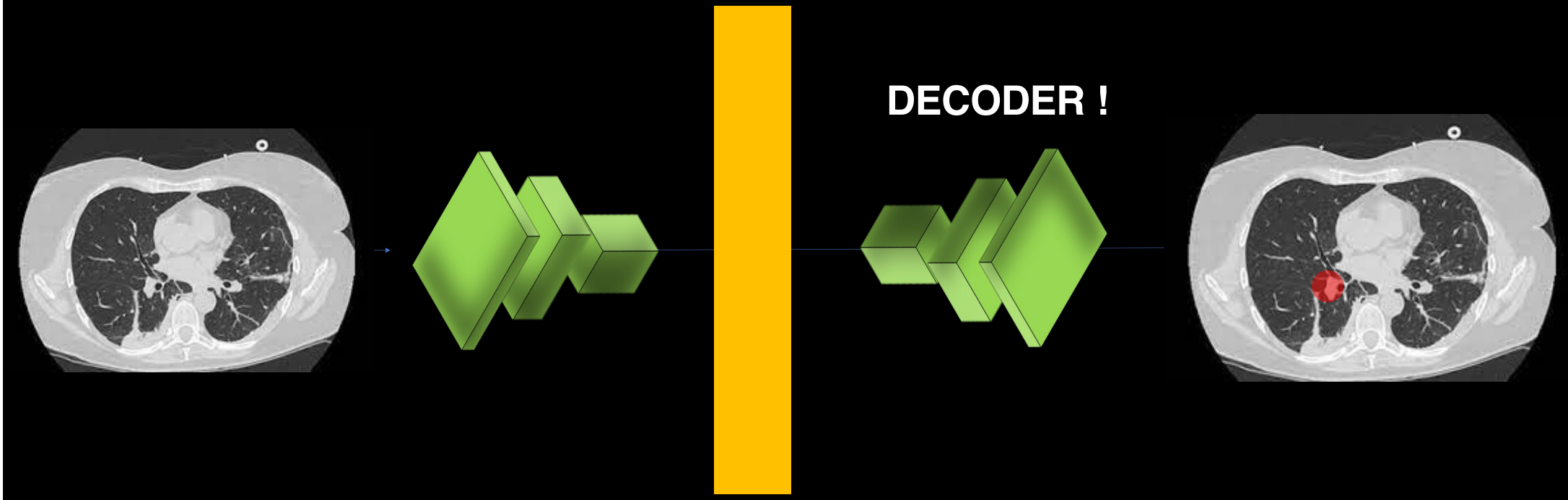




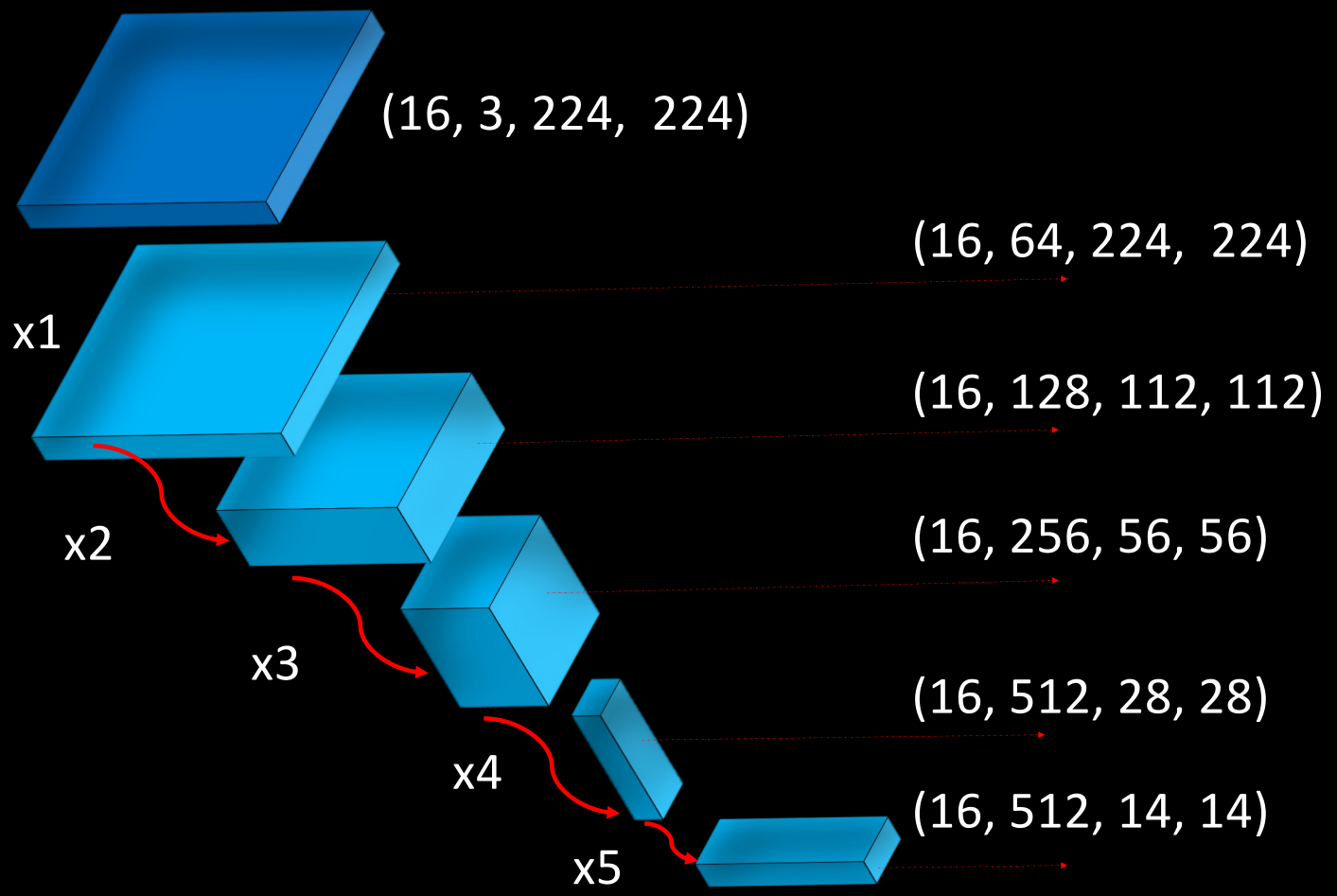
What is the latent space behind all these types of projections?


loss
Input
Class (0, 1, 2…)
Classification
loss
Input
Class +
Location (x_0, y_0..)
Detection
loss
Input
Shape
1
0
2
Segmentation

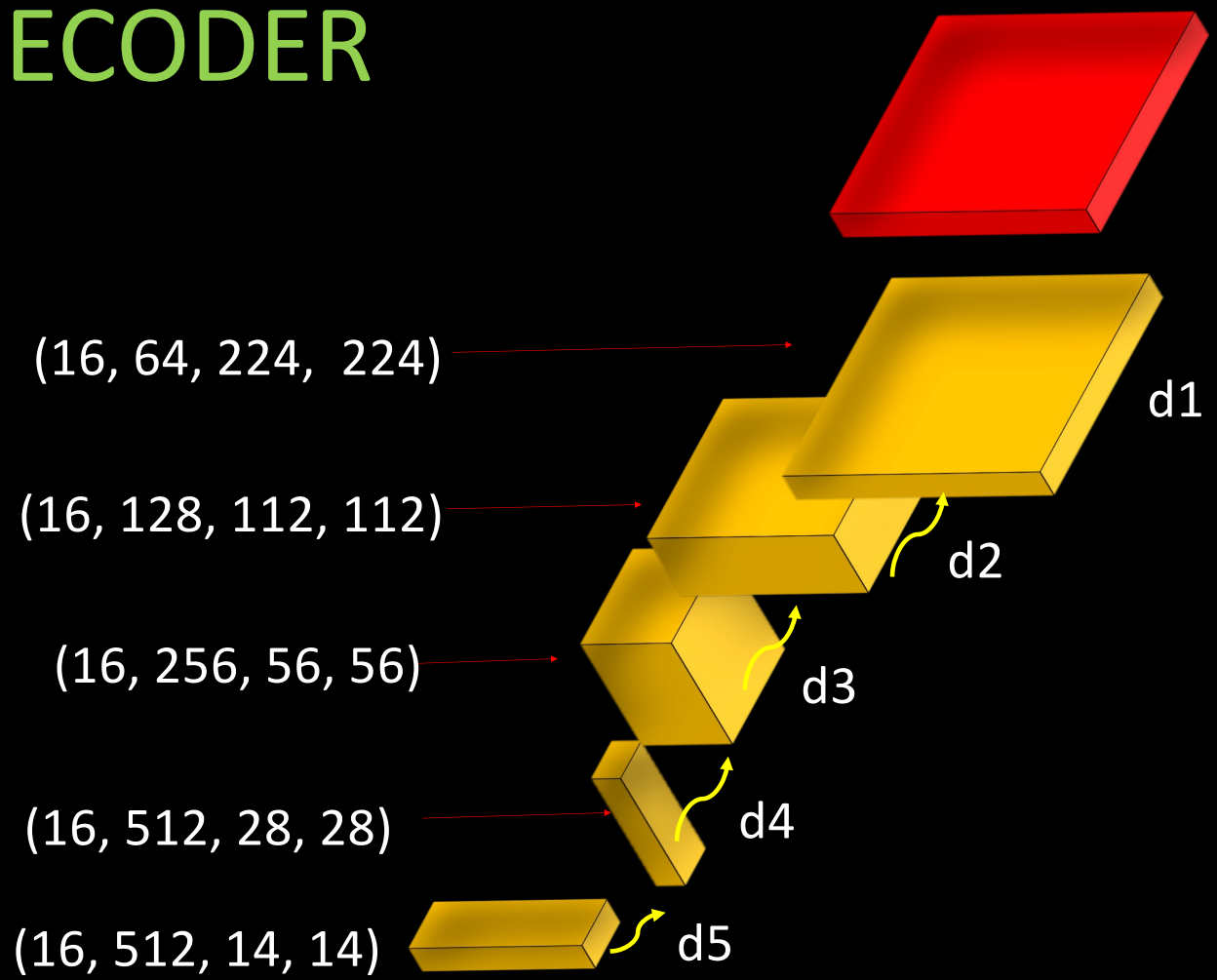
We need one more thing to project the good latent space back to the image we want

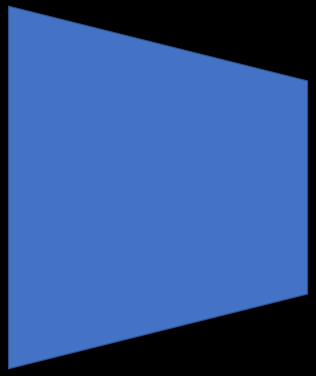


ENCODER

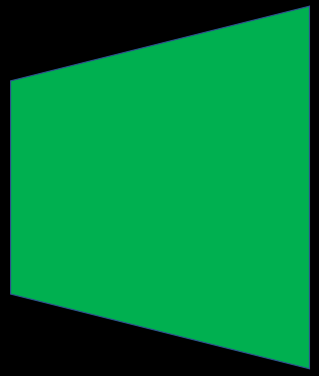

(16, 3, 224, 224)
x1
(16, 64, 224, 224)
x2
(16, 128, 112, 112)
x3
(16, 256, 56, 56)
x4
(16, 512, 28, 28)
x5
(16, 512, 14, 14)

DECODER

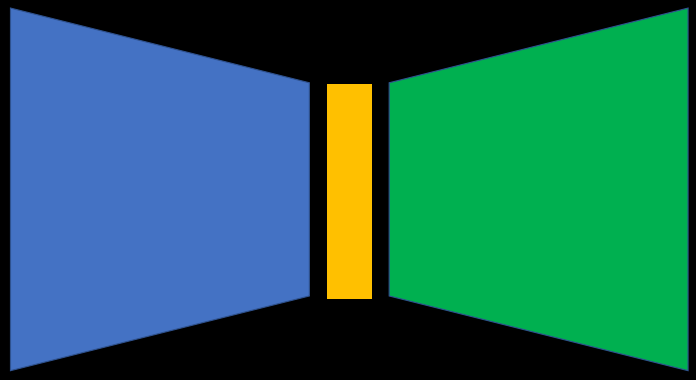

(16, 64, 224, 224)
(16, 128, 112, 112)
(16, 256, 56, 56)
(16, 512, 28, 28)
(16, 512, 14, 14)
d1
d2
d3
d4
d5



Encoder:
Image to features

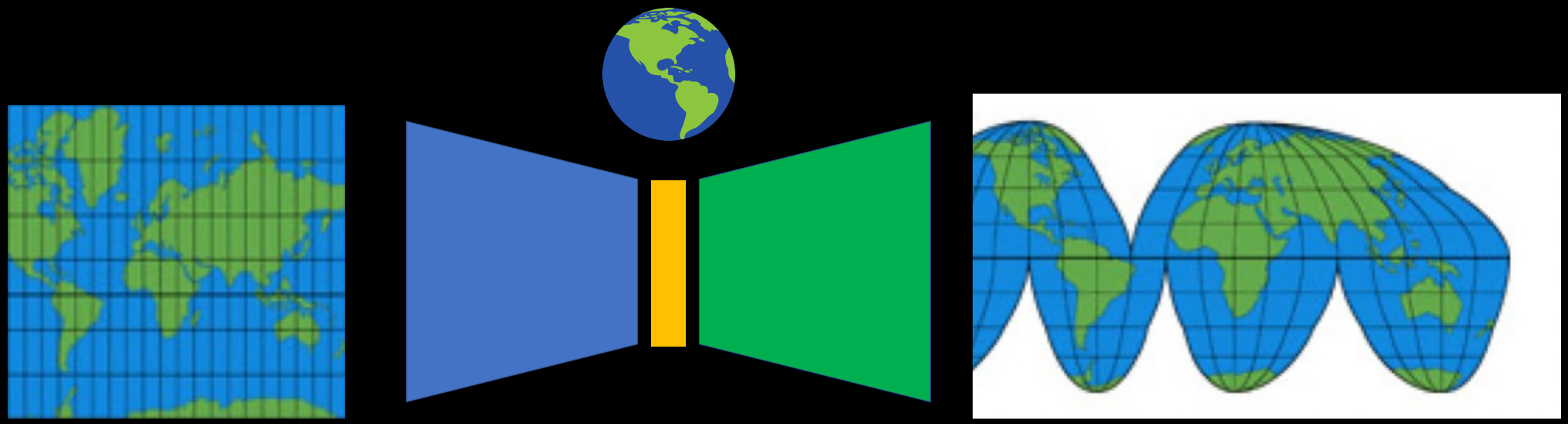


decoder:
Features to image



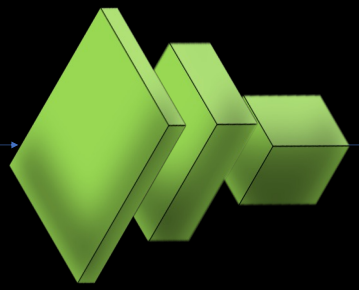
Encoder-decoder:
Image to features then to image
(why?)

Encoder-decoder:
Image to features then to image
(a good latent space contained the essential information)



Multi Tasks

Inputs

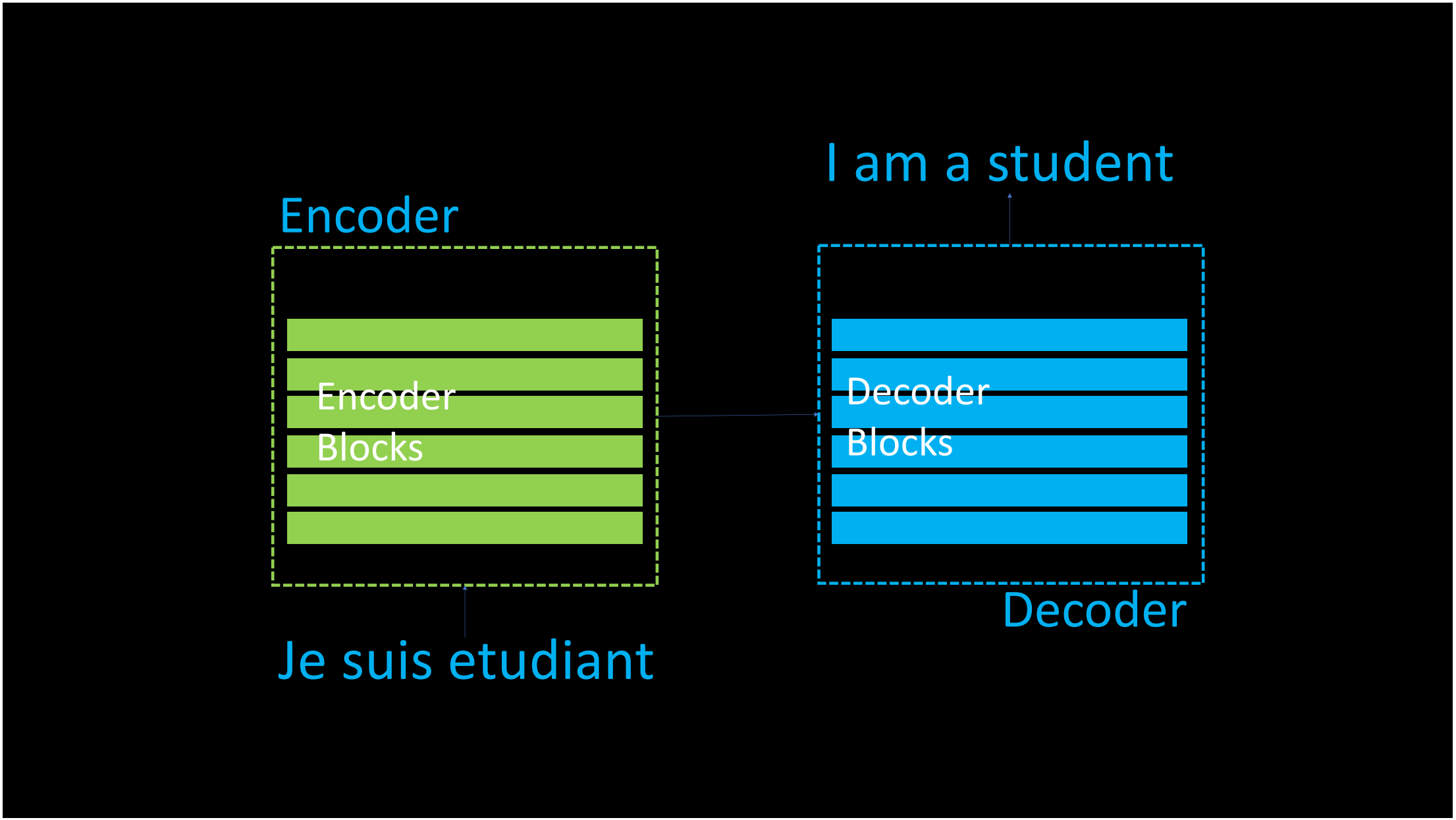


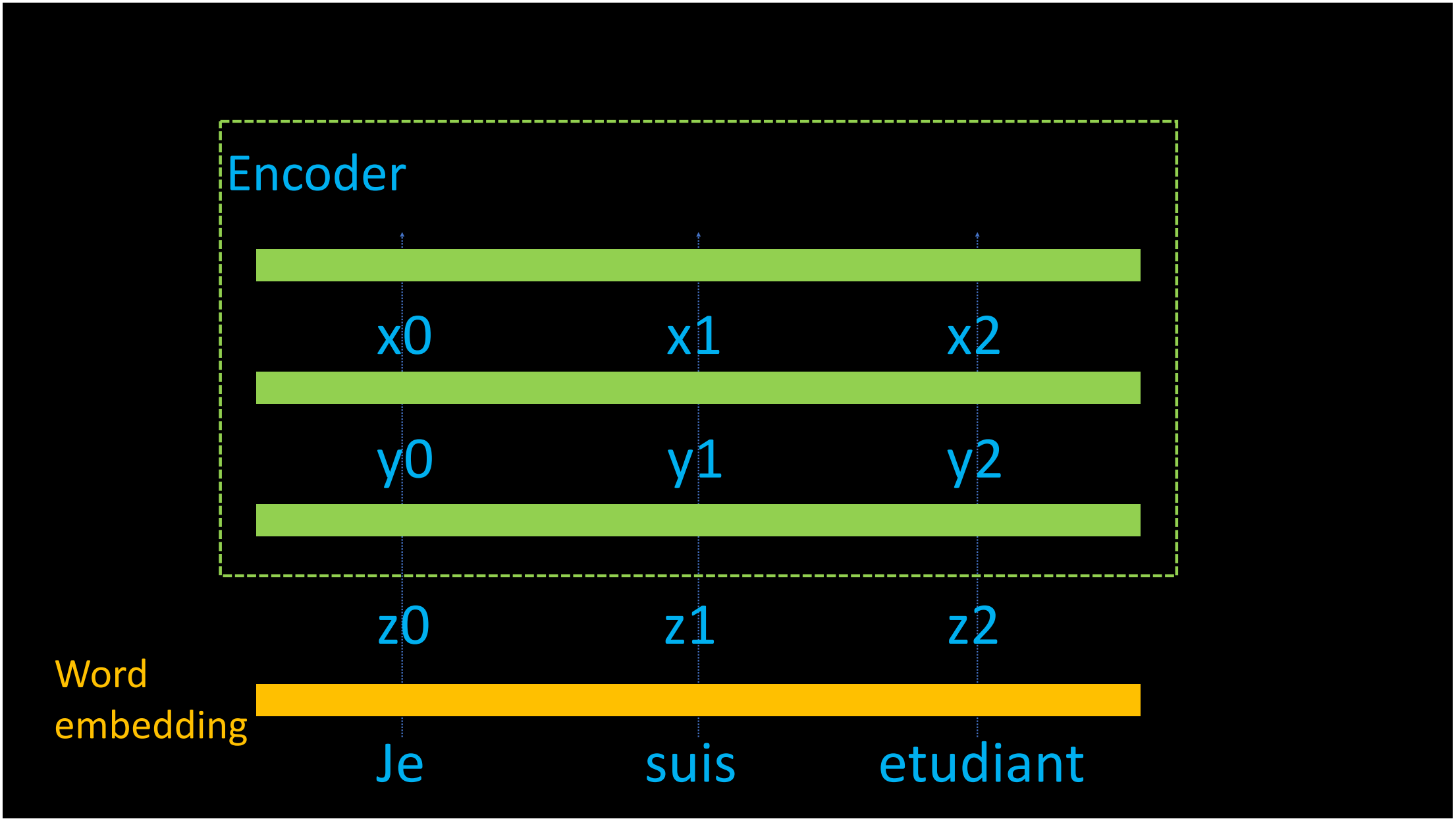
A good latent space

Task A

Task B

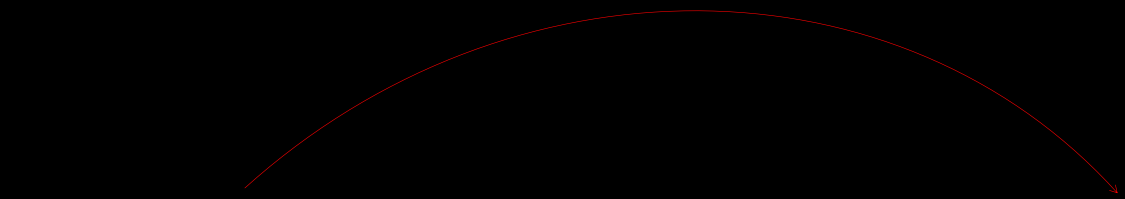
Task C


I am a student
Encoder
Encoder
Blocks
Decoder
Blocks
Decoder
Je suis etudiant


Encoder
x0
x1
x2
y0
y1
y2
z0
z1
z2
Word embedding
Je
suis
etudiant

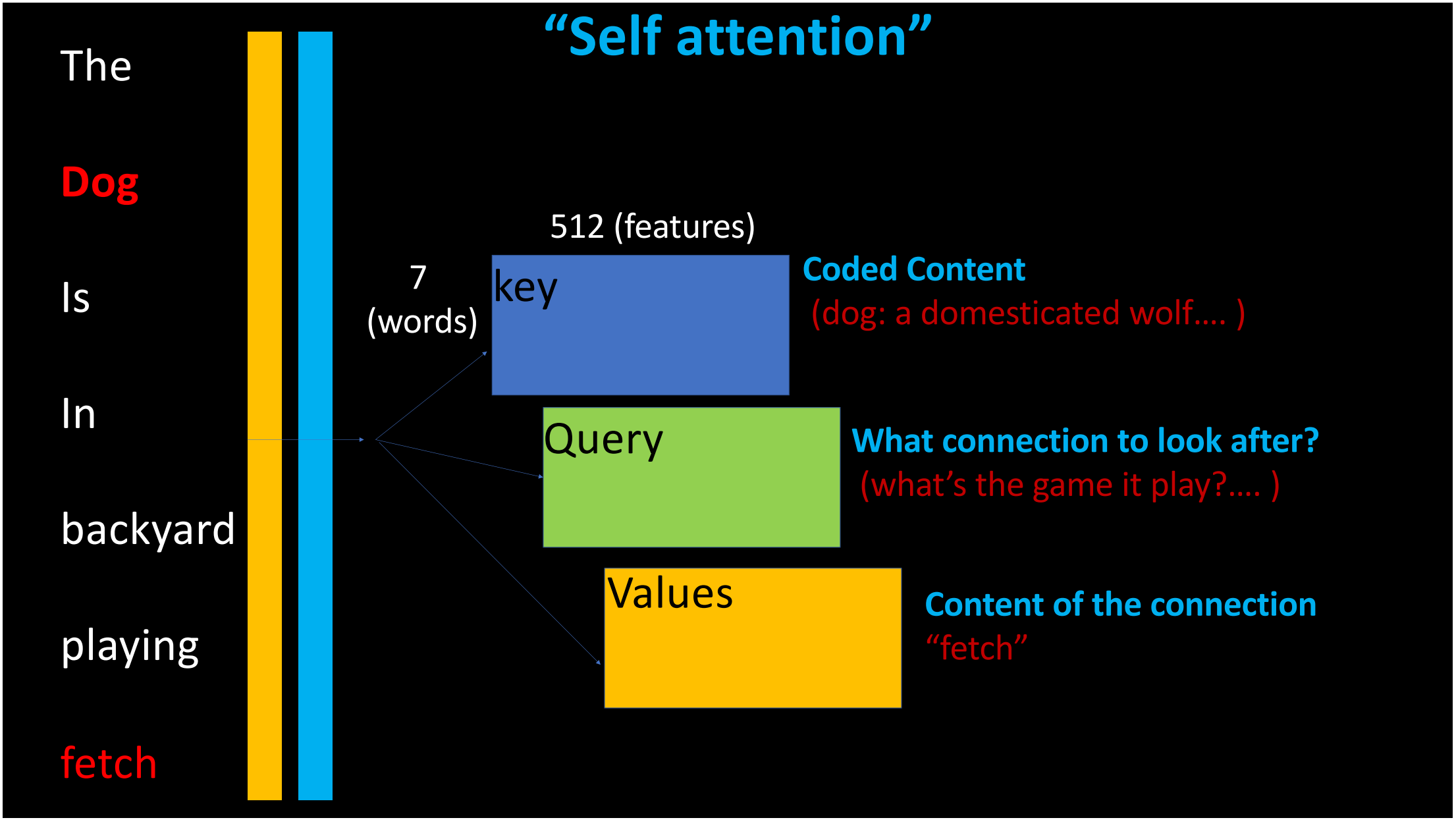
In Each Encoder Blocks:

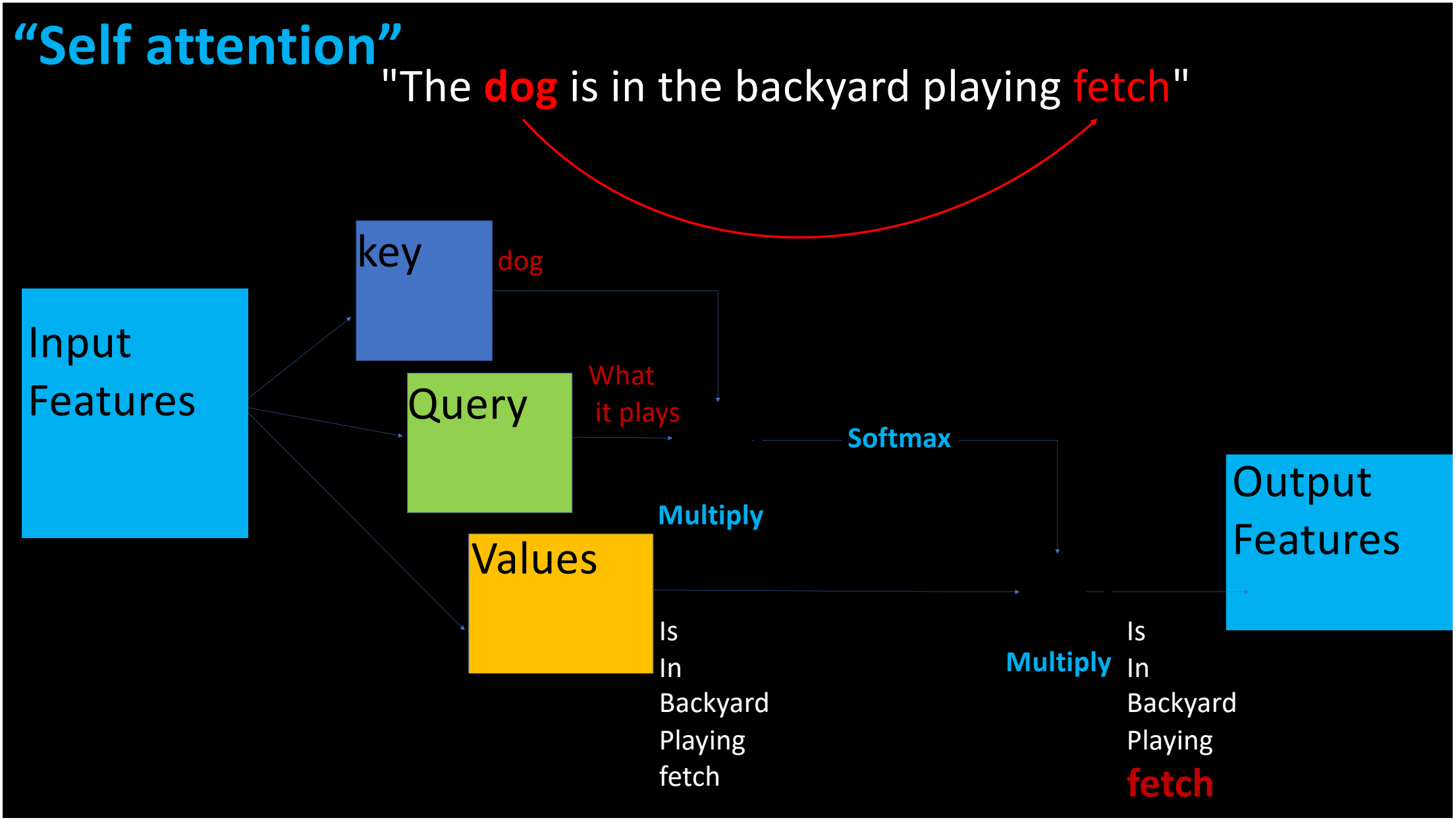
- Self-attention Layer
- Fully-connected (FC)

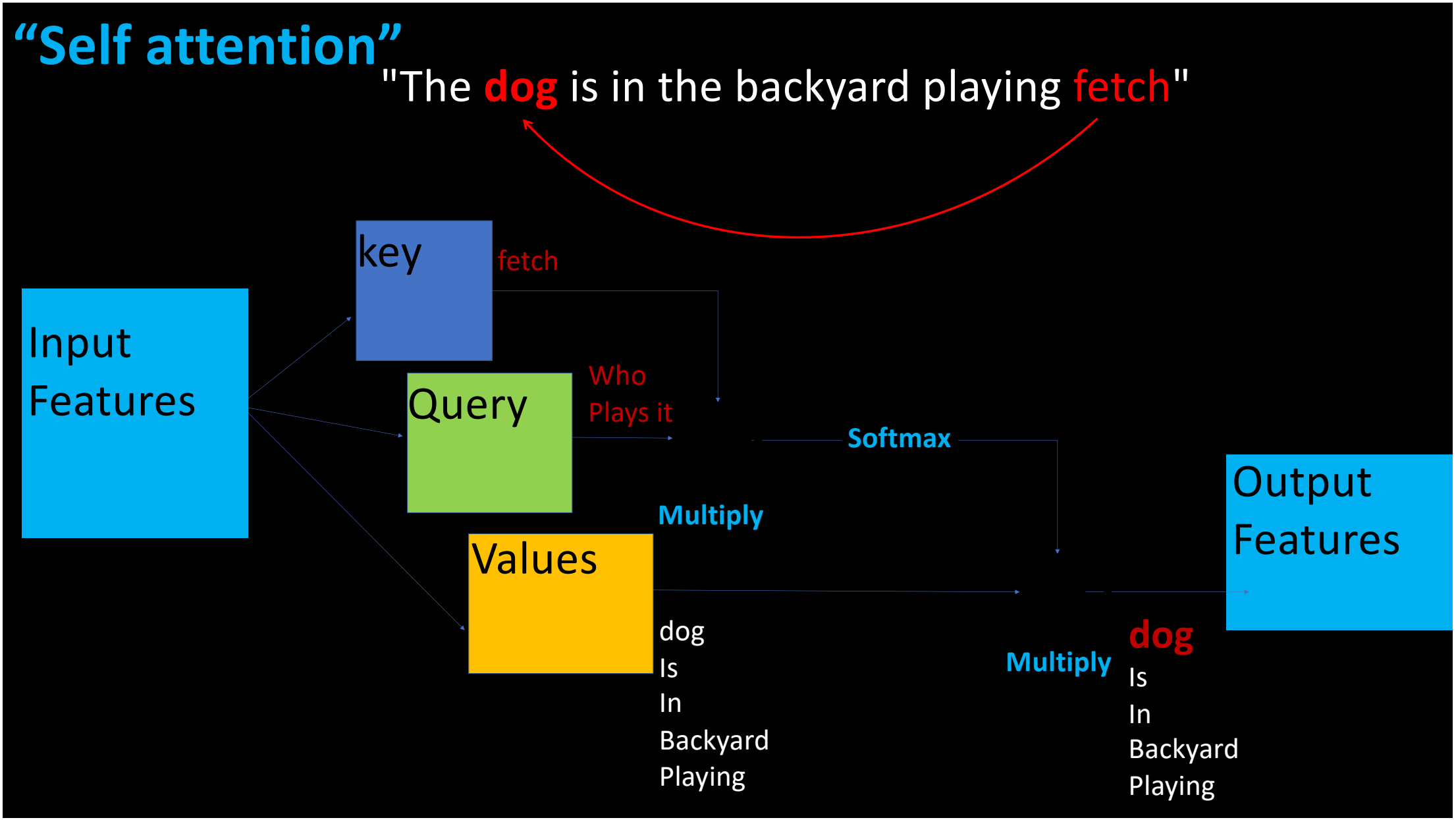


"The **dog** is in the backyard playing fetch"

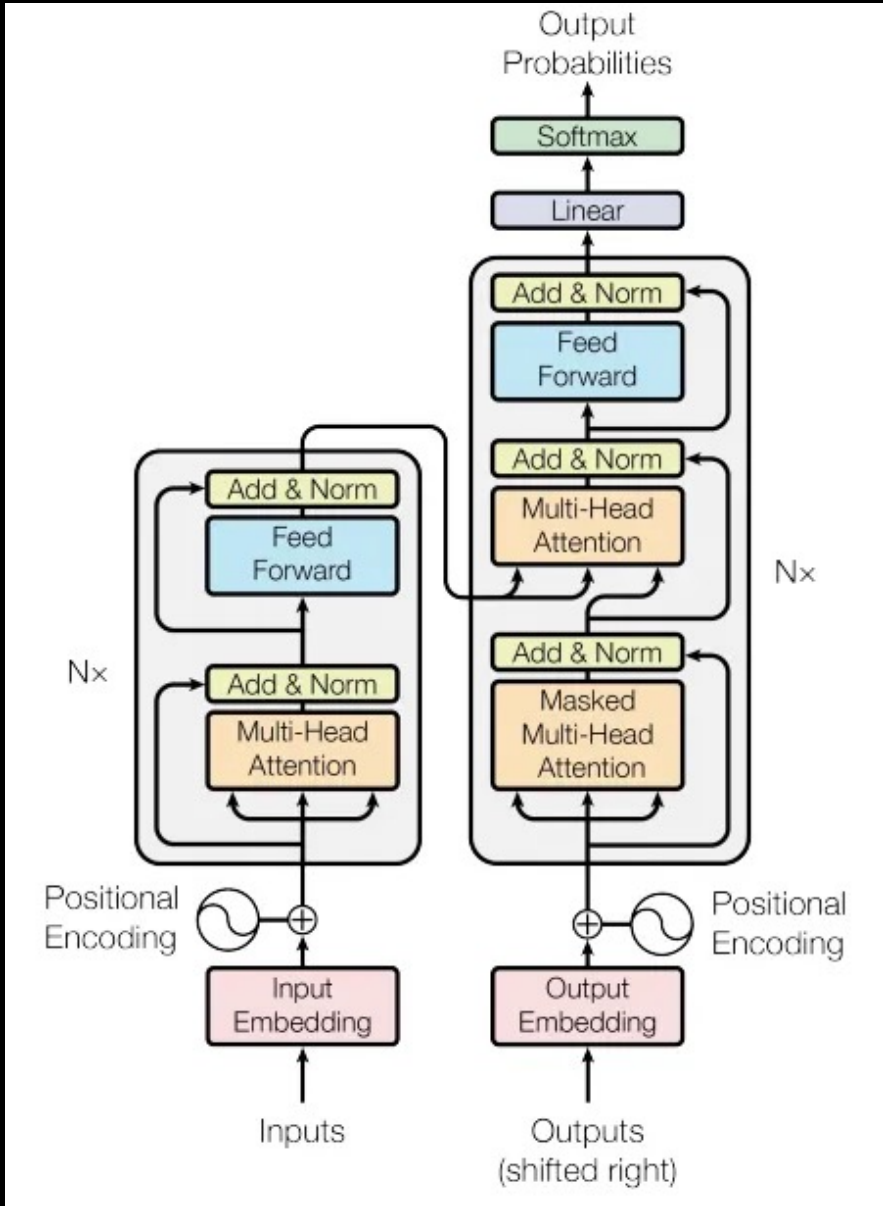
Need a way to connect distant information


“Self attention”
The
Dog
Is
In
backyard
playing
fetch
512 (features)
7
(words)
key
Coded Content
(dog: a domesticated wolf….)
Query
What connection to look after?
(what’s the game it play?….)
Values
Content of the connection
“fetch”


"Self attention"
"The dog is in the backyard playing fetch"
key
dog
Input Features
Query
What it plays
Softmax
Output Features
Multiply
Values
Is In Backyard Playing fetch
Multiply
Is In Backyard Playing fetch

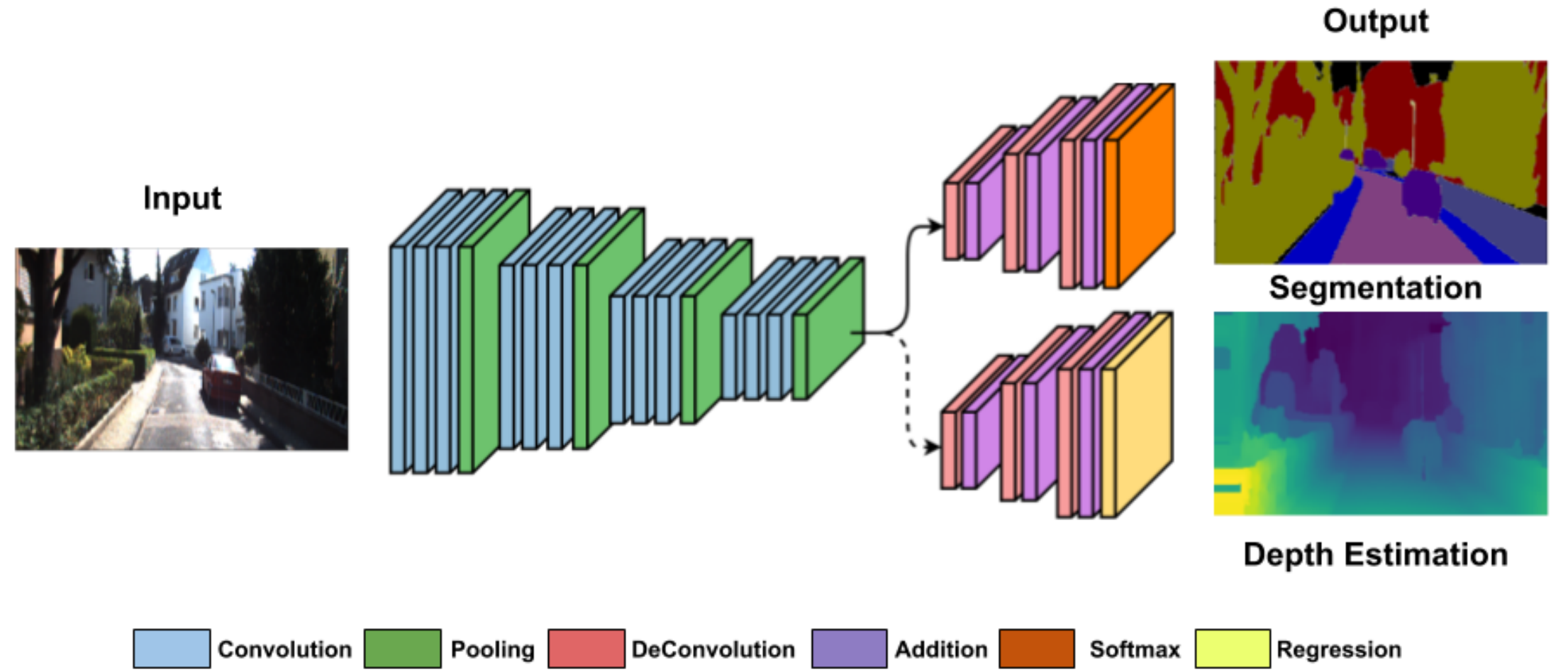

“Self attention”
"The dog is in the backyard playing fetch"
key
fetch
Input Features
Query
Who Plays it
Softmax
Output Features
Multiply
Values
dog
Is
In
Backyard
Playing
Multiply
dog
Is
In
Backyard
Playing

Transformer Architecture

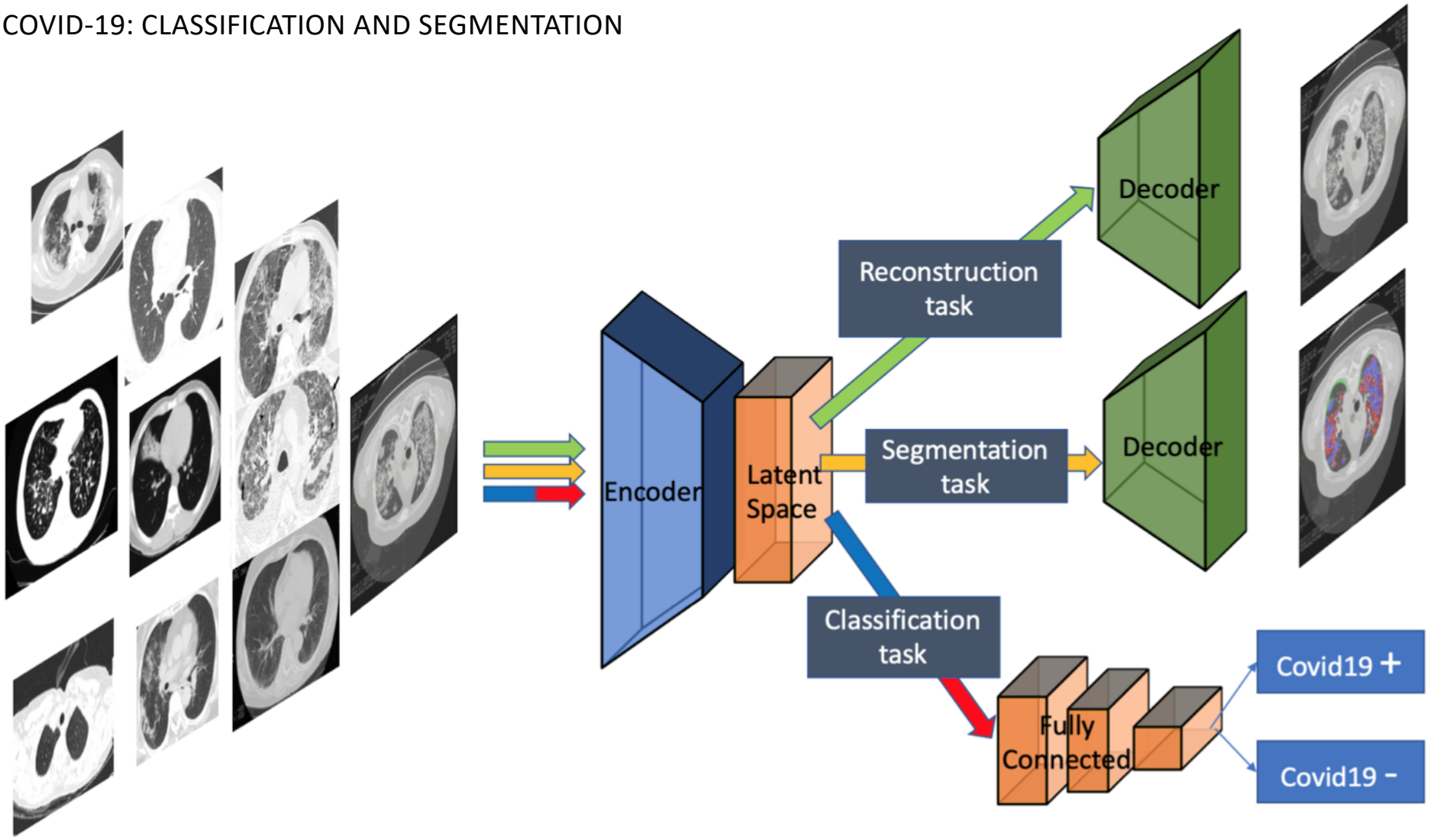


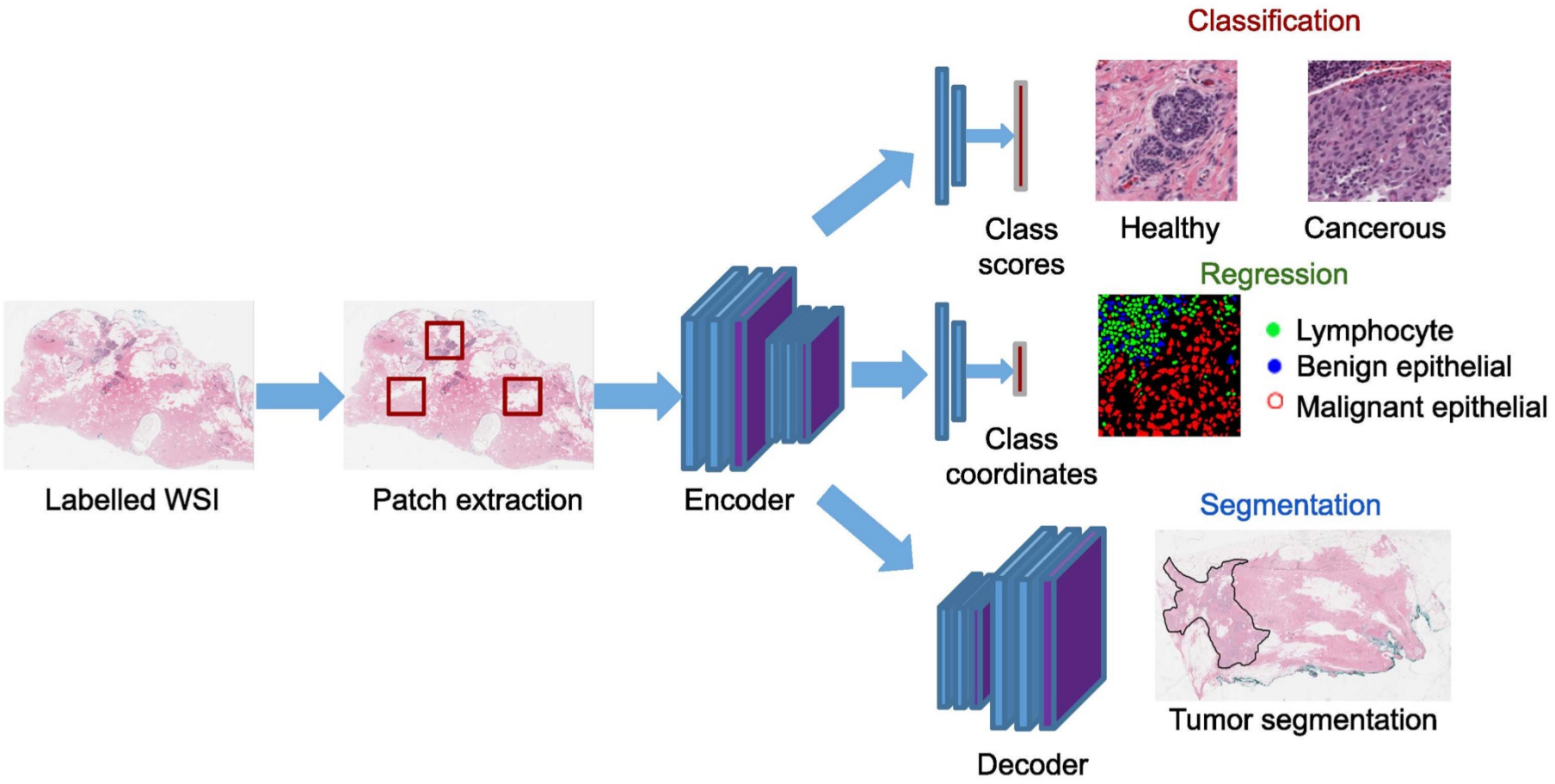
MULTI-TASK

TRAINING

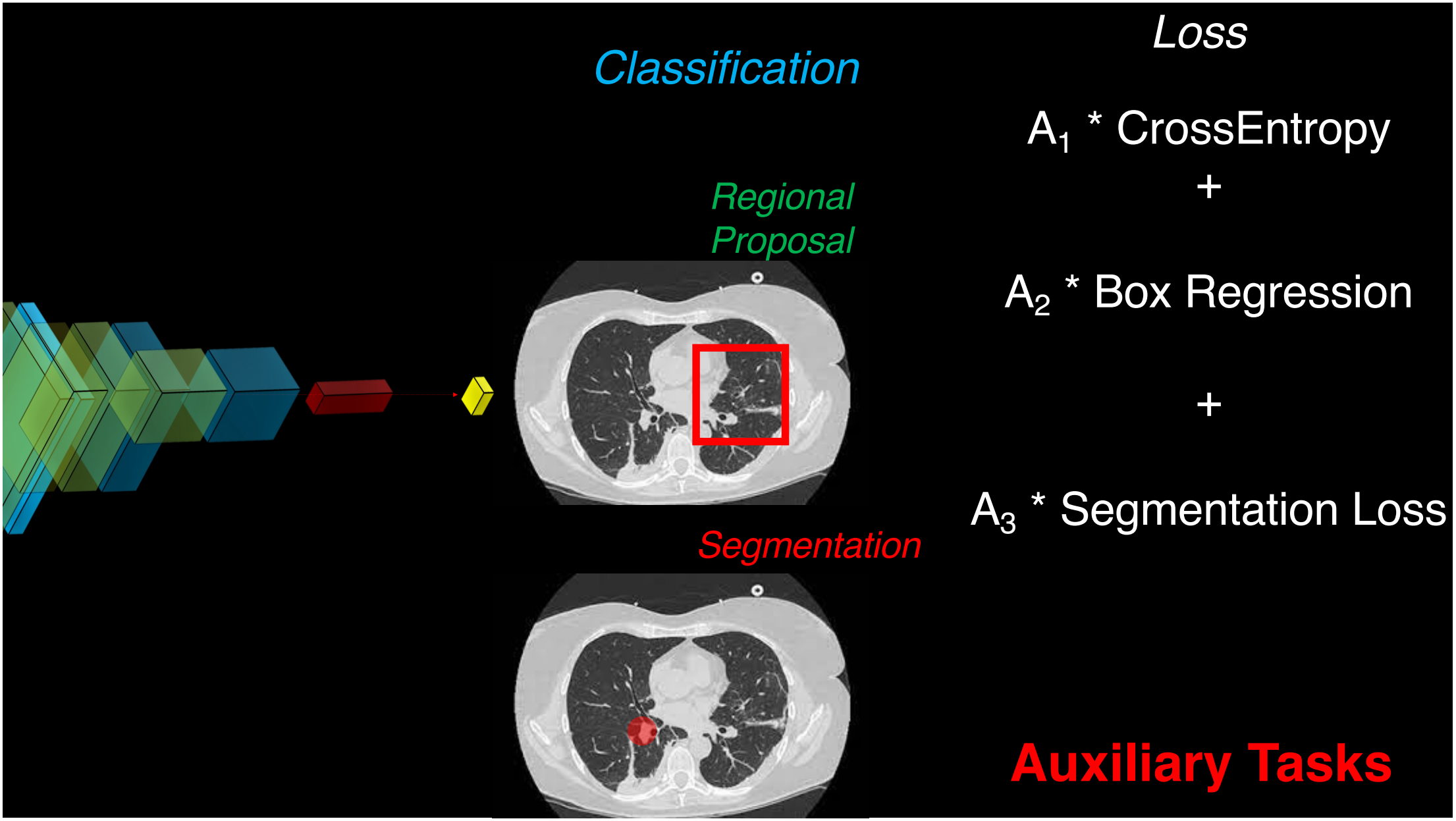

Input
Output
Segmentation
Depth Estimation
Convolution
Pooling
DeConvolution
Addition
Softmax
Regression

MULTI-TASK DEEP LEARNING BASED CT IMAGING ANALYSIS FOR COVID-19: CLASSIFICATION AND SEGMENTATION





Deep neural network models for computational histopathology:
A survey


Classification
Regional Proposal
Segmentation
Loss
A_1 * CrossEntropy
+
A_2 * Box Regression
+
A_3 * Segmentation Loss
Auxiliary Tasks

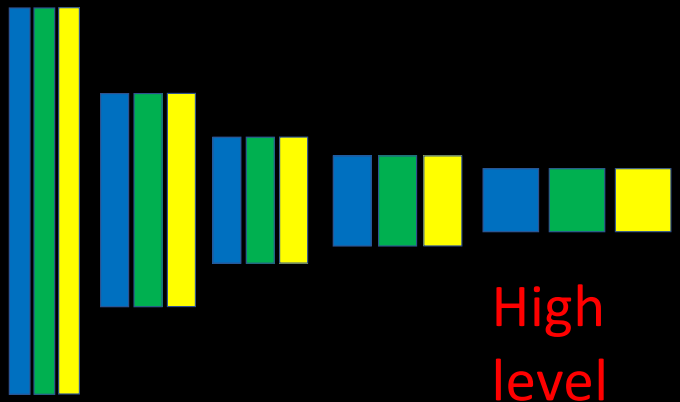
Auxiliary Tasks:

May improve the main task

Achieve multiple objects

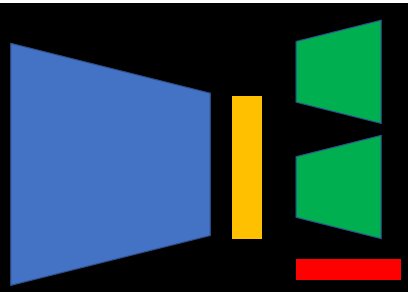
May improve explainability

Summary

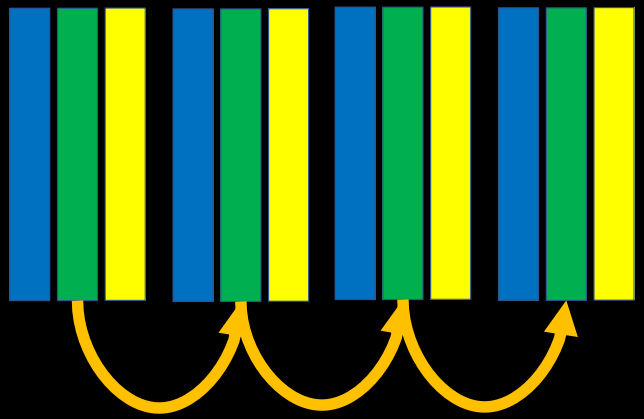


Low level features (edges, textures)

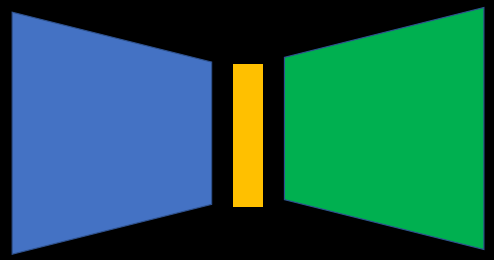
High level features (Overall shapes)



A good latent space allow you to do many tasks simultaneously



Residual give you very deep networks



Use encoder-decoder to go back to image level